SampleSize_n100_10Treatments/table2_nonshrink_10_treatments_MAR_n100

MR	T1vs.T0_none	T2vs.T0_none	T3vs.T0_none	T4vs.T0_none	T5vs.T0_none	T6vs.T0_none	T7vs.T0_none	T8vs.T0_none	T9vs.T0_none	T10vs.T0_none	Joint_none
Scenario I: all means are 0 for T0,T1,...,T10	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
0%	93.10	93.10	96.55	96.55	93.10	96.55	100.00	96.55	93.10	100.00	72.41
Imputation using linear regression	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	89.29	85.71	96.43	92.86	100.00	96.43	85.71	96.43	92.86	92.86	67.86
20%	85.71	96.43	96.43	100.00	85.71	89.29	92.86	92.86	92.86	100.00	64.29
30%	89.66	89.66	96.55	96.55	96.55	96.55	89.66	82.76	96.55	96.55	62.07
40%	93.10	93.10	96.55	100.00	93.10	93.10	93.10	93.10	89.66	93.10	65.52
Imputation using predictive mean matching method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	89.29	82.14	92.86	89.29	100.00	96.43	85.71	96.43	96.43	92.86	60.71
20%	82.14	92.86	96.43	100.00	89.29	89.29	89.29	85.71	89.29	96.43	57.14
30%	89.66	86.21	96.55	96.55	96.55	96.55	86.21	86.21	89.66	96.55	58.62
40%	93.10	93.10	93.10	96.55	93.10	93.10	89.66	82.76	89.66	96.55	58.62
Imputation using propensity−score method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	89.29	89.29	92.86	92.86	100.00	92.86	85.71	96.43	96.43	96.43	64.29
20%	85.71	96.43	96.43	100.00	89.29	92.86	89.29	89.29	92.86	100.00	64.29
30%	89.66	86.21	96.55	96.55	96.55	96.55	89.66	86.21	93.10	96.55	58.62
40%	89.66	93.10	96.55	86.21	89.66	89.66	89.66	86.21	89.66	89.66	51.72
Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu10=6.0	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
0%	93.10	96.55	93.10	93.10	96.55	96.55	100.00	93.10	96.55	100.00	65.52
Imputation using linear regression	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA

SampleSize_n100_10Treatments/table2_nonshrink_10_treatments_MAR_n100

MR	T1vs.T0_none	T2vs.T0_none	T3vs.T0_none	T4vs.T0_none	T5vs.T0_none	T6vs.T0_none	T7vs.T0_none	T8vs.T0_none	T9vs.T0_none	T10vs.T0_none	Joint_none
10%	92.86	92.86	100.00	96.43	96.43	89.29	92.86	96.43	100.00	92.86	67.86
20%	96.55	89.66	100.00	96.55	96.55	86.21	93.10	93.10	86.21	93.10	62.07
30%	89.66	96.55	96.55	93.10	96.55	86.21	86.21	89.66	100.00	89.66	68.97
40%	92.86	89.29	96.43	92.86	96.43	96.43	96.43	96.43	92.86	96.43	64.29
Imputation using predictive mean matching method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	92.86	92.86	100.00	96.43	96.43	92.86	96.43	96.43	100.00	92.86	67.86
20%	93.10	89.66	100.00	96.55	96.55	86.21	93.10	93.10	86.21	93.10	62.07
30%	82.76	96.55	96.55	89.66	93.10	89.66	89.66	86.21	96.55	86.21	65.52
40%	92.86	85.71	96.43	89.29	96.43	96.43	92.86	96.43	89.29	96.43	57.14
Imputation using propensity−score method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	92.86	92.86	100.00	92.86	96.43	89.29	89.29	96.43	100.00	92.86	64.29
20%	93.10	89.66	100.00	96.55	96.55	89.66	93.10	93.10	89.66	93.10	65.52
30%	86.21	96.55	96.55	93.10	96.55	86.21	86.21	82.76	96.55	86.21	62.07
40%	96.43	85.71	96.43	89.29	96.43	96.43	96.43	96.43	92.86	100.00	57.14

MR	T1vs.T0_many	T2vs.T0_many	T3vs.T0_many	T4vs.T0_many	T5vs.T0_many	T6vs.T0_many	T7vs.T0_many	T8vs.T0_many	T9vs.T0_many	T10vs.T0_many	Joint_many
Scenario I: all means are 0 for T0,T1,...,T10	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
0%	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100	100.00
Imputation using linear regression	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	96.43	100.00	100.00	96.43	100.00	100.00	100.00	100.00	100.00	100	92.86
20%	100.00	96.43	100.00	100.00	96.43	100.00	100.00	96.43	100.00	100	96.43
30%	100.00	100.00	100.00	100.00	96.55	100.00	96.55	96.55	100.00	100	89.66
40%	100.00	96.55	100.00	100.00	100.00	100.00	96.55	96.55	100.00	100	93.10
Imputation using predictive mean matching method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	96.43	100.00	100.00	96.43	100.00	100.00	100.00	100.00	100.00	100	92.86
20%	100.00	96.43	100.00	100.00	96.43	100.00	100.00	96.43	100.00	100	96.43
30%	100.00	100.00	100.00	100.00	96.55	100.00	96.55	96.55	100.00	100	89.66
40%	100.00	96.55	100.00	100.00	96.55	93.10	93.10	96.55	96.55	100	82.76
Imputation using propensity−score method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	96.43	100.00	100.00	96.43	100.00	96.43	100.00	100.00	96.43	100	85.71
20%	100.00	96.43	100.00	100.00	96.43	100.00	100.00	96.43	100.00	100	96.43
30%	100.00	100.00	100.00	100.00	96.55	100.00	96.55	96.55	100.00	100	89.66
40%	100.00	93.10	100.00	100.00	100.00	100.00	96.55	96.55	100.00	100	93.10
Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu10=6.0	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
0%	93.10	100.00	96.55	100.00	100.00	100.00	100.00	100.00	100.00	100	93.10
Imputation using linear regression	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA

SampleSize_n100_10Treatments/table2_nonshrink_10_treatments_MAR_n100

MR	T1vs.T0_many	T2vs.T0_many	T3vs.T0_many	T4vs.T0_many	T5vs.T0_many	T6vs.T0_many	T7vs.T0_many	T8vs.T0_many	T9vs.T0_many	T10vs.T0_many	Joint_many
10%	92.86	96.43	100.00	100.00	100	100.00	100.00	100.00	100.00	100.00	92.86
20%	100.00	96.55	100.00	100.00	100	100.00	100.00	96.55	93.10	100.00	86.21
30%	96.55	100.00	100.00	100.00	100	96.55	96.55	100.00	100.00	100.00	93.10
40%	100.00	100.00	96.43	96.43	100	100.00	100.00	100.00	96.43	100.00	89.29
Imputation using predictive mean matching method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	92.86	96.43	100.00	100.00	100	100.00	96.43	100.00	100.00	100.00	89.29
20%	100.00	100.00	100.00	100.00	100	100.00	100.00	96.55	93.10	100.00	89.66
30%	96.55	100.00	100.00	100.00	100	96.55	96.55	100.00	100.00	96.55	89.66
40%	100.00	96.43	96.43	96.43	100	100.00	96.43	100.00	96.43	100.00	85.71
Imputation using propensity−score method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	92.86	96.43	100.00	100.00	100	100.00	100.00	100.00	100.00	100.00	92.86
20%	100.00	100.00	100.00	100.00	100	100.00	100.00	96.55	93.10	100.00	89.66
30%	96.55	100.00	100.00	100.00	100	96.55	96.55	100.00	100.00	100.00	93.10
40%	100.00	92.86	100.00	100.00	100	100.00	96.43	100.00	96.43	100.00	89.29

SampleSize_n100_10Treatments/table2_shrink_10_treatments_MAR_n100

MR	T1vs.T0_none	T2vs.T0_none	T3vs.T0_none	T4vs.T0_none	T5vs.T0_none	T6vs.T0_none	T7vs.T0_none	T8vs.T0_none	T9vs.T0_none	T10vs.T0_none	Joint_none
Scenario I: all means are 0 for T0,T1,...,T10	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
0%	93.10	93.10	96.55	96.55	93.10	96.55	100.00	96.55	93.10	100.00	72.41
Imputation using linear regression	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	89.29	85.71	96.43	92.86	100.00	96.43	85.71	96.43	92.86	92.86	67.86
20%	85.71	96.43	96.43	100.00	85.71	89.29	92.86	92.86	92.86	100.00	64.29
30%	89.66	89.66	96.55	96.55	96.55	96.55	89.66	82.76	96.55	96.55	62.07
40%	93.10	93.10	96.55	100.00	93.10	93.10	93.10	93.10	89.66	93.10	65.52
Imputation using predictive mean matching method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	89.29	82.14	92.86	89.29	100.00	96.43	85.71	96.43	96.43	92.86	60.71
20%	82.14	92.86	96.43	100.00	89.29	89.29	89.29	85.71	89.29	96.43	57.14
30%	89.66	86.21	96.55	96.55	96.55	96.55	86.21	86.21	89.66	96.55	58.62
40%	93.10	93.10	93.10	96.55	93.10	93.10	89.66	82.76	89.66	96.55	58.62
Imputation using propensity−score method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	89.29	89.29	92.86	92.86	100.00	92.86	85.71	96.43	96.43	96.43	64.29
20%	85.71	96.43	96.43	100.00	89.29	92.86	89.29	89.29	92.86	100.00	64.29
30%	89.66	86.21	96.55	96.55	96.55	96.55	89.66	86.21	93.10	96.55	58.62
40%	89.66	93.10	96.55	86.21	89.66	89.66	89.66	86.21	89.66	89.66	51.72
Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu10=6.0	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
0%	93.10	96.55	93.10	93.10	96.55	96.55	100.00	93.10	96.55	100.00	65.52
Imputation using linear regression	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA

SampleSize_n100_10Treatments/table2_shrink_10_treatments_MAR_n100

MR	T1vs.T0_none	T2vs.T0_none	T3vs.T0_none	T4vs.T0_none	T5vs.T0_none	T6vs.T0_none	T7vs.T0_none	T8vs.T0_none	T9vs.T0_none	T10vs.T0_none	Joint_none
10%	92.86	92.86	100.00	96.43	96.43	89.29	92.86	96.43	100.00	92.86	67.86
20%	96.55	89.66	100.00	96.55	96.55	86.21	93.10	93.10	86.21	93.10	62.07
30%	89.66	96.55	96.55	93.10	96.55	86.21	86.21	89.66	100.00	89.66	68.97
40%	92.86	89.29	96.43	92.86	96.43	96.43	96.43	96.43	92.86	96.43	64.29
Imputation using predictive mean matching method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	92.86	92.86	100.00	96.43	96.43	92.86	96.43	96.43	100.00	92.86	67.86
20%	93.10	89.66	100.00	96.55	96.55	86.21	93.10	93.10	86.21	93.10	62.07
30%	82.76	96.55	96.55	89.66	93.10	89.66	89.66	86.21	96.55	86.21	65.52
40%	92.86	85.71	96.43	89.29	96.43	96.43	92.86	96.43	89.29	96.43	57.14
Imputation using propensity−score method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	92.86	92.86	100.00	92.86	96.43	89.29	89.29	96.43	100.00	92.86	64.29
20%	93.10	89.66	100.00	96.55	96.55	89.66	93.10	93.10	89.66	93.10	65.52
30%	86.21	96.55	96.55	93.10	96.55	86.21	86.21	82.76	96.55	86.21	62.07
40%	96.43	85.71	96.43	89.29	96.43	96.43	96.43	96.43	92.86	100.00	57.14

MR	T1vs.T0_many	T2vs.T0_many	T3vs.T0_many	T4vs.T0_many	T5vs.T0_many	T6vs.T0_many	T7vs.T0_many	T8vs.T0_many	T9vs.T0_many	T10vs.T0_many	Joint_many
Scenario I: all means are 0 for T0,T1,...,T10	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
0%	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100	100.00
Imputation using linear regression	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	96.43	100.00	96.43	96.43	100.00	100.00	100.00	100.00	100.00	100	89.29
20%	100.00	96.43	100.00	100.00	96.43	100.00	100.00	96.43	100.00	100	96.43
30%	100.00	100.00	100.00	100.00	96.55	100.00	96.55	96.55	100.00	100	89.66
40%	100.00	96.55	100.00	100.00	100.00	100.00	96.55	96.55	100.00	100	93.10
Imputation using predictive mean matching method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	96.43	100.00	100.00	96.43	100.00	100.00	100.00	100.00	100.00	100	92.86
20%	100.00	96.43	100.00	100.00	96.43	100.00	100.00	96.43	100.00	100	96.43
30%	100.00	100.00	100.00	100.00	96.55	100.00	96.55	96.55	100.00	100	89.66
40%	100.00	96.55	100.00	100.00	96.55	93.10	93.10	96.55	96.55	100	82.76
Imputation using propensity−score method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	96.43	100.00	100.00	96.43	100.00	100.00	100.00	100.00	100.00	100	92.86
20%	100.00	96.43	100.00	100.00	96.43	96.43	100.00	96.43	100.00	100	96.43
30%	100.00	100.00	100.00	100.00	96.55	100.00	96.55	96.55	100.00	100	89.66
40%	100.00	93.10	100.00	100.00	100.00	100.00	96.55	96.55	100.00	100	93.10
Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu10=6.0	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
0%	93.10	100.00	96.55	100.00	100.00	100.00	100.00	96.55	100.00	100	89.66
Imputation using linear regression	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA

SampleSize_n100_10Treatments/table2_shrink_10_treatments_MAR_n100

MR	T1vs.T0_many	T2vs.T0_many	T3vs.T0_many	T4vs.T0_many	T5vs.T0_many	T6vs.T0_many	T7vs.T0_many	T8vs.T0_many	T9vs.T0_many	T10vs.T0_many	Joint_many
10%	96.43	96.43	100.00	96.43	100	100.00	100.00	100.00	100.00	100	92.86
20%	100.00	96.55	100.00	100.00	100	100.00	100.00	96.55	93.10	100	86.21
30%	100.00	100.00	100.00	100.00	100	96.55	96.55	100.00	100.00	100	93.10
40%	100.00	100.00	96.43	96.43	100	100.00	100.00	100.00	96.43	100	89.29
Imputation using predictive mean matching method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	96.43	96.43	100.00	96.43	100	100.00	96.43	100.00	100.00	100	89.29
20%	100.00	100.00	100.00	100.00	100	100.00	100.00	96.55	93.10	100	89.66
30%	96.55	100.00	100.00	100.00	100	96.55	96.55	100.00	100.00	100	93.10
40%	100.00	100.00	96.43	96.43	100	100.00	96.43	100.00	96.43	100	89.29
Imputation using propensity−score method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	96.43	96.43	100.00	96.43	100	100.00	100.00	100.00	100.00	100	92.86
20%	100.00	100.00	100.00	100.00	100	100.00	100.00	96.55	93.10	100	89.66
30%	96.55	100.00	100.00	100.00	100	96.55	96.55	100.00	100.00	100	93.10
40%	100.00	92.86	100.00	100.00	100	100.00	96.43	100.00	96.43	100	89.29

SampleSize_n100_10Treatments/table3_nonshrink_10_treatments_MAR_n100

MR	None_T1vs.T0	None_T2vs.T0	None_T3vs.T0	None_T4vs.T0	None_T5vs.T0	None_T6vs.T0	None_T7vs.T0	None_T8vs.T0	None_T9vs.T0	None_T10vs.T0	Many_T1vs.T0
Scenario I: all means are 0 for T0,T1,...,T10											
0%	1.757	1.761	1.770	1.795	1.795	1.796	1.842	1.813	1.839	1.841	2.417
Imputation using linear regression											
10%	1.836 (4.49)	1.863 (5.77)	1.876 (6.00)	1.857 (3.47)	1.853 (3.20)	1.906 (6.11)	1.869 (1.50)	1.905 (5.06)	1.921 (4.46)	1.909 (3.67)	2.533 (4.80)
20%	1.950 (10.97)	1.946 (10.47)	1.958 (10.62)	1.987 (10.70)	1.966 (9.51)	2.024 (12.69)	2.001 (8.63)	2.018 (11.29)	2.012 (9.45)	2.009 (9.12)	2.689 (11.26)
30%	2.104 (19.71)	2.102 (19.32)	2.133 (20.53)	2.142 (19.32)	2.142 (19.31)	2.172 (20.93)	2.152 (16.85)	2.186 (20.56)	2.153 (17.08)	2.204 (19.70)	2.901 (20.03)
40%	2.324 (32.22)	2.293 (30.16)	2.327 (31.47)	2.328 (29.71)	2.338 (30.22)	2.329 (29.64)	2.318 (25.89)	2.384 (31.51)	2.403 (30.70)	2.406 (30.65)	3.201 (32.46)
Imputation using predictive mean matching method											
10%	1.820 (3.54)	1.850 (5.05)	1.854 (4.73)	1.841 (2.57)	1.847 (2.88)	1.883 (4.81)	1.853 (0.63)	1.890 (4.27)	1.905 (3.63)	1.902 (3.30)	2.509 (3.82)
20%	1.912 (8.81)	1.912 (8.57)	1.913 (8.11)	1.930 (7.52)	1.930 (7.53)	1.984 (10.43)	1.945 (5.63)	1.956 (7.87)	1.969 (7.08)	1.976 (7.29)	2.639 (9.19)
30%	1.987 (13.07)	2.013 (14.28)	2.042 (15.39)	2.062 (14.85)	2.029 (13.03)	2.076 (15.58)	2.036 (10.55)	2.081 (14.78)	2.075 (12.86)	2.084 (13.16)	2.740 (13.36)
40%	2.182 (24.15)	2.180 (23.74)	2.178 (23.08)	2.201 (22.61)	2.228 (24.13)	2.239 (24.67)	2.209 (19.94)	2.211 (21.94)	2.258 (22.80)	2.214 (20.24)	3.007 (24.42)
Imputation using propensity−score method											
10%	1.823 (3.71)	1.857 (5.41)	1.854 (4.73)	1.845 (2.78)	1.842 (2.60)	1.895 (5.50)	1.861 (1.04)	1.892 (4.35)	1.922 (4.52)	1.901 (3.25)	2.513 (3.99)
20%	1.915 (8.98)	1.932 (9.68)	1.928 (8.93)	1.951 (8.71)	1.949 (8.60)	2.005 (11.64)	1.963 (6.60)	1.976 (9.01)	1.989 (8.18)	1.997 (8.43)	2.643 (9.36)
30%	2.041 (16.15)	2.063 (17.10)	2.039 (15.21)	2.085 (16.13)	2.079 (15.84)	2.077 (15.65)	2.048 (11.19)	2.105 (16.09)	2.085 (13.39)	2.127 (15.53)	2.818 (16.59)
40%	2.218 (26.22)	2.230 (26.63)	2.228 (25.89)	2.270 (26.46)	2.259 (25.85)	2.293 (27.66)	2.288 (24.23)	2.279 (25.72)	2.318 (26.07)	2.283 (24.00)	3.056 (26.45)
Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu10=6.0											
0%	1.761	1.765	1.777	1.771	1.808	1.795	1.801	1.821	1.819	1.840	2.418
Imputation using linear regression											

SampleSize_n100_10Treatments/table3_nonshrink_10_treatments_MAR_n100

MR	None_T1vs.T0	None_T2vs.T0	None_T3vs.T0	None_T4vs.T0	None_T5vs.T0	None_T6vs.T0	None_T7vs.T0	None_T8vs.T0	None_T9vs.T0	None_T10vs.T0	Many_T1vs.T0
10%	1.825 (3.63)	1.815 (2.82)	1.850 (4.12)	1.879 (6.09)	1.858 (2.82)	1.877 (4.58)	1.859 (3.21)	1.909 (4.84)	1.890 (3.88)	1.892 (2.81)	2.521 (4.26)
20%	1.960 (11.28)	1.950 (10.43)	1.977 (11.28)	1.970 (11.22)	1.981 (9.62)	1.986 (10.64)	2.000 (11.04)	2.005 (10.13)	2.000 (9.95)	2.030 (10.33)	2.704 (11.83)
30%	2.068 (17.39)	2.073 (17.39)	2.046 (15.16)	2.069 (16.81)	2.101 (16.22)	2.103 (17.14)	2.142 (18.94)	2.156 (18.42)	2.193 (20.57)	2.176 (18.25)	2.858 (18.22)
40%	2.250 (27.76)	2.276 (28.89)	2.230 (25.51)	2.276 (28.50)	2.249 (24.41)	2.363 (31.63)	2.316 (28.61)	2.305 (26.61)	2.329 (28.05)	2.389 (29.85)	3.109 (28.59)
Imputation using predictive mean matching method											
10%	1.804 (2.41)	1.800 (1.97)	1.839 (3.53)	1.860 (5.04)	1.839 (1.72)	1.866 (3.96)	1.838 (2.04)	1.888 (3.69)	1.878 (3.26)	1.879 (2.10)	2.490 (2.99)
20%	1.918 (8.92)	1.917 (8.60)	1.932 (8.74)	1.947 (9.95)	1.957 (8.28)	1.952 (8.74)	1.962 (8.94)	1.961 (7.70)	1.961 (7.80)	1.992 (8.27)	2.645 (9.43)
30%	1.949 (10.66)	1.974 (11.78)	1.947 (9.61)	1.966 (11.01)	2.015 (11.47)	1.984 (10.51)	2.048 (13.69)	2.006 (10.16)	2.072 (13.94)	2.036 (10.64)	2.695 (11.47)
40%	2.136 (21.29)	2.159 (22.32)	2.155 (21.29)	2.146 (21.19)	2.162 (19.61)	2.212 (23.20)	2.201 (22.20)	2.209 (21.32)	2.209 (21.44)	2.238 (21.60)	2.945 (21.81)
Imputation using propensity−score method											
10%	1.808 (2.64)	1.799 (1.89)	1.838 (3.48)	1.870 (5.60)	1.848 (2.25)	1.868 (4.03)	1.843 (2.35)	1.896 (4.16)	1.878 (3.27)	1.883 (2.33)	2.496 (3.24)
20%	1.931 (9.64)	1.935 (9.59)	1.950 (9.77)	1.954 (10.35)	1.975 (9.28)	1.976 (10.08)	1.971 (9.44)	1.993 (9.44)	1.978 (8.77)	2.002 (8.81)	2.664 (10.18)
30%	1.999 (13.46)	2.007 (13.69)	2.005 (12.86)	1.984 (12.05)	2.033 (12.48)	2.030 (13.10)	2.080 (15.49)	2.038 (11.95)	2.125 (16.81)	2.068 (12.38)	2.765 (14.37)
40%	2.171 (23.23)	2.208 (25.05)	2.188 (23.18)	2.261 (27.67)	2.236 (23.69)	2.311 (28.74)	2.265 (25.79)	2.295 (26.05)	2.238 (23.05)	2.282 (24.01)	2.990 (23.68)

SampleSize_n100_10Treatments/table3_nonshrink_10_treatments_MAR_n100

MR	Many_T2vs.T0	Many_T3vs.T0	Many_T4vs.T0	Many_T5vs.T0	Many_T6vs.T0	Many_T7vs.T0	Many_T8vs.T0	Many_T9vs.T0	Many_T10vs.T0
Scenario I: all means are 0 for T0,T1,...,T10									
0%	2.422	2.434	2.469	2.469	2.470	2.533	2.493	2.529	2.532
Imputation using linear regression									
10%	2.570 (6.08)	2.587 (6.30)	2.561 (3.76)	2.555 (3.50)	2.629 (6.42)	2.578 (1.78)	2.627 (5.37)	2.649 (4.75)	2.633 (3.97)
20%	2.684 (10.79)	2.699 (10.91)	2.740 (11.00)	2.711 (9.82)	2.791 (13.00)	2.758 (8.91)	2.781 (11.56)	2.775 (9.73)	2.771 (9.42)
30%	2.898 (19.65)	2.940 (20.82)	2.954 (19.67)	2.953 (19.61)	2.995 (21.23)	2.966 (17.11)	3.013 (20.83)	2.969 (17.41)	3.038 (19.98)
40%	3.159 (30.43)	3.206 (31.74)	3.208 (29.97)	3.221 (30.49)	3.209 (29.90)	3.197 (26.21)	3.285 (31.76)	3.311 (30.93)	3.315 (30.90)
Imputation using predictive mean matching method									
10%	2.551 (5.33)	2.556 (5.00)	2.539 (2.84)	2.546 (3.15)	2.596 (5.09)	2.555 (0.89)	2.607 (4.55)	2.627 (3.90)	2.623 (3.57)
20%	2.639 (8.97)	2.640 (8.49)	2.664 (7.90)	2.664 (7.92)	2.738 (10.83)	2.685 (6.01)	2.699 (8.25)	2.717 (7.44)	2.727 (7.68)
30%	2.775 (14.57)	2.814 (15.63)	2.842 (15.14)	2.798 (13.32)	2.861 (15.84)	2.806 (10.80)	2.868 (15.05)	2.861 (13.14)	2.872 (13.42)
40%	3.003 (23.99)	3.001 (23.33)	3.033 (22.85)	3.071 (24.38)	3.085 (24.90)	3.044 (20.19)	3.047 (22.21)	3.110 (22.99)	3.052 (20.53)
Imputation using propensity−score method									
10%	2.560 (5.69)	2.556 (5.01)	2.544 (3.04)	2.540 (2.87)	2.613 (5.78)	2.566 (1.30)	2.608 (4.62)	2.650 (4.79)	2.622 (3.52)
20%	2.666 (10.05)	2.660 (9.30)	2.693 (9.09)	2.690 (8.98)	2.767 (12.01)	2.709 (6.97)	2.727 (9.37)	2.745 (8.54)	2.755 (8.80)
30%	2.848 (17.56)	2.815 (15.68)	2.879 (16.61)	2.870 (16.27)	2.869 (16.14)	2.828 (11.64)	2.906 (16.55)	2.880 (13.88)	2.937 (15.98)
40%	3.072 (26.83)	3.070 (26.14)	3.126 (26.64)	3.112 (26.08)	3.158 (27.83)	3.151 (24.42)	3.140 (25.95)	3.193 (26.26)	3.146 (24.23)
Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu10=6.0									
0%	2.423	2.439	2.431	2.481	2.464	2.472	2.499	2.497	2.526
Imputation using linear regression									

SampleSize_n100_10Treatments/table3_nonshrink_10_treatments_MAR_n100

MR	Many_T2vs.T0	Many_T3vs.T0	Many_T4vs.T0	Many_T5vs.T0	Many_T6vs.T0	Many_T7vs.T0	Many_T8vs.T0	Many_T9vs.T0	Many_T10vs.T0
10%	2.506 (3.44)	2.554 (4.74)	2.594 (6.73)	2.566 (3.44)	2.593 (5.21)	2.567 (3.83)	2.636 (5.47)	2.609 (4.51)	2.612 (3.43)
20%	2.690 (11.00)	2.728 (11.86)	2.717 (11.78)	2.734 (10.18)	2.740 (11.20)	2.758 (11.58)	2.766 (10.68)	2.759 (10.51)	2.801 (10.90)
30%	2.865 (18.22)	2.828 (15.97)	2.859 (17.63)	2.904 (17.04)	2.906 (17.94)	2.961 (19.78)	2.980 (19.22)	3.031 (21.40)	3.006 (19.02)
40%	3.145 (29.77)	3.083 (26.42)	3.145 (29.38)	3.109 (25.32)	3.267 (32.57)	3.201 (29.49)	3.186 (27.48)	3.219 (28.94)	3.302 (30.72)
Imputation using predictive mean matching method									
10%	2.485 (2.54)	2.539 (4.11)	2.568 (5.63)	2.538 (2.30)	2.576 (4.55)	2.537 (2.62)	2.606 (4.28)	2.593 (3.84)	2.593 (2.68)
20%	2.644 (9.12)	2.665 (9.27)	2.685 (10.47)	2.699 (8.78)	2.692 (9.26)	2.706 (9.44)	2.705 (8.23)	2.704 (8.32)	2.748 (8.80)
30%	2.729 (12.61)	2.693 (10.42)	2.718 (11.81)	2.785 (12.25)	2.743 (11.32)	2.831 (14.50)	2.774 (10.98)	2.865 (14.77)	2.815 (11.43)
40%	2.977 (22.86)	2.971 (21.82)	2.960 (21.76)	2.981 (20.16)	3.051 (23.81)	3.035 (22.77)	3.046 (21.87)	3.046 (22.02)	3.086 (22.18)
Imputation using propensity−score method									
10%	2.483 (2.47)	2.538 (4.07)	2.581 (6.20)	2.552 (2.84)	2.578 (4.63)	2.545 (2.94)	2.618 (4.76)	2.593 (3.86)	2.600 (2.92)
20%	2.668 (10.12)	2.690 (10.31)	2.695 (10.88)	2.724 (9.80)	2.726 (10.61)	2.719 (9.97)	2.748 (9.97)	2.729 (9.30)	2.762 (9.35)
30%	2.777 (14.60)	2.774 (13.75)	2.746 (12.96)	2.813 (13.37)	2.809 (14.00)	2.877 (16.39)	2.821 (12.87)	2.939 (17.72)	2.861 (13.28)
40%	3.042 (25.54)	3.016 (23.67)	3.114 (28.10)	3.080 (24.13)	3.184 (29.23)	3.122 (26.29)	3.162 (26.51)	3.085 (23.57)	3.145 (24.52)

MR	None_T1vs.T0	None_T2vs.T0	None_T3vs.T0	None_T4vs.T0	None_T5vs.T0	None_T6vs.T0	None_T7vs.T0	None_T8vs.T0	None_T9vs.T0	None_T10vs.T0	Many_T1vs.T0
Scenario I: all means are 0 for T0,T1,...,T10											
0%	1.757	1.761	1.770	1.795	1.795	1.796	1.842	1.813	1.839	1.841	2.443
Imputation using linear regression											
10%	1.836 (4.49)	1.863 (5.77)	1.876 (6.00)	1.857 (3.47)	1.853 (3.20)	1.906 (6.11)	1.869 (1.50)	1.905 (5.06)	1.921 (4.46)	1.909 (3.67)	2.542 (4.06)
20%	1.950 (10.97)	1.946 (10.47)	1.958 (10.62)	1.987 (10.70)	1.966 (9.51)	2.024 (12.69)	2.001 (8.63)	2.018 (11.29)	2.012 (9.45)	2.009 (9.12)	2.695 (10.34)
30%	2.104 (19.71)	2.102 (19.32)	2.133 (20.53)	2.142 (19.32)	2.142 (19.31)	2.172 (20.93)	2.152 (16.85)	2.186 (20.56)	2.153 (17.08)	2.204 (19.70)	2.907 (19.02)
40%	2.324 (32.22)	2.293 (30.16)	2.327 (31.47)	2.328 (29.71)	2.338 (30.22)	2.329 (29.64)	2.318 (25.89)	2.384 (31.51)	2.403 (30.70)	2.406 (30.65)	3.197 (30.90)
Imputation using predictive mean matching method											
10%	1.820 (3.54)	1.850 (5.05)	1.854 (4.73)	1.841 (2.57)	1.847 (2.88)	1.883 (4.81)	1.853 (0.63)	1.890 (4.27)	1.905 (3.63)	1.902 (3.30)	2.521 (3.21)
20%	1.912 (8.81)	1.912 (8.57)	1.913 (8.11)	1.930 (7.52)	1.930 (7.53)	1.984 (10.43)	1.945 (5.63)	1.956 (7.87)	1.969 (7.08)	1.976 (7.29)	2.645 (8.29)
30%	1.987 (13.07)	2.013 (14.28)	2.042 (15.39)	2.062 (14.85)	2.029 (13.03)	2.076 (15.58)	2.036 (10.55)	2.081 (14.78)	2.075 (12.86)	2.084 (13.16)	2.745 (12.37)
40%	2.182 (24.15)	2.180 (23.74)	2.178 (23.08)	2.201 (22.61)	2.228 (24.13)	2.239 (24.67)	2.209 (19.94)	2.211 (21.94)	2.258 (22.80)	2.214 (20.24)	3.001 (22.85)
Imputation using propensity−score method											
10%	1.823 (3.71)	1.857 (5.41)	1.854 (4.73)	1.845 (2.78)	1.842 (2.60)	1.895 (5.50)	1.861 (1.04)	1.892 (4.35)	1.922 (4.52)	1.901 (3.25)	2.524 (3.34)
20%	1.915 (8.98)	1.932 (9.68)	1.928 (8.93)	1.951 (8.71)	1.949 (8.60)	2.005 (11.64)	1.963 (6.60)	1.976 (9.01)	1.989 (8.18)	1.997 (8.43)	2.651 (8.52)
30%	2.041 (16.15)	2.063 (17.10)	2.039 (15.21)	2.085 (16.13)	2.079 (15.84)	2.077 (15.65)	2.048 (11.19)	2.105 (16.09)	2.085 (13.39)	2.127 (15.53)	2.828 (15.77)
40%	2.218 (26.22)	2.230 (26.63)	2.228 (25.89)	2.270 (26.46)	2.259 (25.85)	2.293 (27.66)	2.288 (24.23)	2.279 (25.72)	2.318 (26.07)	2.283 (24.00)	3.054 (25.01)
Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu10=6.0											
0%	1.761	1.765	1.777	1.771	1.808	1.795	1.801	1.821	1.819	1.840	2.418
Imputation using linear regression											

SampleSize_n100_10Treatments/table3_shrink_10_treatments_MAR_n100

MR	None_T1vs.T0	None_T2vs.T0	None_T3vs.T0	None_T4vs.T0	None_T5vs.T0	None_T6vs.T0	None_T7vs.T0	None_T8vs.T0	None_T9vs.T0	None_T10vs.T0	Many_T1vs.T0
10%	1.825 (3.63)	1.815 (2.82)	1.850 (4.12)	1.879 (6.09)	1.858 (2.82)	1.877 (4.58)	1.859 (3.21)	1.909 (4.84)	1.890 (3.88)	1.892 (2.81)	2.535 (4.81)
20%	1.960 (11.28)	1.950 (10.43)	1.977 (11.28)	1.970 (11.22)	1.981 (9.62)	1.986 (10.64)	2.000 (11.04)	2.005 (10.13)	2.000 (9.95)	2.030 (10.33)	2.715 (12.25)
30%	2.068 (17.39)	2.073 (17.39)	2.046 (15.16)	2.069 (16.81)	2.101 (16.22)	2.103 (17.14)	2.142 (18.94)	2.156 (18.42)	2.193 (20.57)	2.176 (18.25)	2.867 (18.53)
40%	2.250 (27.76)	2.276 (28.89)	2.230 (25.51)	2.276 (28.50)	2.249 (24.41)	2.363 (31.63)	2.316 (28.61)	2.305 (26.61)	2.329 (28.05)	2.389 (29.85)	3.133 (29.54)
Imputation using predictive mean matching method											
10%	1.804 (2.41)	1.800 (1.97)	1.839 (3.53)	1.860 (5.04)	1.839 (1.72)	1.866 (3.96)	1.838 (2.04)	1.888 (3.69)	1.878 (3.26)	1.879 (2.10)	2.505 (3.59)
20%	1.918 (8.92)	1.917 (8.60)	1.932 (8.74)	1.947 (9.95)	1.957 (8.28)	1.952 (8.74)	1.962 (8.94)	1.961 (7.70)	1.961 (7.80)	1.992 (8.27)	2.657 (9.87)
30%	1.949 (10.66)	1.974 (11.78)	1.947 (9.61)	1.966 (11.01)	2.015 (11.47)	1.984 (10.51)	2.048 (13.69)	2.006 (10.16)	2.072 (13.94)	2.036 (10.64)	2.710 (12.04)
40%	2.136 (21.29)	2.159 (22.32)	2.155 (21.29)	2.146 (21.19)	2.162 (19.61)	2.212 (23.20)	2.201 (22.20)	2.209 (21.32)	2.209 (21.44)	2.238 (21.60)	2.972 (22.87)
Imputation using propensity−score method											
10%	1.808 (2.64)	1.799 (1.89)	1.838 (3.48)	1.870 (5.60)	1.848 (2.25)	1.868 (4.03)	1.843 (2.35)	1.896 (4.16)	1.878 (3.27)	1.883 (2.33)	2.512 (3.86)
20%	1.931 (9.64)	1.935 (9.59)	1.950 (9.77)	1.954 (10.35)	1.975 (9.28)	1.976 (10.08)	1.971 (9.44)	1.993 (9.44)	1.978 (8.77)	2.002 (8.81)	2.676 (10.66)
30%	1.999 (13.46)	2.007 (13.69)	2.005 (12.86)	1.984 (12.05)	2.033 (12.48)	2.030 (13.10)	2.080 (15.49)	2.038 (11.95)	2.125 (16.81)	2.068 (12.38)	2.775 (14.75)
40%	2.171 (23.23)	2.208 (25.05)	2.188 (23.18)	2.261 (27.67)	2.236 (23.69)	2.311 (28.74)	2.265 (25.79)	2.295 (26.05)	2.238 (23.05)	2.282 (24.01)	3.013 (24.60)

MR	Many_T2vs.T0	Many_T3vs.T0	Many_T4vs.T0	Many_T5vs.T0	Many_T6vs.T0	Many_T7vs.T0	Many_T8vs.T0	Many_T9vs.T0	Many_T10vs.T0
Scenario I: all means are 0 for T0,T1,...,T10									
0%	2.445	2.442	2.451	2.454	2.450	2.463	2.460	2.468	2.468
Imputation using linear regression									
10%	2.547 (4.20)	2.568 (5.15)	2.550 (4.02)	2.542 (3.61)	2.581 (5.36)	2.547 (3.45)	2.577 (4.76)	2.582 (4.62)	2.576 (4.40)
20%	2.688 (9.95)	2.705 (10.76)	2.722 (11.04)	2.696 (9.90)	2.747 (12.10)	2.736 (11.10)	2.758 (12.10)	2.743 (11.11)	2.727 (10.51)
30%	2.894 (18.38)	2.934 (20.14)	2.924 (19.32)	2.938 (19.73)	2.969 (21.19)	2.955 (20.00)	2.987 (21.44)	2.935 (18.90)	3.002 (21.65)
40%	3.156 (29.11)	3.198 (30.94)	3.195 (30.36)	3.204 (30.59)	3.197 (30.49)	3.177 (29.01)	3.259 (32.46)	3.284 (33.05)	3.277 (32.80)
Imputation using predictive mean matching method									
10%	2.531 (3.53)	2.540 (3.98)	2.532 (3.30)	2.533 (3.23)	2.547 (3.97)	2.526 (2.57)	2.557 (3.96)	2.558 (3.62)	2.565 (3.92)
20%	2.640 (8.01)	2.650 (8.49)	2.648 (8.05)	2.649 (7.98)	2.695 (10.00)	2.661 (8.06)	2.675 (8.76)	2.689 (8.93)	2.680 (8.59)
30%	2.768 (13.22)	2.812 (15.14)	2.810 (14.64)	2.777 (13.16)	2.838 (15.83)	2.798 (13.63)	2.836 (15.29)	2.822 (14.34)	2.835 (14.86)
40%	2.996 (22.57)	2.995 (22.63)	3.017 (23.09)	3.047 (24.18)	3.063 (25.03)	3.019 (22.61)	3.016 (22.60)	3.083 (24.90)	3.016 (22.20)
Imputation using propensity−score method									
10%	2.538 (3.83)	2.539 (3.97)	2.535 (3.41)	2.527 (3.00)	2.565 (4.71)	2.536 (2.98)	2.560 (4.05)	2.580 (4.52)	2.565 (3.94)
20%	2.669 (9.16)	2.669 (9.29)	2.679 (9.28)	2.675 (9.02)	2.725 (11.22)	2.685 (9.05)	2.705 (9.95)	2.716 (10.03)	2.710 (9.79)
30%	2.845 (16.36)	2.814 (15.21)	2.849 (16.24)	2.855 (16.38)	2.848 (16.25)	2.821 (14.56)	2.878 (17.00)	2.843 (15.18)	2.903 (17.62)
40%	3.068 (25.49)	3.064 (25.46)	3.115 (27.10)	3.096 (26.19)	3.141 (28.19)	3.132 (27.17)	3.114 (26.61)	3.167 (28.29)	3.116 (26.24)
Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu10=6.0									
0%	2.420	2.421	2.424	2.435	2.428	2.436	2.436	2.439	2.445
Imputation using linear regression									

MR	Many_T2vs.T0	Many_T3vs.T0	Many_T4vs.T0	Many_T5vs.T0	Many_T6vs.T0	Many_T7vs.T0	Many_T8vs.T0	Many_T9vs.T0	Many_T10vs.T0
10%	2.530 (4.55)	2.549 (5.27)	2.559 (5.58)	2.552 (4.82)	2.553 (5.17)	2.547 (4.56)	2.584 (6.05)	2.561 (5.01)	2.568 (5.02)
20%	2.701 (11.61)	2.719 (12.32)	2.714 (11.98)	2.722 (11.77)	2.724 (12.21)	2.736 (12.32)	2.736 (12.31)	2.729 (11.89)	2.752 (12.54)
30%	2.866 (18.44)	2.842 (17.39)	2.863 (18.15)	2.894 (18.85)	2.899 (19.41)	2.934 (20.47)	2.959 (21.46)	2.994 (22.75)	2.980 (21.87)
40%	3.151 (30.22)	3.091 (27.68)	3.146 (29.80)	3.108 (27.63)	3.237 (33.32)	3.190 (30.96)	3.171 (30.14)	3.199 (31.17)	3.274 (33.88)
Imputation using predictive mean matching method									
10%	2.508 (3.65)	2.531 (4.53)	2.532 (4.47)	2.523 (3.63)	2.537 (4.48)	2.519 (3.42)	2.555 (4.88)	2.543 (4.27)	2.546 (4.14)
20%	2.658 (9.82)	2.662 (9.94)	2.684 (10.75)	2.686 (10.32)	2.678 (10.31)	2.682 (10.11)	2.677 (9.89)	2.676 (9.72)	2.701 (10.45)
30%	2.727 (12.69)	2.705 (11.73)	2.726 (12.48)	2.777 (14.06)	2.735 (12.66)	2.808 (15.28)	2.750 (12.88)	2.823 (15.74)	2.786 (13.94)
40%	2.991 (23.58)	2.978 (23.02)	2.962 (22.23)	2.981 (22.41)	3.018 (24.32)	3.021 (24.04)	3.027 (24.25)	3.025 (24.01)	3.056 (24.99)
Imputation using propensity−score method									
10%	2.507 (3.61)	2.531 (4.53)	2.548 (5.14)	2.538 (4.24)	2.538 (4.55)	2.525 (3.68)	2.569 (5.45)	2.546 (4.38)	2.554 (4.46)
20%	2.681 (10.79)	2.686 (10.93)	2.690 (11.01)	2.714 (11.44)	2.710 (11.62)	2.694 (10.59)	2.720 (11.67)	2.699 (10.67)	2.714 (10.99)
30%	2.777 (14.74)	2.787 (15.11)	2.752 (13.55)	2.803 (15.11)	2.799 (15.28)	2.854 (17.16)	2.799 (14.91)	2.899 (18.88)	2.833 (15.84)
40%	3.054 (26.18)	3.022 (24.81)	3.113 (28.45)	3.077 (26.36)	3.153 (29.88)	3.108 (27.62)	3.143 (29.02)	3.065 (25.68)	3.116 (27.44)

SampleSize_n150_10Treatments/table2_nonshrink_10_treatments_MAR_n150

MR	T1vs.T0_none	T2vs.T0_none	T3vs.T0_none	T4vs.T0_none	T5vs.T0_none	T6vs.T0_none	T7vs.T0_none	T8vs.T0_none	T9vs.T0_none	T10vs.T0_none	Joint_none
Scenario I: all means are 0 for T0,T1,...,T10	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
0%	86.21	89.66	89.66	93.10	93.10	93.10	93.10	96.55	93.10	96.55	65.52
Imputation using linear regression	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	92.86	92.86	92.86	100.00	100.00	92.86	100.00	96.43	96.43	96.43	78.57
20%	96.55	100.00	100.00	96.55	96.55	93.10	93.10	93.10	96.55	93.10	68.97
30%	86.21	100.00	79.31	89.66	93.10	89.66	96.55	89.66	86.21	96.55	48.28
40%	89.29	96.43	85.71	100.00	89.29	96.43	89.29	92.86	92.86	92.86	57.14
Imputation using predictive mean matching method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	92.86	92.86	89.29	100.00	96.43	92.86	100.00	96.43	96.43	96.43	78.57
20%	96.55	100.00	100.00	96.55	96.55	93.10	96.55	93.10	96.55	89.66	72.41
30%	82.76	100.00	75.86	89.66	89.66	93.10	100.00	86.21	86.21	86.21	44.83
40%	85.71	100.00	85.71	96.43	85.71	96.43	89.29	92.86	92.86	89.29	53.57
Imputation using propensity−score method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	92.86	92.86	89.29	100.00	100.00	92.86	100.00	96.43	96.43	96.43	78.57
20%	93.10	96.55	100.00	96.55	96.55	93.10	96.55	89.66	96.55	89.66	72.41
30%	79.31	96.55	82.76	86.21	93.10	89.66	100.00	89.66	86.21	93.10	55.17
40%	82.14	96.43	92.86	96.43	85.71	100.00	85.71	92.86	96.43	89.29	46.43
Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu10=6.0	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
0%	92.86	89.29	96.43	89.29	89.29	89.29	92.86	89.29	89.29	92.86	67.86
Imputation using linear regression	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA

SampleSize_n150_10Treatments/table2_nonshrink_10_treatments_MAR_n150

MR	T1vs.T0_none	T2vs.T0_none	T3vs.T0_none	T4vs.T0_none	T5vs.T0_none	T6vs.T0_none	T7vs.T0_none	T8vs.T0_none	T9vs.T0_none	T10vs.T0_none	Joint_none
10%	92.86	100.00	96.43	92.86	100.00	100.00	100.00	100.00	100.00	100.00	82.14
20%	86.21	96.55	96.55	89.66	89.66	96.55	89.66	86.21	93.10	86.21	55.17
30%	100.00	96.55	96.55	100.00	96.55	96.55	100.00	100.00	96.55	96.55	79.31
40%	92.86	92.86	92.86	96.43	89.29	82.14	89.29	92.86	92.86	89.29	64.29
Imputation using predictive mean matching method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	92.86	100.00	96.43	92.86	100.00	100.00	100.00	100.00	100.00	100.00	82.14
20%	86.21	89.66	93.10	89.66	89.66	89.66	89.66	86.21	93.10	82.76	48.28
30%	100.00	89.66	93.10	96.55	96.55	93.10	100.00	100.00	100.00	96.55	72.41
40%	92.86	92.86	92.86	96.43	89.29	78.57	89.29	89.29	100.00	92.86	50.00
Imputation using propensity−score method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	89.29	100.00	96.43	92.86	100.00	100.00	100.00	96.43	100.00	100.00	75.00
20%	86.21	93.10	93.10	86.21	89.66	93.10	89.66	89.66	93.10	82.76	51.72
30%	100.00	93.10	100.00	100.00	96.55	100.00	100.00	100.00	100.00	93.10	82.76
40%	96.43	92.86	92.86	96.43	89.29	82.14	92.86	89.29	100.00	92.86	60.71

SampleSize_n150_10Treatments/table2_nonshrink_10_treatments_MAR_n150

MR	T1vs.T0_many	T2vs.T0_many	T3vs.T0_many	T4vs.T0_many	T5vs.T0_many	T6vs.T0_many	T7vs.T0_many	T8vs.T0_many	T9vs.T0_many	T10vs.T0_many	Joint_many
Scenario I: all means are 0 for T0,T1,...,T10	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
0%	96.55	96.55	100.00	96.55	100.00	96.55	100.00	100.00	93.10	100.00	82.76
Imputation using linear regression	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	96.43	96.43
20%	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	96.55	100.00	96.55
30%	96.55	100.00	100.00	100.00	100.00	93.10	100.00	100.00	100.00	100.00	93.10
40%	96.43	100.00	100.00	100.00	100.00	100.00	96.43	100.00	100.00	100.00	92.86
Imputation using predictive mean matching method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	96.43	96.43
20%	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	96.55	100.00	96.55
30%	89.66	100.00	96.55	100.00	100.00	96.55	100.00	100.00	96.55	96.55	89.66
40%	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00
Imputation using propensity−score method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00
20%	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	96.55	100.00	96.55
30%	96.55	100.00	96.55	100.00	96.55	96.55	100.00	100.00	100.00	96.55	89.66
40%	96.43	100.00	100.00	100.00	96.43	100.00	96.43	100.00	100.00	100.00	89.29
Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu10=6.0	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
0%	100.00	100.00	100.00	100.00	96.43	100.00	100.00	96.43	100.00	96.43	89.29
Imputation using linear regression	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA

SampleSize_n150_10Treatments/table2_nonshrink_10_treatments_MAR_n150

MR	T1vs.T0_many	T2vs.T0_many	T3vs.T0_many	T4vs.T0_many	T5vs.T0_many	T6vs.T0_many	T7vs.T0_many	T8vs.T0_many	T9vs.T0_many	T10vs.T0_many	Joint_many
10%	100.00	100.00	100.00	100.00	100	100.00	100.00	100.00	100.00	100.00	100.00
20%	96.55	100.00	96.55	100.00	100	100.00	96.55	96.55	100.00	100.00	89.66
30%	100.00	100.00	100.00	100.00	100	100.00	100.00	100.00	100.00	100.00	100.00
40%	96.43	100.00	100.00	96.43	100	96.43	100.00	100.00	100.00	96.43	92.86
Imputation using predictive mean matching method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	100.00	100.00	100.00	100.00	100	100.00	100.00	100.00	100.00	100.00	100.00
20%	96.55	100.00	100.00	100.00	100	100.00	96.55	89.66	96.55	96.55	86.21
30%	100.00	100.00	100.00	100.00	100	100.00	100.00	100.00	100.00	100.00	100.00
40%	96.43	100.00	96.43	96.43	100	96.43	92.86	96.43	100.00	100.00	85.71
Imputation using propensity−score method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	100.00	100.00	100.00	96.43	100	100.00	100.00	100.00	100.00	100.00	96.43
20%	96.55	100.00	96.55	100.00	100	100.00	96.55	93.10	96.55	96.55	86.21
30%	100.00	100.00	100.00	100.00	100	100.00	100.00	100.00	100.00	100.00	100.00
40%	96.43	96.43	100.00	96.43	100	96.43	96.43	96.43	100.00	100.00	89.29

SampleSize_n150_10Treatments/table2_shrink_10_treatments_MAR_n150

MR	T1vs.T0_none	T2vs.T0_none	T3vs.T0_none	T4vs.T0_none	T5vs.T0_none	T6vs.T0_none	T7vs.T0_none	T8vs.T0_none	T9vs.T0_none	T10vs.T0_none	Joint_none
Scenario I: all means are 0 for T0,T1,...,T10	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
0%	86.21	89.66	89.66	93.10	93.10	93.10	93.10	96.55	93.10	96.55	65.52
Imputation using linear regression	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	92.86	92.86	92.86	100.00	100.00	92.86	100.00	96.43	96.43	96.43	78.57
20%	96.55	100.00	100.00	96.55	96.55	93.10	93.10	93.10	96.55	93.10	68.97
30%	86.21	100.00	79.31	89.66	93.10	89.66	96.55	89.66	86.21	96.55	48.28
40%	89.29	96.43	85.71	100.00	89.29	96.43	89.29	92.86	92.86	92.86	57.14
Imputation using predictive mean matching method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	92.86	92.86	89.29	100.00	96.43	92.86	100.00	96.43	96.43	96.43	78.57
20%	96.55	100.00	100.00	96.55	96.55	93.10	96.55	93.10	96.55	89.66	72.41
30%	82.76	100.00	75.86	89.66	89.66	93.10	100.00	86.21	86.21	86.21	44.83
40%	85.71	100.00	85.71	96.43	85.71	96.43	89.29	92.86	92.86	89.29	53.57
Imputation using propensity−score method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	92.86	92.86	89.29	100.00	100.00	92.86	100.00	96.43	96.43	96.43	78.57
20%	93.10	96.55	100.00	96.55	96.55	93.10	96.55	89.66	96.55	89.66	72.41
30%	79.31	96.55	82.76	86.21	93.10	89.66	100.00	89.66	86.21	93.10	55.17
40%	82.14	96.43	92.86	96.43	85.71	100.00	85.71	92.86	96.43	89.29	46.43
Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu10=6.0	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
0%	92.86	89.29	96.43	89.29	89.29	89.29	92.86	89.29	89.29	92.86	67.86
Imputation using linear regression	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA

SampleSize_n150_10Treatments/table2_shrink_10_treatments_MAR_n150

MR	T1vs.T0_none	T2vs.T0_none	T3vs.T0_none	T4vs.T0_none	T5vs.T0_none	T6vs.T0_none	T7vs.T0_none	T8vs.T0_none	T9vs.T0_none	T10vs.T0_none	Joint_none
10%	92.86	100.00	96.43	92.86	100.00	100.00	100.00	100.00	100.00	100.00	82.14
20%	86.21	96.55	96.55	89.66	89.66	96.55	89.66	86.21	93.10	86.21	55.17
30%	100.00	96.55	96.55	100.00	96.55	96.55	100.00	100.00	96.55	96.55	79.31
40%	92.86	92.86	92.86	96.43	89.29	82.14	89.29	92.86	92.86	89.29	64.29
Imputation using predictive mean matching method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	92.86	100.00	96.43	92.86	100.00	100.00	100.00	100.00	100.00	100.00	82.14
20%	86.21	89.66	93.10	89.66	89.66	89.66	89.66	86.21	93.10	82.76	48.28
30%	100.00	89.66	93.10	96.55	96.55	93.10	100.00	100.00	100.00	96.55	72.41
40%	92.86	92.86	92.86	96.43	89.29	78.57	89.29	89.29	100.00	92.86	50.00
Imputation using propensity−score method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	89.29	100.00	96.43	92.86	100.00	100.00	100.00	96.43	100.00	100.00	75.00
20%	86.21	93.10	93.10	86.21	89.66	93.10	89.66	89.66	93.10	82.76	51.72
30%	100.00	93.10	100.00	100.00	96.55	100.00	100.00	100.00	100.00	93.10	82.76
40%	96.43	92.86	92.86	96.43	89.29	82.14	92.86	89.29	100.00	92.86	60.71

MR	T1vs.T0_many	T2vs.T0_many	T3vs.T0_many	T4vs.T0_many	T5vs.T0_many	T6vs.T0_many	T7vs.T0_many	T8vs.T0_many	T9vs.T0_many	T10vs.T0_many	Joint_many
Scenario I: all means are 0 for T0,T1,...,T10	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
0%	96.55	96.55	100.00	96.55	100.00	96.55	100.00	100.00	93.10	100.00	82.76
Imputation using linear regression	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00
20%	100.00	100.00	100.00	100.00	100.00	100.00	100.00	96.55	96.55	100.00	93.10
30%	96.55	100.00	96.55	100.00	100.00	93.10	100.00	96.55	100.00	100.00	86.21
40%	96.43	100.00	100.00	100.00	100.00	100.00	96.43	100.00	100.00	100.00	92.86
Imputation using predictive mean matching method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00
20%	100.00	100.00	100.00	100.00	100.00	100.00	100.00	96.55	96.55	100.00	93.10
30%	89.66	100.00	96.55	100.00	100.00	96.55	100.00	100.00	96.55	96.55	89.66
40%	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00
Imputation using propensity−score method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00
20%	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	96.55	100.00	96.55
30%	96.55	100.00	96.55	100.00	96.55	93.10	100.00	100.00	100.00	96.55	86.21
40%	96.43	100.00	100.00	100.00	96.43	100.00	96.43	100.00	100.00	100.00	89.29
Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu10=6.0	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
0%	100.00	100.00	100.00	100.00	96.43	100.00	100.00	96.43	100.00	96.43	89.29
Imputation using linear regression	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA

SampleSize_n150_10Treatments/table2_shrink_10_treatments_MAR_n150

MR	T1vs.T0_many	T2vs.T0_many	T3vs.T0_many	T4vs.T0_many	T5vs.T0_many	T6vs.T0_many	T7vs.T0_many	T8vs.T0_many	T9vs.T0_many	T10vs.T0_many	Joint_many
10%	100.00	100.00	100.00	100.00	100	100.00	100.00	100.00	100.00	100.00	100.00
20%	96.55	100.00	100.00	100.00	100	100.00	96.55	96.55	100.00	100.00	93.10
30%	100.00	100.00	100.00	100.00	100	100.00	100.00	100.00	100.00	100.00	100.00
40%	96.43	100.00	96.43	96.43	100	96.43	100.00	100.00	100.00	96.43	89.29
Imputation using predictive mean matching method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	100.00	100.00	100.00	100.00	100	100.00	100.00	100.00	100.00	100.00	100.00
20%	96.55	100.00	100.00	100.00	100	100.00	96.55	89.66	96.55	96.55	86.21
30%	100.00	100.00	100.00	100.00	100	100.00	100.00	100.00	100.00	100.00	100.00
40%	96.43	100.00	96.43	96.43	100	96.43	92.86	96.43	100.00	100.00	85.71
Imputation using propensity−score method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	100.00	100.00	100.00	100.00	100	100.00	100.00	100.00	100.00	100.00	100.00
20%	96.55	100.00	96.55	100.00	100	100.00	96.55	93.10	96.55	96.55	86.21
30%	100.00	100.00	100.00	100.00	100	100.00	100.00	100.00	100.00	100.00	100.00
40%	96.43	96.43	100.00	96.43	100	96.43	96.43	96.43	100.00	100.00	89.29

SampleSize_n150_10Treatments/table3_nonshrink_10_treatments_MAR_n150

MR	None_T1vs.T0	None_T2vs.T0	None_T3vs.T0	None_T4vs.T0	None_T5vs.T0	None_T6vs.T0	None_T7vs.T0	None_T8vs.T0	None_T9vs.T0	None_T10vs.T0	Many_T1vs.T0
Scenario I: all means are 0 for T0,T1,...,T10											
0%	1.449	1.445	1.453	1.462	1.485	1.493	1.480	1.489	1.504	1.515	1.992
Imputation using linear regression											
10%	1.502 (3.68)	1.498 (3.69)	1.497 (2.99)	1.523 (4.21)	1.520 (2.35)	1.534 (2.74)	1.537 (3.86)	1.544 (3.69)	1.545 (2.72)	1.557 (2.80)	2.074 (4.11)
20%	1.594 (10.08)	1.584 (9.65)	1.571 (8.12)	1.589 (8.69)	1.609 (8.31)	1.626 (8.87)	1.626 (9.87)	1.619 (8.71)	1.617 (7.50)	1.646 (8.64)	2.201 (10.53)
30%	1.707 (17.84)	1.705 (18.03)	1.721 (18.45)	1.740 (19.04)	1.759 (18.42)	1.729 (15.77)	1.768 (19.50)	1.787 (20.00)	1.753 (16.60)	1.807 (19.26)	2.358 (18.40)
40%	1.852 (27.89)	1.876 (29.86)	1.875 (29.01)	1.867 (27.71)	1.893 (27.43)	1.909 (27.84)	1.909 (29.05)	1.934 (29.89)	1.927 (28.14)	1.971 (30.08)	2.557 (28.39)
Imputation using predictive mean matching method											
10%	1.493 (3.09)	1.483 (2.67)	1.491 (2.63)	1.524 (4.26)	1.521 (2.39)	1.527 (2.24)	1.537 (3.91)	1.544 (3.71)	1.534 (2.02)	1.559 (2.93)	2.061 (3.46)
20%	1.555 (7.34)	1.552 (7.45)	1.543 (6.20)	1.575 (7.74)	1.591 (7.12)	1.601 (7.20)	1.610 (8.79)	1.594 (7.02)	1.590 (5.74)	1.632 (7.71)	2.147 (7.79)
30%	1.628 (12.41)	1.619 (12.05)	1.663 (14.44)	1.677 (14.73)	1.664 (12.05)	1.671 (11.92)	1.712 (15.74)	1.705 (14.53)	1.704 (13.31)	1.743 (15.03)	2.250 (12.94)
40%	1.809 (24.86)	1.797 (24.35)	1.827 (25.73)	1.836 (25.62)	1.835 (23.55)	1.822 (22.04)	1.858 (25.57)	1.830 (22.87)	1.849 (22.98)	1.904 (25.66)	2.489 (24.99)
Imputation using propensity−score method											
10%	1.490 (2.87)	1.487 (2.92)	1.490 (2.52)	1.520 (3.99)	1.516 (2.07)	1.530 (2.46)	1.531 (3.51)	1.539 (3.33)	1.532 (1.88)	1.558 (2.83)	2.058 (3.31)
20%	1.553 (7.25)	1.557 (7.75)	1.534 (5.58)	1.577 (7.92)	1.585 (6.69)	1.619 (8.39)	1.614 (9.11)	1.601 (7.53)	1.592 (5.87)	1.633 (7.81)	2.147 (7.82)
30%	1.661 (14.64)	1.633 (13.05)	1.691 (16.38)	1.690 (15.63)	1.690 (13.77)	1.706 (14.23)	1.728 (16.77)	1.726 (15.91)	1.731 (15.14)	1.737 (14.65)	2.297 (15.33)
40%	1.823 (25.83)	1.840 (27.36)	1.863 (28.18)	1.850 (26.55)	1.841 (23.95)	1.876 (25.64)	1.909 (29.03)	1.919 (28.86)	1.912 (27.17)	1.910 (26.05)	2.515 (26.27)
Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu10=6.0											
0%	1.421	1.426	1.446	1.446	1.452	1.454	1.473	1.468	1.473	1.466	1.957
Imputation using linear regression											

SampleSize_n150_10Treatments/table3_nonshrink_10_treatments_MAR_n150

MR	None_T1vs.T0	None_T2vs.T0	None_T3vs.T0	None_T4vs.T0	None_T5vs.T0	None_T6vs.T0	None_T7vs.T0	None_T8vs.T0	None_T9vs.T0	None_T10vs.T0	Many_T1vs.T0
10%	1.498 (5.41)	1.524 (6.82)	1.512 (4.57)	1.521 (5.21)	1.519 (4.63)	1.534 (5.47)	1.526 (3.62)	1.550 (5.56)	1.563 (6.13)	1.564 (6.66)	2.066 (5.56)
20%	1.570 (10.45)	1.582 (10.88)	1.601 (10.75)	1.604 (10.93)	1.615 (11.23)	1.623 (11.59)	1.635 (11.02)	1.630 (11.05)	1.648 (11.93)	1.646 (12.25)	2.164 (10.58)
30%	1.687 (18.76)	1.693 (18.68)	1.673 (15.71)	1.681 (16.27)	1.707 (17.59)	1.727 (18.75)	1.707 (15.93)	1.749 (19.16)	1.753 (19.03)	1.750 (19.37)	2.335 (19.31)
40%	1.876 (32.02)	1.860 (30.42)	1.904 (31.71)	1.845 (27.62)	1.904 (31.16)	1.875 (28.92)	1.895 (28.64)	1.931 (31.51)	1.973 (34.00)	1.954 (33.25)	2.588 (32.20)
Imputation using predictive mean matching method											
10%	1.484 (4.41)	1.513 (6.10)	1.502 (3.92)	1.508 (4.27)	1.505 (3.67)	1.529 (5.12)	1.518 (3.07)	1.541 (4.96)	1.547 (5.05)	1.555 (6.06)	2.047 (4.56)
20%	1.544 (8.67)	1.555 (8.99)	1.565 (8.23)	1.579 (9.22)	1.593 (9.73)	1.596 (9.73)	1.607 (9.12)	1.615 (10.03)	1.620 (10.01)	1.617 (10.27)	2.129 (8.75)
30%	1.658 (16.68)	1.649 (15.61)	1.642 (13.55)	1.639 (13.36)	1.656 (14.10)	1.698 (16.75)	1.674 (13.65)	1.713 (16.68)	1.738 (17.99)	1.702 (16.05)	2.291 (17.07)
40%	1.755 (23.50)	1.777 (24.57)	1.774 (22.70)	1.761 (21.76)	1.796 (23.74)	1.774 (22.01)	1.783 (21.07)	1.800 (22.63)	1.847 (25.45)	1.824 (24.38)	2.425 (23.89)
Imputation using propensity−score method											
10%	1.479 (4.11)	1.512 (5.97)	1.504 (4.01)	1.511 (4.49)	1.504 (3.64)	1.528 (5.09)	1.520 (3.22)	1.541 (4.98)	1.549 (5.16)	1.554 (5.98)	2.041 (4.28)
20%	1.554 (9.34)	1.562 (9.49)	1.582 (9.39)	1.596 (10.36)	1.605 (10.60)	1.616 (11.09)	1.629 (10.62)	1.613 (9.89)	1.629 (10.61)	1.628 (10.99)	2.142 (9.43)
30%	1.647 (15.94)	1.674 (17.37)	1.660 (14.84)	1.649 (14.02)	1.690 (16.41)	1.701 (16.98)	1.692 (14.87)	1.739 (18.44)	1.723 (17.02)	1.731 (18.07)	2.278 (16.38)
40%	1.846 (29.88)	1.849 (29.65)	1.867 (29.12)	1.819 (25.82)	1.859 (28.08)	1.850 (27.22)	1.875 (27.31)	1.874 (27.68)	1.900 (29.00)	1.910 (30.24)	2.546 (30.07)

SampleSize_n150_10Treatments/table3_nonshrink_10_treatments_MAR_n150

MR	Many_T2vs.T0	Many_T3vs.T0	Many_T4vs.T0	Many_T5vs.T0	Many_T6vs.T0	Many_T7vs.T0	Many_T8vs.T0	Many_T9vs.T0	Many_T10vs.T0
Scenario I: all means are 0 for T0,T1,...,T10									
0%	1.987	1.998	2.010	2.042	2.053	2.034	2.047	2.068	2.083
Imputation using linear regression									
10%	2.068 (4.11)	2.067 (3.41)	2.103 (4.64)	2.099 (2.77)	2.118 (3.16)	2.122 (4.29)	2.132 (4.13)	2.133 (3.14)	2.150 (3.22)
20%	2.187 (10.09)	2.170 (8.57)	2.194 (9.14)	2.221 (8.75)	2.245 (9.31)	2.245 (10.33)	2.235 (9.15)	2.232 (7.93)	2.272 (9.09)
30%	2.356 (18.57)	2.378 (19.03)	2.404 (19.60)	2.429 (18.95)	2.389 (16.36)	2.443 (20.07)	2.468 (20.56)	2.423 (17.17)	2.495 (19.78)
40%	2.589 (30.32)	2.588 (29.52)	2.577 (28.23)	2.611 (27.88)	2.633 (28.25)	2.635 (29.51)	2.669 (30.33)	2.660 (28.62)	2.720 (30.56)
Imputation using predictive mean matching method									
10%	2.047 (3.03)	2.058 (3.00)	2.103 (4.63)	2.099 (2.76)	2.107 (2.60)	2.122 (4.28)	2.131 (4.09)	2.117 (2.38)	2.152 (3.29)
20%	2.143 (7.89)	2.131 (6.64)	2.174 (8.16)	2.196 (7.53)	2.210 (7.64)	2.223 (9.25)	2.200 (7.46)	2.196 (6.17)	2.253 (8.14)
30%	2.237 (12.59)	2.298 (14.98)	2.316 (15.24)	2.299 (12.59)	2.309 (12.44)	2.365 (16.25)	2.356 (15.07)	2.354 (13.82)	2.406 (15.51)
40%	2.474 (24.50)	2.515 (25.87)	2.528 (25.77)	2.526 (23.67)	2.508 (22.15)	2.557 (25.71)	2.519 (23.04)	2.547 (23.16)	2.620 (25.78)
Imputation using propensity−score method									
10%	2.053 (3.35)	2.057 (2.96)	2.099 (4.44)	2.093 (2.51)	2.113 (2.90)	2.115 (3.96)	2.125 (3.78)	2.116 (2.32)	2.151 (3.27)
20%	2.152 (8.31)	2.121 (6.14)	2.180 (8.46)	2.190 (7.24)	2.237 (8.93)	2.231 (9.67)	2.213 (8.08)	2.200 (6.41)	2.257 (8.37)
30%	2.260 (13.75)	2.339 (17.07)	2.338 (16.32)	2.337 (14.46)	2.359 (14.90)	2.390 (17.48)	2.388 (16.62)	2.395 (15.81)	2.402 (15.33)
40%	2.538 (27.77)	2.569 (28.58)	2.552 (26.99)	2.539 (24.34)	2.587 (26.00)	2.632 (29.37)	2.646 (29.21)	2.638 (27.58)	2.635 (26.51)
Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu10=6.0									
0%	1.965	1.992	1.992	2.000	2.003	2.029	2.022	2.029	2.020
Imputation using linear regression									

SampleSize_n150_10Treatments/table3_nonshrink_10_treatments_MAR_n150

MR	Many_T2vs.T0	Many_T3vs.T0	Many_T4vs.T0	Many_T5vs.T0	Many_T6vs.T0	Many_T7vs.T0	Many_T8vs.T0	Many_T9vs.T0	Many_T10vs.T0
10%	2.102 (6.97)	2.086 (4.72)	2.099 (5.37)	2.095 (4.78)	2.116 (5.63)	2.105 (3.76)	2.138 (5.71)	2.156 (6.27)	2.158 (6.82)
20%	2.181 (11.00)	2.208 (10.87)	2.212 (11.06)	2.227 (11.37)	2.238 (11.70)	2.255 (11.15)	2.248 (11.16)	2.273 (12.04)	2.269 (12.36)
30%	2.343 (19.23)	2.315 (16.25)	2.327 (16.84)	2.362 (18.14)	2.390 (19.30)	2.363 (16.48)	2.421 (19.72)	2.426 (19.58)	2.422 (19.93)
40%	2.567 (30.66)	2.626 (31.85)	2.546 (27.83)	2.626 (31.32)	2.587 (29.15)	2.613 (28.80)	2.663 (31.67)	2.722 (34.16)	2.695 (33.44)
Imputation using predictive mean matching method									
10%	2.088 (6.25)	2.073 (4.07)	2.080 (4.43)	2.076 (3.82)	2.109 (5.27)	2.094 (3.21)	2.125 (5.10)	2.134 (5.20)	2.145 (6.22)
20%	2.143 (9.07)	2.157 (8.32)	2.177 (9.32)	2.196 (9.82)	2.200 (9.82)	2.216 (9.21)	2.226 (10.09)	2.233 (10.08)	2.229 (10.35)
30%	2.279 (15.99)	2.269 (13.91)	2.266 (13.74)	2.290 (14.50)	2.346 (17.12)	2.313 (14.02)	2.367 (17.05)	2.401 (18.34)	2.352 (16.45)
40%	2.455 (24.95)	2.451 (23.07)	2.433 (22.14)	2.481 (24.10)	2.453 (22.42)	2.464 (21.43)	2.488 (23.01)	2.552 (25.81)	2.521 (24.81)
Imputation using propensity−score method									
10%	2.086 (6.15)	2.075 (4.19)	2.085 (4.67)	2.076 (3.82)	2.109 (5.27)	2.097 (3.39)	2.126 (5.15)	2.137 (5.33)	2.144 (6.17)
20%	2.153 (9.57)	2.180 (9.47)	2.200 (10.43)	2.213 (10.68)	2.227 (11.16)	2.246 (10.69)	2.224 (9.97)	2.245 (10.68)	2.243 (11.07)
30%	2.314 (17.76)	2.295 (15.22)	2.279 (14.43)	2.336 (16.81)	2.352 (17.38)	2.339 (15.27)	2.403 (18.82)	2.383 (17.44)	2.393 (18.46)
40%	2.551 (29.85)	2.575 (29.29)	2.511 (26.05)	2.565 (28.28)	2.553 (27.44)	2.586 (27.46)	2.586 (27.87)	2.621 (29.22)	2.635 (30.46)

SampleSize_n150_10Treatments/table3_shrink_10_treatments_MAR_n150

MR	None_T1vs.T0	None_T2vs.T0	None_T3vs.T0	None_T4vs.T0	None_T5vs.T0	None_T6vs.T0	None_T7vs.T0	None_T8vs.T0	None_T9vs.T0	None_T10vs.T0	Many_T1vs.T0
Scenario I: all means are 0 for T0,T1,...,T10											
0%	1.449	1.445	1.453	1.462	1.485	1.493	1.480	1.489	1.504	1.515	1.993
Imputation using linear regression											
10%	1.502 (3.68)	1.498 (3.69)	1.497 (2.99)	1.523 (4.21)	1.520 (2.35)	1.534 (2.74)	1.537 (3.86)	1.544 (3.69)	1.545 (2.72)	1.557 (2.80)	2.079 (4.31)
20%	1.594 (10.08)	1.584 (9.65)	1.571 (8.12)	1.589 (8.69)	1.609 (8.31)	1.626 (8.87)	1.626 (9.87)	1.619 (8.71)	1.617 (7.50)	1.646 (8.64)	2.195 (10.16)
30%	1.707 (17.84)	1.705 (18.03)	1.721 (18.45)	1.740 (19.04)	1.759 (18.42)	1.729 (15.77)	1.768 (19.50)	1.787 (20.00)	1.753 (16.60)	1.807 (19.26)	2.367 (18.76)
40%	1.852 (27.89)	1.876 (29.86)	1.875 (29.01)	1.867 (27.71)	1.893 (27.43)	1.909 (27.84)	1.909 (29.05)	1.934 (29.89)	1.927 (28.14)	1.971 (30.08)	2.556 (28.26)
Imputation using predictive mean matching method											
10%	1.493 (3.09)	1.483 (2.67)	1.491 (2.63)	1.524 (4.26)	1.521 (2.39)	1.527 (2.24)	1.537 (3.91)	1.544 (3.71)	1.534 (2.02)	1.559 (2.93)	2.068 (3.78)
20%	1.555 (7.34)	1.552 (7.45)	1.543 (6.20)	1.575 (7.74)	1.591 (7.12)	1.601 (7.20)	1.610 (8.79)	1.594 (7.02)	1.590 (5.74)	1.632 (7.71)	2.140 (7.38)
30%	1.628 (12.41)	1.619 (12.05)	1.663 (14.44)	1.677 (14.73)	1.664 (12.05)	1.671 (11.92)	1.712 (15.74)	1.705 (14.53)	1.704 (13.31)	1.743 (15.03)	2.259 (13.35)
40%	1.809 (24.86)	1.797 (24.35)	1.827 (25.73)	1.836 (25.62)	1.835 (23.55)	1.822 (22.04)	1.858 (25.57)	1.830 (22.87)	1.849 (22.98)	1.904 (25.66)	2.489 (24.92)
Imputation using propensity−score method											
10%	1.490 (2.87)	1.487 (2.92)	1.490 (2.52)	1.520 (3.99)	1.516 (2.07)	1.530 (2.46)	1.531 (3.51)	1.539 (3.33)	1.532 (1.88)	1.558 (2.83)	2.064 (3.57)
20%	1.553 (7.25)	1.557 (7.75)	1.534 (5.58)	1.577 (7.92)	1.585 (6.69)	1.619 (8.39)	1.614 (9.11)	1.601 (7.53)	1.592 (5.87)	1.633 (7.81)	2.141 (7.46)
30%	1.661 (14.64)	1.633 (13.05)	1.691 (16.38)	1.690 (15.63)	1.690 (13.77)	1.706 (14.23)	1.728 (16.77)	1.726 (15.91)	1.731 (15.14)	1.737 (14.65)	2.307 (15.79)
40%	1.823 (25.83)	1.840 (27.36)	1.863 (28.18)	1.850 (26.55)	1.841 (23.95)	1.876 (25.64)	1.909 (29.03)	1.919 (28.86)	1.912 (27.17)	1.910 (26.05)	2.515 (26.19)
Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu10=6.0											
0%	1.421	1.426	1.446	1.446	1.452	1.454	1.473	1.468	1.473	1.466	1.976
Imputation using linear regression											

SampleSize_n150_10Treatments/table3_shrink_10_treatments_MAR_n150

MR	None_T1vs.T0	None_T2vs.T0	None_T3vs.T0	None_T4vs.T0	None_T5vs.T0	None_T6vs.T0	None_T7vs.T0	None_T8vs.T0	None_T9vs.T0	None_T10vs.T0	Many_T1vs.T0
10%	1.498 (5.41)	1.524 (6.82)	1.512 (4.57)	1.521 (5.21)	1.519 (4.63)	1.534 (5.47)	1.526 (3.62)	1.550 (5.56)	1.563 (6.13)	1.564 (6.66)	2.077 (5.13)
20%	1.570 (10.45)	1.582 (10.88)	1.601 (10.75)	1.604 (10.93)	1.615 (11.23)	1.623 (11.59)	1.635 (11.02)	1.630 (11.05)	1.648 (11.93)	1.646 (12.25)	2.172 (9.94)
30%	1.687 (18.76)	1.693 (18.68)	1.673 (15.71)	1.681 (16.27)	1.707 (17.59)	1.727 (18.75)	1.707 (15.93)	1.749 (19.16)	1.753 (19.03)	1.750 (19.37)	2.344 (18.68)
40%	1.876 (32.02)	1.860 (30.42)	1.904 (31.71)	1.845 (27.62)	1.904 (31.16)	1.875 (28.92)	1.895 (28.64)	1.931 (31.51)	1.973 (34.00)	1.954 (33.25)	2.585 (30.85)
Imputation using predictive mean matching method											
10%	1.484 (4.41)	1.513 (6.10)	1.502 (3.92)	1.508 (4.27)	1.505 (3.67)	1.529 (5.12)	1.518 (3.07)	1.541 (4.96)	1.547 (5.05)	1.555 (6.06)	2.060 (4.26)
20%	1.544 (8.67)	1.555 (8.99)	1.565 (8.23)	1.579 (9.22)	1.593 (9.73)	1.596 (9.73)	1.607 (9.12)	1.615 (10.03)	1.620 (10.01)	1.617 (10.27)	2.139 (8.28)
30%	1.658 (16.68)	1.649 (15.61)	1.642 (13.55)	1.639 (13.36)	1.656 (14.10)	1.698 (16.75)	1.674 (13.65)	1.713 (16.68)	1.738 (17.99)	1.702 (16.05)	2.296 (16.22)
40%	1.755 (23.50)	1.777 (24.57)	1.774 (22.70)	1.761 (21.76)	1.796 (23.74)	1.774 (22.01)	1.783 (21.07)	1.800 (22.63)	1.847 (25.45)	1.824 (24.38)	2.425 (22.76)
Imputation using propensity−score method											
10%	1.479 (4.11)	1.512 (5.97)	1.504 (4.01)	1.511 (4.49)	1.504 (3.64)	1.528 (5.09)	1.520 (3.22)	1.541 (4.98)	1.549 (5.16)	1.554 (5.98)	2.054 (3.97)
20%	1.554 (9.34)	1.562 (9.49)	1.582 (9.39)	1.596 (10.36)	1.605 (10.60)	1.616 (11.09)	1.629 (10.62)	1.613 (9.89)	1.629 (10.61)	1.628 (10.99)	2.149 (8.77)
30%	1.647 (15.94)	1.674 (17.37)	1.660 (14.84)	1.649 (14.02)	1.690 (16.41)	1.701 (16.98)	1.692 (14.87)	1.739 (18.44)	1.723 (17.02)	1.731 (18.07)	2.283 (15.58)
40%	1.846 (29.88)	1.849 (29.65)	1.867 (29.12)	1.819 (25.82)	1.859 (28.08)	1.850 (27.22)	1.875 (27.31)	1.874 (27.68)	1.900 (29.00)	1.910 (30.24)	2.544 (28.79)

MR	Many_T2vs.T0	Many_T3vs.T0	Many_T4vs.T0	Many_T5vs.T0	Many_T6vs.T0	Many_T7vs.T0	Many_T8vs.T0	Many_T9vs.T0	Many_T10vs.T0
Scenario I: all means are 0 for T0,T1,...,T10									
0%	1.992	1.994	1.999	2.003	2.009	2.002	2.004	2.007	2.020
Imputation using linear regression									
10%	2.084 (4.64)	2.069 (3.79)	2.086 (4.36)	2.086 (4.13)	2.099 (4.49)	2.096 (4.66)	2.093 (4.42)	2.103 (4.76)	2.099 (3.94)
20%	2.185 (9.73)	2.176 (9.16)	2.189 (9.53)	2.202 (9.91)	2.214 (10.24)	2.209 (10.31)	2.213 (10.41)	2.209 (10.02)	2.229 (10.35)
30%	2.368 (18.91)	2.373 (19.04)	2.394 (19.81)	2.422 (20.93)	2.373 (18.16)	2.421 (20.92)	2.443 (21.91)	2.400 (19.54)	2.468 (22.22)
40%	2.587 (29.90)	2.580 (29.42)	2.566 (28.41)	2.604 (29.99)	2.627 (30.78)	2.617 (30.69)	2.653 (32.37)	2.635 (31.26)	2.687 (33.05)
Imputation using predictive mean matching method									
10%	2.064 (3.63)	2.061 (3.39)	2.086 (4.40)	2.085 (4.10)	2.088 (3.93)	2.095 (4.65)	2.092 (4.40)	2.089 (4.05)	2.102 (4.09)
20%	2.143 (7.62)	2.138 (7.24)	2.171 (8.61)	2.178 (8.75)	2.182 (8.64)	2.186 (9.19)	2.176 (8.55)	2.171 (8.13)	2.209 (9.40)
30%	2.251 (13.01)	2.295 (15.09)	2.310 (15.60)	2.294 (14.50)	2.296 (14.33)	2.345 (17.10)	2.332 (16.34)	2.329 (16.03)	2.378 (17.76)
40%	2.471 (24.05)	2.506 (25.68)	2.514 (25.80)	2.518 (25.69)	2.500 (24.46)	2.536 (26.66)	2.502 (24.82)	2.517 (25.38)	2.584 (27.93)
Imputation using propensity−score method									
10%	2.070 (3.94)	2.060 (3.35)	2.083 (4.20)	2.081 (3.87)	2.093 (4.22)	2.091 (4.42)	2.087 (4.13)	2.088 (4.03)	2.101 (4.05)
20%	2.151 (8.02)	2.128 (6.74)	2.178 (8.96)	2.173 (8.47)	2.208 (9.94)	2.197 (9.73)	2.190 (9.28)	2.179 (8.52)	2.214 (9.61)
30%	2.274 (14.18)	2.338 (17.29)	2.333 (16.74)	2.333 (16.48)	2.348 (16.90)	2.371 (18.40)	2.366 (18.04)	2.374 (18.26)	2.377 (17.71)
40%	2.536 (27.34)	2.561 (28.45)	2.540 (27.07)	2.533 (26.44)	2.580 (28.45)	2.616 (30.65)	2.630 (31.20)	2.613 (30.14)	2.601 (28.79)
Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu10=6.0									
0%	1.974	1.988	1.985	1.984	1.983	1.991	1.988	1.990	1.992
Imputation using linear regression									

SampleSize_n150_10Treatments/table3_shrink_10_treatments_MAR_n150

MR	Many_T2vs.T0	Many_T3vs.T0	Many_T4vs.T0	Many_T5vs.T0	Many_T6vs.T0	Many_T7vs.T0	Many_T8vs.T0	Many_T9vs.T0	Many_T10vs.T0
10%	2.088 (5.77)	2.077 (4.45)	2.083 (4.96)	2.090 (5.34)	2.090 (5.38)	2.088 (4.90)	2.101 (5.65)	2.110 (6.04)	2.105 (5.66)
20%	2.185 (10.67)	2.198 (10.52)	2.199 (10.80)	2.202 (11.01)	2.218 (11.83)	2.222 (11.65)	2.220 (11.67)	2.242 (12.64)	2.235 (12.19)
30%	2.351 (19.09)	2.327 (17.04)	2.332 (17.49)	2.359 (18.91)	2.382 (20.11)	2.354 (18.26)	2.397 (20.54)	2.400 (20.60)	2.403 (20.61)
40%	2.563 (29.81)	2.621 (31.81)	2.545 (28.22)	2.615 (31.83)	2.577 (29.96)	2.605 (30.87)	2.645 (33.04)	2.694 (35.35)	2.668 (33.89)
Imputation using predictive mean matching method									
10%	2.076 (5.14)	2.063 (3.73)	2.065 (4.05)	2.074 (4.55)	2.082 (5.01)	2.077 (4.35)	2.087 (4.97)	2.091 (5.05)	2.091 (4.97)
20%	2.148 (8.82)	2.148 (8.04)	2.164 (9.04)	2.174 (9.57)	2.178 (9.81)	2.184 (9.73)	2.199 (10.60)	2.203 (10.69)	2.198 (10.32)
30%	2.289 (15.96)	2.281 (14.71)	2.272 (14.48)	2.285 (15.18)	2.337 (17.83)	2.303 (15.72)	2.341 (17.76)	2.373 (19.24)	2.332 (17.07)
40%	2.456 (24.41)	2.449 (23.14)	2.435 (22.66)	2.469 (24.44)	2.445 (23.27)	2.454 (23.31)	2.471 (24.26)	2.526 (26.94)	2.493 (25.12)
Imputation using propensity−score method									
10%	2.074 (5.04)	2.066 (3.88)	2.069 (4.26)	2.072 (4.46)	2.083 (5.03)	2.079 (4.45)	2.088 (5.04)	2.092 (5.13)	2.091 (4.96)
20%	2.156 (9.24)	2.169 (9.07)	2.185 (10.11)	2.191 (10.43)	2.205 (11.17)	2.214 (11.21)	2.196 (10.45)	2.213 (11.18)	2.209 (10.86)
30%	2.320 (17.50)	2.305 (15.93)	2.283 (15.04)	2.332 (17.55)	2.342 (18.12)	2.328 (16.95)	2.378 (19.61)	2.355 (18.33)	2.373 (19.10)
40%	2.548 (29.07)	2.571 (29.31)	2.512 (26.55)	2.555 (28.80)	2.545 (28.32)	2.579 (29.55)	2.572 (29.37)	2.595 (30.39)	2.610 (30.99)

SampleSize_n20_10Treatments/table2_nonshrink_10_treatments_MAR_n20

MR	T1vs.T0_none	T2vs.T0_none	T3vs.T0_none	T4vs.T0_none	T5vs.T0_none	T6vs.T0_none	T7vs.T0_none	T8vs.T0_none	T9vs.T0_none	T10vs.T0_none	Joint_none
Scenario I: all means are 0 for T0,T1,...,T10	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
0%	93.10	93.10	93.10	93.10	93.10	100.00	96.55	93.10	100.00	100.00	68.97
Imputation using linear regression	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	100.00	93.10	93.10	96.55	93.10	89.66	93.10	93.10	100.00	96.55	72.41
20%	92.86	92.86	92.86	89.29	100.00	89.29	96.43	100.00	89.29	96.43	71.43
30%	96.43	92.86	96.43	96.43	100.00	100.00	92.86	92.86	96.43	96.43	75.00
40%	93.10	89.66	100.00	100.00	96.55	89.66	96.55	100.00	93.10	93.10	65.52
Imputation using predictive mean matching method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	100.00	86.21	93.10	96.55	93.10	89.66	93.10	89.66	100.00	96.55	68.97
20%	85.71	92.86	89.29	85.71	100.00	89.29	96.43	96.43	85.71	96.43	64.29
30%	92.86	92.86	85.71	85.71	82.14	92.86	89.29	92.86	85.71	92.86	39.29
40%	86.21	75.86	89.66	82.76	75.86	86.21	89.66	89.66	75.86	82.76	48.28
Imputation using propensity−score method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	100.00	93.10	93.10	96.55	93.10	89.66	93.10	89.66	100.00	93.10	72.41
20%	82.14	92.86	89.29	85.71	100.00	89.29	96.43	96.43	92.86	96.43	64.29
30%	92.86	92.86	92.86	92.86	78.57	92.86	85.71	85.71	82.14	92.86	39.29
40%	89.66	79.31	86.21	89.66	79.31	93.10	89.66	93.10	82.76	79.31	41.38
Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu10=6.0	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
0%	96.55	96.55	96.55	96.55	93.10	96.55	96.55	96.55	89.66	93.10	68.97
Imputation using linear regression	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA

SampleSize_n20_10Treatments/table2_nonshrink_10_treatments_MAR_n20

MR	T1vs.T0_none	T2vs.T0_none	T3vs.T0_none	T4vs.T0_none	T5vs.T0_none	T6vs.T0_none	T7vs.T0_none	T8vs.T0_none	T9vs.T0_none	T10vs.T0_none	Joint_none
10%	96.55	93.10	96.55	89.66	96.55	100.00	100.00	100.00	100.00	82.76	65.52
20%	96.43	100.00	96.43	96.43	89.29	100.00	92.86	100.00	96.43	85.71	75.00
30%	93.10	96.55	96.55	96.55	96.55	93.10	89.66	89.66	96.55	96.55	75.86
40%	96.55	100.00	100.00	93.10	96.55	96.55	96.55	100.00	96.55	96.55	75.86
Imputation using predictive mean matching method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	96.55	86.21	93.10	86.21	93.10	100.00	96.55	96.55	100.00	82.76	62.07
20%	85.71	100.00	89.29	92.86	92.86	96.43	92.86	92.86	92.86	82.14	57.14
30%	86.21	93.10	86.21	93.10	86.21	86.21	89.66	86.21	93.10	89.66	58.62
40%	89.66	86.21	93.10	93.10	86.21	86.21	89.66	79.31	93.10	93.10	48.28
Imputation using propensity−score method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	96.55	89.66	89.66	89.66	89.66	100.00	96.55	96.55	100.00	82.76	62.07
20%	92.86	100.00	89.29	92.86	92.86	96.43	92.86	92.86	92.86	82.14	60.71
30%	93.10	89.66	93.10	93.10	86.21	86.21	89.66	82.76	89.66	89.66	55.17
40%	86.21	93.10	93.10	86.21	86.21	93.10	79.31	82.76	96.55	89.66	51.72

MR	T1vs.T0_many	T2vs.T0_many	T3vs.T0_many	T4vs.T0_many	T5vs.T0_many	T6vs.T0_many	T7vs.T0_many	T8vs.T0_many	T9vs.T0_many	T10vs.T0_many	Joint_many
Scenario I: all means are 0 for T0,T1,...,T10	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
0%	100.00	100.00	96.55	100.00	96.55	100.00	100.00	100.00	100.00	100.00	93.10
Imputation using linear regression	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	100.00	96.55	100.00	100.00	96.55	96.55	96.55	100.00	100.00	100.00	93.10
20%	100.00	96.43	100.00	100.00	100.00	100.00	100.00	100.00	100.00	96.43	92.86
30%	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00
40%	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00
Imputation using predictive mean matching method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	100.00	96.55	100.00	100.00	96.55	96.55	96.55	100.00	100.00	100.00	93.10
20%	100.00	96.43	96.43	96.43	100.00	96.43	100.00	100.00	96.43	96.43	85.71
30%	96.43	100.00	100.00	100.00	100.00	100.00	100.00	96.43	96.43	96.43	89.29
40%	96.55	93.10	96.55	100.00	100.00	96.55	96.55	100.00	100.00	96.55	79.31
Imputation using propensity−score method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	100.00	96.55	100.00	100.00	96.55	96.55	96.55	100.00	100.00	100.00	93.10
20%	100.00	96.43	96.43	96.43	100.00	100.00	100.00	100.00	96.43	96.43	89.29
30%	96.43	96.43	100.00	96.43	100.00	96.43	96.43	100.00	92.86	92.86	82.14
40%	100.00	96.55	96.55	100.00	100.00	96.55	100.00	100.00	100.00	93.10	86.21
Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu10=6.0	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
0%	96.55	96.55	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	96.55
Imputation using linear regression	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA

SampleSize_n20_10Treatments/table2_nonshrink_10_treatments_MAR_n20

MR	T1vs.T0_many	T2vs.T0_many	T3vs.T0_many	T4vs.T0_many	T5vs.T0_many	T6vs.T0_many	T7vs.T0_many	T8vs.T0_many	T9vs.T0_many	T10vs.T0_many	Joint_many
10%	100.00	100.00	100	93.1	96.55	100.00	100.00	100.00	100.00	96.55	89.66
20%	96.43	100.00	100	100.0	100.00	100.00	100.00	100.00	100.00	96.43	92.86
30%	96.55	96.55	100	100.0	100.00	100.00	96.55	100.00	100.00	100.00	96.55
40%	100.00	100.00	100	100.0	100.00	100.00	100.00	100.00	100.00	100.00	100.00
Imputation using predictive mean matching method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	100.00	100.00	100	93.1	96.55	100.00	100.00	100.00	100.00	96.55	89.66
20%	96.43	100.00	100	100.0	96.43	100.00	96.43	100.00	100.00	96.43	89.29
30%	93.10	96.55	100	100.0	100.00	100.00	96.55	100.00	96.55	100.00	89.66
40%	96.55	96.55	100	93.1	96.55	96.55	96.55	96.55	96.55	96.55	79.31
Imputation using propensity−score method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	100.00	100.00	100	93.1	96.55	100.00	100.00	100.00	100.00	96.55	89.66
20%	96.43	100.00	100	100.0	96.43	100.00	92.86	100.00	100.00	96.43	85.71
30%	93.10	96.55	100	100.0	100.00	100.00	96.55	96.55	100.00	96.55	86.21
40%	96.55	100.00	100	93.1	100.00	100.00	93.10	96.55	96.55	93.10	82.76

SampleSize_n20_10Treatments/table2_shrink_10_treatments_MAR_n20

MR	T1vs.T0_none	T2vs.T0_none	T3vs.T0_none	T4vs.T0_none	T5vs.T0_none	T6vs.T0_none	T7vs.T0_none	T8vs.T0_none	T9vs.T0_none	T10vs.T0_none	Joint_none
Scenario I: all means are 0 for T0,T1,...,T10	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
0%	93.10	93.10	93.10	93.10	93.10	100.00	96.55	93.10	100.00	100.00	68.97
Imputation using linear regression	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	100.00	93.10	93.10	96.55	93.10	89.66	93.10	93.10	100.00	96.55	72.41
20%	92.86	92.86	92.86	89.29	100.00	89.29	96.43	100.00	89.29	96.43	71.43
30%	96.43	92.86	96.43	96.43	100.00	100.00	92.86	92.86	96.43	96.43	75.00
40%	93.10	89.66	100.00	100.00	96.55	89.66	96.55	100.00	93.10	93.10	65.52
Imputation using predictive mean matching method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	100.00	86.21	93.10	96.55	93.10	89.66	93.10	89.66	100.00	96.55	68.97
20%	85.71	92.86	89.29	85.71	100.00	89.29	96.43	96.43	85.71	96.43	64.29
30%	92.86	92.86	85.71	85.71	82.14	92.86	89.29	92.86	85.71	92.86	39.29
40%	86.21	75.86	89.66	82.76	75.86	86.21	89.66	89.66	75.86	82.76	48.28
Imputation using propensity−score method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	100.00	93.10	93.10	96.55	93.10	89.66	93.10	89.66	100.00	93.10	72.41
20%	82.14	92.86	89.29	85.71	100.00	89.29	96.43	96.43	92.86	96.43	64.29
30%	92.86	92.86	92.86	92.86	78.57	92.86	85.71	85.71	82.14	92.86	39.29
40%	89.66	79.31	86.21	89.66	79.31	93.10	89.66	93.10	82.76	79.31	41.38
Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu10=6.0	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
0%	96.55	96.55	96.55	96.55	93.10	96.55	96.55	96.55	89.66	93.10	68.97
Imputation using linear regression	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA

SampleSize_n20_10Treatments/table2_shrink_10_treatments_MAR_n20

MR	T1vs.T0_none	T2vs.T0_none	T3vs.T0_none	T4vs.T0_none	T5vs.T0_none	T6vs.T0_none	T7vs.T0_none	T8vs.T0_none	T9vs.T0_none	T10vs.T0_none	Joint_none
10%	96.55	93.10	96.55	89.66	96.55	100.00	100.00	100.00	100.00	82.76	65.52
20%	96.43	100.00	96.43	96.43	89.29	100.00	92.86	100.00	96.43	85.71	75.00
30%	93.10	96.55	96.55	96.55	96.55	93.10	89.66	89.66	96.55	96.55	75.86
40%	96.55	100.00	100.00	93.10	96.55	96.55	96.55	100.00	96.55	96.55	75.86
Imputation using predictive mean matching method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	96.55	86.21	93.10	86.21	93.10	100.00	96.55	96.55	100.00	82.76	62.07
20%	85.71	100.00	89.29	92.86	92.86	96.43	92.86	92.86	92.86	82.14	57.14
30%	86.21	93.10	86.21	93.10	86.21	86.21	89.66	86.21	93.10	89.66	58.62
40%	89.66	86.21	93.10	93.10	86.21	86.21	89.66	79.31	93.10	93.10	48.28
Imputation using propensity−score method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	96.55	89.66	89.66	89.66	89.66	100.00	96.55	96.55	100.00	82.76	62.07
20%	92.86	100.00	89.29	92.86	92.86	96.43	92.86	92.86	92.86	82.14	60.71
30%	93.10	89.66	93.10	93.10	86.21	86.21	89.66	82.76	89.66	89.66	55.17
40%	86.21	93.10	93.10	86.21	86.21	93.10	79.31	82.76	96.55	89.66	51.72

SampleSize_n20_10Treatments/table2_shrink_10_treatments_MAR_n20

MR	T1vs.T0_many	T2vs.T0_many	T3vs.T0_many	T4vs.T0_many	T5vs.T0_many	T6vs.T0_many	T7vs.T0_many	T8vs.T0_many	T9vs.T0_many	T10vs.T0_many	Joint_many
Scenario I: all means are 0 for T0,T1,...,T10	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
0%	100.00	100.00	96.55	100.00	96.55	100.00	100.00	100.00	100.00	100.00	93.10
Imputation using linear regression	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	100.00	96.55	100.00	100.00	96.55	96.55	96.55	100.00	100.00	100.00	93.10
20%	100.00	96.43	100.00	100.00	100.00	100.00	100.00	100.00	100.00	96.43	92.86
30%	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00
40%	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00
Imputation using predictive mean matching method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	100.00	96.55	100.00	100.00	96.55	96.55	100.00	100.00	100.00	100.00	93.10
20%	100.00	96.43	96.43	96.43	100.00	96.43	100.00	100.00	96.43	96.43	85.71
30%	96.43	100.00	100.00	100.00	100.00	100.00	100.00	96.43	100.00	92.86	89.29
40%	96.55	100.00	96.55	100.00	96.55	96.55	96.55	100.00	100.00	96.55	86.21
Imputation using propensity−score method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	100.00	96.55	100.00	100.00	96.55	96.55	96.55	100.00	100.00	100.00	93.10
20%	100.00	96.43	96.43	96.43	100.00	100.00	100.00	100.00	96.43	100.00	92.86
30%	96.43	96.43	100.00	96.43	100.00	96.43	96.43	100.00	92.86	92.86	82.14
40%	100.00	100.00	96.55	100.00	100.00	96.55	96.55	100.00	100.00	93.10	86.21
Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu10=6.0	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
0%	96.55	96.55	100.00	100.00	100.00	100.00	96.55	100.00	100.00	100.00	96.55
Imputation using linear regression	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA

SampleSize_n20_10Treatments/table2_shrink_10_treatments_MAR_n20

MR	T1vs.T0_many	T2vs.T0_many	T3vs.T0_many	T4vs.T0_many	T5vs.T0_many	T6vs.T0_many	T7vs.T0_many	T8vs.T0_many	T9vs.T0_many	T10vs.T0_many	Joint_many
10%	100.00	100.00	100	100.00	96.55	100	100.00	100.00	100.00	96.55	96.55
20%	96.43	100.00	100	100.00	100.00	100	100.00	100.00	100.00	96.43	92.86
30%	96.55	96.55	100	100.00	100.00	100	96.55	100.00	100.00	100.00	96.55
40%	100.00	100.00	100	100.00	100.00	100	100.00	100.00	100.00	100.00	100.00
Imputation using predictive mean matching method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	100.00	100.00	100	93.10	96.55	100	100.00	100.00	100.00	96.55	89.66
20%	96.43	100.00	100	100.00	96.43	100	96.43	100.00	100.00	96.43	89.29
30%	93.10	96.55	100	96.55	100.00	100	96.55	96.55	96.55	100.00	89.66
40%	96.55	96.55	100	93.10	96.55	100	96.55	93.10	96.55	96.55	79.31
Imputation using propensity−score method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	100.00	100.00	100	93.10	96.55	100	100.00	100.00	100.00	96.55	89.66
20%	96.43	100.00	100	100.00	96.43	100	100.00	100.00	100.00	96.43	89.29
30%	93.10	96.55	100	100.00	100.00	100	96.55	96.55	100.00	96.55	86.21
40%	96.55	100.00	100	93.10	100.00	100	96.55	100.00	96.55	89.66	79.31

SampleSize_n20_10Treatments/table3_nonshrink_10_treatments_MAR_n20

MR	None_T1vs.T0	None_T2vs.T0	None_T3vs.T0	None_T4vs.T0	None_T5vs.T0	None_T6vs.T0	None_T7vs.T0	None_T8vs.T0	None_T9vs.T0	None_T10vs.T0	Many_T1vs.T0
Scenario I: all means are 0 for T0,T1,...,T10											
0%	3.911	4.030	3.924	4.010	4.047	4.112	4.267	4.109	4.036	4.241	5.284
Imputation using linear regression											
10%	4.195 (7.24)	4.237 (5.14)	4.316 (10.00)	4.345 (8.34)	4.433 (9.56)	4.412 (7.30)	4.335 (1.60)	4.408 (7.28)	4.384 (8.62)	4.568 (7.71)	5.755 (8.91)
20%	4.661 (19.17)	4.759 (18.09)	4.664 (18.87)	4.727 (17.87)	4.644 (14.77)	4.678 (13.76)	4.822 (13.02)	4.828 (17.50)	4.822 (19.48)	4.756 (12.15)	6.423 (21.55)
30%	4.857 (24.17)	5.396 (33.90)	5.013 (27.77)	5.155 (28.54)	5.017 (23.98)	5.057 (22.98)	5.054 (18.45)	4.987 (21.36)	5.143 (27.44)	5.143 (21.28)	6.678 (26.37)
40%	5.807 (48.46)	5.781 (43.45)	5.890 (50.12)	5.890 (46.87)	5.623 (38.95)	5.932 (44.27)	5.896 (38.19)	6.024 (46.60)	6.142 (52.19)	6.316 (48.94)	7.975 (50.92)
Imputation using predictive mean matching method											
10%	4.028 (2.97)	4.095 (1.60)	4.127 (5.18)	4.147 (3.42)	4.222 (4.33)	4.216 (2.54)	4.194 (−1.70)	4.233 (3.02)	4.194 (3.92)	4.388 (3.46)	5.521 (4.47)
20%	4.175 (6.75)	4.312 (6.99)	4.193 (6.86)	4.283 (6.79)	4.280 (5.76)	4.277 (4.02)	4.416 (3.49)	4.344 (5.73)	4.365 (8.16)	4.295 (1.27)	5.752 (8.85)
30%	4.109 (5.04)	4.338 (7.63)	4.213 (7.37)	4.211 (4.99)	4.241 (4.80)	4.208 (2.34)	4.239 (−0.65)	4.184 (1.82)	4.241 (5.10)	4.341 (2.35)	5.653 (6.98)
40%	4.396 (12.39)	4.383 (8.76)	4.430 (12.90)	4.424 (10.31)	4.308 (6.46)	4.548 (10.60)	4.645 (8.87)	4.532 (10.29)	4.658 (15.41)	4.636 (9.31)	6.036 (14.24)
Imputation using propensity−score method											
10%	4.024 (2.87)	4.086 (1.39)	4.091 (4.27)	4.167 (3.90)	4.246 (4.93)	4.236 (3.02)	4.175 (−2.15)	4.227 (2.88)	4.170 (3.33)	4.376 (3.19)	5.516 (4.39)
20%	4.193 (7.20)	4.386 (8.83)	4.140 (5.52)	4.263 (6.30)	4.246 (4.92)	4.284 (4.17)	4.428 (3.78)	4.443 (8.13)	4.374 (8.40)	4.348 (2.52)	5.774 (9.27)
30%	4.064 (3.90)	4.201 (4.23)	4.199 (7.02)	4.371 (8.98)	4.248 (4.98)	4.159 (1.15)	4.263 (−0.10)	4.193 (2.05)	4.273 (5.89)	4.413 (4.05)	5.587 (5.74)
40%	4.251 (8.69)	4.286 (6.35)	4.200 (7.05)	4.333 (8.05)	4.196 (3.70)	4.423 (7.57)	4.529 (6.14)	4.343 (5.69)	4.503 (11.59)	4.484 (5.74)	5.856 (10.82)
Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu10=6.0											
0%	3.949	3.853	3.952	3.981	4.063	4.035	4.010	3.895	4.095	4.041	5.320
Imputation using linear regression											

MR	None_T1vs.T0	None_T2vs.T0	None_T3vs.T0	None_T4vs.T0	None_T5vs.T0	None_T6vs.T0	None_T7vs.T0	None_T8vs.T0	None_T9vs.T0	None_T10vs.T0	Many_T1vs.T0
10%	4.130 (4.59)	4.230 (9.78)	4.319 (9.28)	4.271 (7.27)	4.282 (5.40)	4.376 (8.44)	4.403 (9.81)	4.353 (11.75)	4.465 (9.04)	4.372 (8.21)	5.690 (6.96)
20%	4.557 (15.38)	4.476 (16.17)	4.361 (10.35)	4.709 (18.29)	4.479 (10.24)	4.615 (14.37)	4.575 (14.09)	4.582 (17.64)	4.621 (12.85)	4.627 (14.50)	6.280 (18.05)
30%	5.278 (33.66)	5.225 (35.58)	5.282 (33.64)	5.570 (39.90)	5.301 (30.46)	5.460 (35.32)	5.249 (30.91)	5.510 (41.45)	5.509 (34.53)	5.445 (34.76)	7.196 (35.27)
40%	5.737 (45.27)	5.712 (48.23)	5.891 (49.04)	5.528 (38.84)	5.638 (38.77)	6.005 (48.83)	6.169 (53.85)	6.552 (68.23)	5.826 (42.27)	5.847 (44.70)	7.861 (47.77)
Imputation using predictive mean matching method											
10%	3.893 (−1.42)	4.054 (5.22)	4.066 (2.89)	4.045 (1.60)	4.049 (−0.35)	4.162 (3.14)	4.190 (4.51)	4.160 (6.81)	4.241 (3.56)	4.141 (2.48)	5.371 (0.96)
20%	4.156 (5.24)	4.003 (3.87)	3.948 (−0.11)	4.214 (5.84)	4.063 (0.01)	4.179 (3.57)	4.191 (4.53)	4.153 (6.62)	4.224 (3.16)	4.248 (5.13)	5.726 (7.64)
30%	4.245 (7.49)	4.287 (11.25)	4.400 (11.32)	4.418 (10.96)	4.271 (5.11)	4.501 (11.55)	4.325 (7.86)	4.531 (16.33)	4.449 (8.64)	4.512 (11.67)	5.826 (9.52)
40%	4.207 (6.53)	4.278 (11.01)	4.345 (9.94)	4.301 (8.04)	4.301 (5.85)	4.361 (8.09)	4.426 (10.39)	4.458 (14.44)	4.478 (9.35)	4.463 (10.44)	5.804 (9.10)
Imputation using propensity−score method											
10%	3.912 (−0.93)	4.071 (5.64)	4.126 (4.39)	4.106 (3.13)	4.064 (0.02)	4.191 (3.86)	4.204 (4.84)	4.203 (7.90)	4.275 (4.39)	4.197 (3.88)	5.391 (1.33)
20%	4.234 (7.21)	4.074 (5.72)	4.004 (1.32)	4.209 (5.72)	4.175 (2.75)	4.291 (6.35)	4.224 (5.34)	4.239 (8.83)	4.242 (3.59)	4.337 (7.33)	5.827 (9.54)
30%	4.372 (10.71)	4.393 (14.01)	4.518 (14.32)	4.544 (14.14)	4.355 (7.19)	4.561 (13.03)	4.427 (10.40)	4.786 (22.88)	4.559 (11.34)	4.545 (12.47)	5.972 (12.26)
40%	4.163 (5.41)	4.230 (9.77)	4.195 (6.13)	4.230 (6.24)	4.282 (5.39)	4.243 (5.15)	4.331 (8.02)	4.437 (13.91)	4.492 (9.70)	4.444 (9.99)	5.750 (8.08)

MR	Many_T2vs.T0	Many_T3vs.T0	Many_T4vs.T0	Many_T5vs.T0	Many_T6vs.T0	Many_T7vs.T0	Many_T8vs.T0	Many_T9vs.T0	Many_T10vs.T0
Scenario I: all means are 0 for T0,T1,...,T10									
0%	5.444	5.299	5.418	5.468	5.553	5.764	5.550	5.453	5.729
Imputation using linear regression									
10%	5.817 (6.84)	5.923 (11.76)	5.961 (10.03)	6.086 (11.30)	6.055 (9.03)	5.949 (3.22)	6.049 (8.99)	6.017 (10.35)	6.273 (9.50)
20%	6.561 (20.51)	6.423 (21.20)	6.514 (20.23)	6.400 (17.04)	6.447 (16.08)	6.645 (15.28)	6.657 (19.96)	6.643 (21.83)	6.556 (14.44)
30%	7.414 (36.17)	6.891 (30.03)	7.089 (30.86)	6.902 (26.22)	6.951 (25.16)	6.949 (20.56)	6.859 (23.59)	7.064 (29.54)	7.075 (23.50)
40%	7.936 (45.77)	8.087 (52.60)	8.085 (49.24)	7.719 (41.16)	8.146 (46.68)	8.098 (40.51)	8.263 (48.89)	8.433 (54.64)	8.668 (51.31)
Imputation using predictive mean matching method									
10%	5.615 (3.13)	5.658 (6.77)	5.687 (4.96)	5.790 (5.89)	5.781 (4.09)	5.750 (−0.23)	5.804 (4.57)	5.751 (5.46)	6.019 (5.07)
20%	5.942 (9.14)	5.773 (8.94)	5.899 (8.89)	5.894 (7.78)	5.891 (6.09)	6.083 (5.54)	5.987 (7.88)	6.012 (10.26)	5.918 (3.31)
30%	5.963 (9.53)	5.792 (9.30)	5.794 (6.95)	5.836 (6.73)	5.788 (4.23)	5.835 (1.23)	5.754 (3.68)	5.835 (7.00)	5.971 (4.23)
40%	6.014 (10.47)	6.081 (14.76)	6.072 (12.08)	5.916 (8.19)	6.246 (12.47)	6.379 (10.68)	6.216 (12.01)	6.398 (17.32)	6.365 (11.10)
Imputation using propensity−score method									
10%	5.604 (2.94)	5.611 (5.88)	5.714 (5.46)	5.825 (6.53)	5.809 (4.59)	5.726 (−0.66)	5.796 (4.44)	5.720 (4.90)	6.005 (4.83)
20%	6.042 (10.97)	5.700 (7.57)	5.872 (8.38)	5.846 (6.90)	5.898 (6.21)	6.099 (5.82)	6.121 (10.29)	6.024 (10.47)	5.987 (4.51)
30%	5.772 (6.02)	5.771 (8.89)	6.009 (10.91)	5.843 (6.86)	5.719 (2.97)	5.863 (1.73)	5.764 (3.87)	5.875 (7.74)	6.064 (5.86)
40%	5.899 (8.35)	5.784 (9.14)	5.965 (10.09)	5.785 (5.79)	6.100 (9.85)	6.244 (8.34)	5.979 (7.73)	6.207 (13.83)	6.178 (7.84)
Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu10=6.0									
0%	5.193	5.325	5.365	5.475	5.438	5.405	5.248	5.520	5.444
Imputation using linear regression									

MR	Many_T2vs.T0	Many_T3vs.T0	Many_T4vs.T0	Many_T5vs.T0	Many_T6vs.T0	Many_T7vs.T0	Many_T8vs.T0	Many_T9vs.T0	Many_T10vs.T0
10%	5.831 (12.29)	5.952 (11.77)	5.884 (9.68)	5.900 (7.76)	6.033 (10.94)	6.068 (12.27)	5.998 (14.29)	6.154 (11.48)	6.025 (10.67)
20%	6.168 (18.77)	6.011 (12.88)	6.492 (21.01)	6.176 (12.80)	6.361 (16.98)	6.310 (16.74)	6.316 (20.35)	6.369 (15.38)	6.378 (17.14)
30%	7.119 (37.10)	7.204 (35.28)	7.589 (41.46)	7.220 (31.87)	7.445 (36.91)	7.150 (32.28)	7.511 (43.12)	7.503 (35.92)	7.423 (36.35)
40%	7.826 (50.71)	8.076 (51.66)	7.576 (41.23)	7.725 (41.10)	8.217 (51.11)	8.453 (56.40)	8.993 (71.35)	7.988 (44.72)	8.016 (47.23)
Imputation using predictive mean matching method									
10%	5.595 (7.76)	5.612 (5.39)	5.582 (4.04)	5.586 (2.04)	5.745 (5.66)	5.783 (6.98)	5.741 (9.39)	5.851 (5.99)	5.715 (4.97)
20%	5.514 (6.19)	5.440 (2.16)	5.806 (8.23)	5.602 (2.32)	5.757 (5.87)	5.779 (6.91)	5.722 (9.03)	5.822 (5.47)	5.854 (7.52)
30%	5.881 (13.26)	6.040 (13.42)	6.062 (12.99)	5.860 (7.03)	6.176 (13.58)	5.934 (9.79)	6.218 (18.48)	6.104 (10.58)	6.192 (13.73)
40%	5.902 (13.67)	5.995 (12.59)	5.935 (10.63)	5.930 (8.32)	6.016 (10.63)	6.105 (12.95)	6.146 (17.10)	6.177 (11.90)	6.156 (13.07)
Imputation using propensity−score method									
10%	5.612 (8.08)	5.688 (6.81)	5.658 (5.47)	5.600 (2.29)	5.778 (6.26)	5.794 (7.20)	5.793 (10.38)	5.891 (6.73)	5.786 (6.28)
20%	5.604 (7.92)	5.511 (3.48)	5.794 (8.00)	5.748 (4.99)	5.904 (8.57)	5.815 (7.59)	5.832 (11.12)	5.838 (5.77)	5.968 (9.62)
30%	5.996 (15.48)	6.173 (15.92)	6.209 (15.74)	5.947 (8.63)	6.233 (14.62)	6.047 (11.87)	6.533 (24.47)	6.227 (12.82)	6.211 (14.09)
40%	5.847 (12.60)	5.798 (8.89)	5.850 (9.05)	5.921 (8.15)	5.865 (7.86)	5.990 (10.83)	6.133 (16.86)	6.211 (12.52)	6.145 (12.87)

SampleSize_n20_10Treatments/table3_shrink_10_treatments_MAR_n20

MR	None_T1vs.T0	None_T2vs.T0	None_T3vs.T0	None_T4vs.T0	None_T5vs.T0	None_T6vs.T0	None_T7vs.T0	None_T8vs.T0	None_T9vs.T0	None_T10vs.T0	Many_T1vs.T0
Scenario I: all means are 0 for T0,T1,...,T10											
0%	3.911	4.030	3.924	4.010	4.047	4.112	4.267	4.109	4.036	4.241	5.553
Imputation using linear regression											
10%	4.195 (7.24)	4.237 (5.14)	4.316 (10.00)	4.345 (8.34)	4.433 (9.56)	4.412 (7.30)	4.335 (1.60)	4.408 (7.28)	4.384 (8.62)	4.568 (7.71)	5.827 (4.95)
20%	4.661 (19.17)	4.759 (18.09)	4.664 (18.87)	4.727 (17.87)	4.644 (14.77)	4.678 (13.76)	4.822 (13.02)	4.828 (17.50)	4.822 (19.48)	4.756 (12.15)	6.494 (16.96)
30%	4.857 (24.17)	5.396 (33.90)	5.013 (27.77)	5.155 (28.54)	5.017 (23.98)	5.057 (22.98)	5.054 (18.45)	4.987 (21.36)	5.143 (27.44)	5.143 (21.28)	6.773 (21.99)
40%	5.807 (48.46)	5.781 (43.45)	5.890 (50.12)	5.890 (46.87)	5.623 (38.95)	5.932 (44.27)	5.896 (38.19)	6.024 (46.60)	6.142 (52.19)	6.316 (48.94)	7.996 (44.01)
Imputation using predictive mean matching method											
10%	4.028 (2.97)	4.095 (1.60)	4.127 (5.18)	4.147 (3.42)	4.222 (4.33)	4.216 (2.54)	4.194 (−1.70)	4.233 (3.02)	4.194 (3.92)	4.388 (3.46)	5.579 (0.47)
20%	4.175 (6.75)	4.312 (6.99)	4.193 (6.86)	4.283 (6.79)	4.280 (5.76)	4.277 (4.02)	4.416 (3.49)	4.344 (5.73)	4.365 (8.16)	4.295 (1.27)	5.821 (4.83)
30%	4.109 (5.04)	4.338 (7.63)	4.213 (7.37)	4.211 (4.99)	4.241 (4.80)	4.208 (2.34)	4.239 (−0.65)	4.184 (1.82)	4.241 (5.10)	4.341 (2.35)	5.747 (3.50)
40%	4.396 (12.39)	4.383 (8.76)	4.430 (12.90)	4.424 (10.31)	4.308 (6.46)	4.548 (10.60)	4.645 (8.87)	4.532 (10.29)	4.658 (15.41)	4.636 (9.31)	6.049 (8.94)
Imputation using propensity−score method											
10%	4.024 (2.87)	4.086 (1.39)	4.091 (4.27)	4.167 (3.90)	4.246 (4.93)	4.236 (3.02)	4.175 (−2.15)	4.227 (2.88)	4.170 (3.33)	4.376 (3.19)	5.573 (0.37)
20%	4.193 (7.20)	4.386 (8.83)	4.140 (5.52)	4.263 (6.30)	4.246 (4.92)	4.284 (4.17)	4.428 (3.78)	4.443 (8.13)	4.374 (8.40)	4.348 (2.52)	5.838 (5.14)
30%	4.064 (3.90)	4.201 (4.23)	4.199 (7.02)	4.371 (8.98)	4.248 (4.98)	4.159 (1.15)	4.263 (−0.10)	4.193 (2.05)	4.273 (5.89)	4.413 (4.05)	5.710 (2.84)
40%	4.251 (8.69)	4.286 (6.35)	4.200 (7.05)	4.333 (8.05)	4.196 (3.70)	4.423 (7.57)	4.529 (6.14)	4.343 (5.69)	4.503 (11.59)	4.484 (5.74)	5.901 (6.27)
Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu10=6.0											
0%	3.949	3.853	3.952	3.981	4.063	4.035	4.010	3.895	4.095	4.041	5.396
Imputation using linear regression											

MR	None_T1vs.T0	None_T2vs.T0	None_T3vs.T0	None_T4vs.T0	None_T5vs.T0	None_T6vs.T0	None_T7vs.T0	None_T8vs.T0	None_T9vs.T0	None_T10vs.T0	Many_T1vs.T0
10%	4.130 (4.59)	4.230 (9.78)	4.319 (9.28)	4.271 (7.27)	4.282 (5.40)	4.376 (8.44)	4.403 (9.81)	4.353 (11.75)	4.465 (9.04)	4.372 (8.21)	5.880 (8.96)
20%	4.557 (15.38)	4.476 (16.17)	4.361 (10.35)	4.709 (18.29)	4.479 (10.24)	4.615 (14.37)	4.575 (14.09)	4.582 (17.64)	4.621 (12.85)	4.627 (14.50)	6.273 (16.25)
30%	5.278 (33.66)	5.225 (35.58)	5.282 (33.64)	5.570 (39.90)	5.301 (30.46)	5.460 (35.32)	5.249 (30.91)	5.510 (41.45)	5.509 (34.53)	5.445 (34.76)	7.212 (33.65)
40%	5.737 (45.27)	5.712 (48.23)	5.891 (49.04)	5.528 (38.84)	5.638 (38.77)	6.005 (48.83)	6.169 (53.85)	6.552 (68.23)	5.826 (42.27)	5.847 (44.70)	7.920 (46.77)
Imputation using predictive mean matching method											
10%	3.893 (−1.42)	4.054 (5.22)	4.066 (2.89)	4.045 (1.60)	4.049 (−0.35)	4.162 (3.14)	4.190 (4.51)	4.160 (6.81)	4.241 (3.56)	4.141 (2.48)	5.598 (3.73)
20%	4.156 (5.24)	4.003 (3.87)	3.948 (−0.11)	4.214 (5.84)	4.063 (0.01)	4.179 (3.57)	4.191 (4.53)	4.153 (6.62)	4.224 (3.16)	4.248 (5.13)	5.726 (6.11)
30%	4.245 (7.49)	4.287 (11.25)	4.400 (11.32)	4.418 (10.96)	4.271 (5.11)	4.501 (11.55)	4.325 (7.86)	4.531 (16.33)	4.449 (8.64)	4.512 (11.67)	5.889 (9.14)
40%	4.207 (6.53)	4.278 (11.01)	4.345 (9.94)	4.301 (8.04)	4.301 (5.85)	4.361 (8.09)	4.426 (10.39)	4.458 (14.44)	4.478 (9.35)	4.463 (10.44)	5.910 (9.52)
Imputation using propensity−score method											
10%	3.912 (−0.93)	4.071 (5.64)	4.126 (4.39)	4.106 (3.13)	4.064 (0.02)	4.191 (3.86)	4.204 (4.84)	4.203 (7.90)	4.275 (4.39)	4.197 (3.88)	5.628 (4.29)
20%	4.234 (7.21)	4.074 (5.72)	4.004 (1.32)	4.209 (5.72)	4.175 (2.75)	4.291 (6.35)	4.224 (5.34)	4.239 (8.83)	4.242 (3.59)	4.337 (7.33)	5.788 (7.25)
30%	4.372 (10.71)	4.393 (14.01)	4.518 (14.32)	4.544 (14.14)	4.355 (7.19)	4.561 (13.03)	4.427 (10.40)	4.786 (22.88)	4.559 (11.34)	4.545 (12.47)	6.026 (11.67)
40%	4.163 (5.41)	4.230 (9.77)	4.195 (6.13)	4.230 (6.24)	4.282 (5.39)	4.243 (5.15)	4.331 (8.02)	4.437 (13.91)	4.492 (9.70)	4.444 (9.99)	5.877 (8.91)

MR	Many_T2vs.T0	Many_T3vs.T0	Many_T4vs.T0	Many_T5vs.T0	Many_T6vs.T0	Many_T7vs.T0	Many_T8vs.T0	Many_T9vs.T0	Many_T10vs.T0
Scenario I: all means are 0 for T0,T1,...,T10									
0%	5.554	5.565	5.563	5.584	5.578	5.595	5.582	5.569	5.608
Imputation using linear regression									
10%	5.829 (4.95)	5.907 (6.16)	5.908 (6.20)	5.986 (7.20)	5.926 (6.24)	5.879 (5.07)	5.962 (6.81)	5.889 (5.75)	6.051 (7.90)
20%	6.479 (16.64)	6.491 (16.64)	6.501 (16.86)	6.433 (15.21)	6.428 (15.24)	6.573 (17.47)	6.524 (16.89)	6.565 (17.89)	6.515 (16.17)
30%	7.394 (33.13)	6.899 (23.98)	7.080 (27.27)	6.855 (22.77)	6.974 (25.03)	6.946 (24.14)	6.854 (22.79)	7.049 (26.58)	7.021 (25.19)
40%	7.980 (43.67)	8.115 (45.83)	8.101 (45.62)	7.781 (39.33)	8.121 (45.60)	8.053 (43.93)	8.274 (48.23)	8.355 (50.03)	8.579 (52.97)
Imputation using predictive mean matching method									
10%	5.611 (1.02)	5.630 (1.17)	5.623 (1.08)	5.675 (1.64)	5.653 (1.34)	5.657 (1.11)	5.695 (2.03)	5.615 (0.82)	5.758 (2.68)
20%	5.848 (5.29)	5.860 (5.31)	5.872 (5.55)	5.887 (5.43)	5.869 (5.22)	5.950 (6.35)	5.861 (5.00)	5.880 (5.59)	5.865 (4.58)
30%	5.942 (6.98)	5.848 (5.10)	5.766 (3.64)	5.800 (3.88)	5.804 (4.05)	5.840 (4.37)	5.743 (2.89)	5.804 (4.22)	5.905 (5.28)
40%	6.049 (8.91)	6.079 (9.25)	6.101 (9.66)	5.984 (7.16)	6.186 (10.89)	6.264 (11.96)	6.227 (11.56)	6.251 (12.25)	6.258 (11.58)
Imputation using propensity−score method									
10%	5.606 (0.94)	5.587 (0.40)	5.645 (1.47)	5.706 (2.19)	5.678 (1.80)	5.631 (0.63)	5.693 (1.99)	5.588 (0.33)	5.762 (2.74)
20%	5.918 (6.54)	5.802 (4.27)	5.850 (5.15)	5.850 (4.77)	5.881 (5.44)	5.942 (6.20)	5.976 (7.05)	5.885 (5.67)	5.943 (5.98)
30%	5.792 (4.27)	5.836 (4.88)	5.966 (7.25)	5.804 (3.95)	5.751 (3.09)	5.853 (4.61)	5.759 (3.17)	5.858 (5.19)	6.018 (7.31)
40%	5.961 (7.32)	5.837 (4.89)	6.023 (8.26)	5.836 (4.52)	6.072 (8.86)	6.132 (9.60)	6.017 (7.80)	6.100 (9.53)	6.123 (9.18)
Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu10=6.0									
0%	5.358	5.378	5.365	5.399	5.392	5.382	5.380	5.376	5.398
Imputation using linear regression									

MR	Many_T2vs.T0	Many_T3vs.T0	Many_T4vs.T0	Many_T5vs.T0	Many_T6vs.T0	Many_T7vs.T0	Many_T8vs.T0	Many_T9vs.T0	Many_T10vs.T0
10%	5.941 (10.90)	5.979 (11.16)	5.934 (10.61)	5.961 (10.40)	5.973 (10.76)	5.988 (11.25)	5.944 (10.49)	6.009 (11.76)	6.025 (11.60)
20%	6.252 (16.70)	6.132 (14.01)	6.443 (20.09)	6.221 (15.22)	6.328 (17.34)	6.263 (16.36)	6.237 (15.93)	6.281 (16.83)	6.277 (16.28)
30%	7.137 (33.21)	7.137 (32.70)	7.511 (40.01)	7.236 (34.03)	7.340 (36.11)	7.112 (32.14)	7.356 (36.74)	7.392 (37.50)	7.320 (35.59)
40%	7.891 (47.30)	8.098 (50.56)	7.638 (42.37)	7.796 (44.39)	8.271 (53.38)	8.426 (56.56)	8.955 (66.47)	7.994 (48.70)	7.998 (48.15)
Imputation using predictive mean matching method									
10%	5.697 (6.34)	5.656 (5.15)	5.643 (5.19)	5.648 (4.61)	5.678 (5.30)	5.704 (5.99)	5.670 (5.39)	5.709 (6.19)	5.703 (5.64)
20%	5.636 (5.19)	5.570 (3.57)	5.764 (7.45)	5.646 (4.57)	5.761 (6.83)	5.717 (6.22)	5.645 (4.92)	5.705 (6.11)	5.751 (6.53)
30%	5.909 (10.28)	5.951 (10.65)	6.014 (12.09)	5.935 (9.93)	6.088 (12.91)	5.906 (9.73)	6.056 (12.57)	5.998 (11.56)	6.096 (12.91)
40%	6.002 (12.03)	6.042 (12.34)	6.002 (11.87)	6.011 (11.33)	6.103 (13.18)	6.106 (13.46)	6.149 (14.29)	6.154 (14.47)	6.137 (13.68)
Imputation using propensity−score method									
10%	5.707 (6.52)	5.719 (6.34)	5.706 (6.36)	5.666 (4.95)	5.711 (5.91)	5.718 (6.25)	5.718 (6.29)	5.751 (6.96)	5.761 (6.72)
20%	5.721 (6.78)	5.643 (4.93)	5.776 (7.66)	5.789 (7.21)	5.880 (9.03)	5.758 (6.99)	5.762 (7.11)	5.746 (6.89)	5.871 (8.75)
30%	6.028 (12.51)	6.094 (13.31)	6.144 (14.52)	6.031 (11.70)	6.145 (13.96)	6.020 (11.85)	6.387 (18.72)	6.125 (13.93)	6.113 (13.24)
40%	5.960 (11.25)	5.886 (9.44)	5.924 (10.42)	5.997 (11.08)	5.991 (11.10)	5.990 (11.29)	6.123 (13.81)	6.196 (15.25)	6.117 (13.30)

SampleSize_n200_10Treatments/table2_nonshrink_10_treatments_MAR_n200

MR	T1vs.T0_none	T2vs.T0_none	T3vs.T0_none	T4vs.T0_none	T5vs.T0_none	T6vs.T0_none	T7vs.T0_none	T8vs.T0_none	T9vs.T0_none	T10vs.T0_none	Joint_none
Scenario I: all means are 0 for T0,T1,...,T10	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
0%	96.43	96.43	100.00	85.71	100.00	100.00	82.14	96.43	96.43	92.86	67.86
Imputation using linear regression	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	92.86	96.43	92.86	96.43	96.43	96.43	89.29	100.00	100.00	89.29	64.29
20%	89.66	82.76	96.55	96.55	93.10	100.00	93.10	100.00	96.55	93.10	65.52
30%	86.21	93.10	93.10	96.55	89.66	96.55	96.55	93.10	86.21	100.00	62.07
40%	96.43	96.43	96.43	96.43	96.43	92.86	92.86	92.86	96.43	96.43	75.00
Imputation using predictive mean matching method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	92.86	96.43	92.86	96.43	96.43	96.43	89.29	100.00	100.00	89.29	64.29
20%	86.21	82.76	96.55	96.55	93.10	100.00	93.10	100.00	96.55	93.10	62.07
30%	89.66	93.10	86.21	89.66	89.66	96.55	96.55	89.66	93.10	96.55	62.07
40%	96.43	92.86	85.71	100.00	92.86	85.71	85.71	96.43	96.43	96.43	60.71
Imputation using propensity−score method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	92.86	96.43	92.86	100.00	96.43	96.43	89.29	100.00	100.00	89.29	67.86
20%	89.66	79.31	96.55	96.55	93.10	100.00	93.10	100.00	96.55	93.10	65.52
30%	89.66	93.10	89.66	89.66	89.66	96.55	96.55	93.10	86.21	96.55	62.07
40%	92.86	96.43	96.43	100.00	96.43	92.86	92.86	96.43	100.00	96.43	67.86
Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu10=6.0	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
0%	100.00	100.00	96.43	96.43	96.43	92.86	92.86	100.00	96.43	100.00	82.14
Imputation using linear regression	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA

SampleSize_n200_10Treatments/table2_nonshrink_10_treatments_MAR_n200

MR	T1vs.T0_none	T2vs.T0_none	T3vs.T0_none	T4vs.T0_none	T5vs.T0_none	T6vs.T0_none	T7vs.T0_none	T8vs.T0_none	T9vs.T0_none	T10vs.T0_none	Joint_none
10%	100.00	100.00	96.55	100.00	100.00	100.00	93.10	93.10	100.00	100.00	82.76
20%	93.10	93.10	93.10	86.21	96.55	79.31	100.00	93.10	96.55	93.10	68.97
30%	96.43	100.00	96.43	100.00	96.43	92.86	100.00	92.86	96.43	96.43	78.57
40%	92.86	82.14	96.43	100.00	92.86	92.86	100.00	89.29	92.86	100.00	60.71
Imputation using predictive mean matching method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	100.00	100.00	96.55	100.00	100.00	100.00	93.10	96.55	96.55	100.00	82.76
20%	93.10	96.55	96.55	86.21	93.10	82.76	96.55	93.10	96.55	93.10	68.97
30%	96.43	100.00	100.00	100.00	100.00	92.86	96.43	96.43	92.86	96.43	82.14
40%	89.29	82.14	96.43	100.00	92.86	92.86	100.00	96.43	92.86	96.43	71.43
Imputation using propensity−score method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	96.55	100.00	93.10	100.00	100.00	100.00	96.55	93.10	100.00	100.00	82.76
20%	93.10	93.10	93.10	86.21	96.55	86.21	100.00	93.10	96.55	93.10	72.41
30%	96.43	100.00	100.00	100.00	96.43	92.86	96.43	92.86	92.86	96.43	78.57
40%	92.86	85.71	96.43	100.00	89.29	96.43	96.43	96.43	96.43	96.43	75.00

SampleSize_n200_10Treatments/table2_nonshrink_10_treatments_MAR_n200

MR	T1vs.T0_many	T2vs.T0_many	T3vs.T0_many	T4vs.T0_many	T5vs.T0_many	T6vs.T0_many	T7vs.T0_many	T8vs.T0_many	T9vs.T0_many	T10vs.T0_many	Joint_many
Scenario I: all means are 0 for T0,T1,...,T10	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
0%	100.00	100.00	100	100.00	100.00	100.00	100.00	100.00	96.43	100	96.43
Imputation using linear regression	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	100.00	96.43	100	100.00	100.00	100.00	100.00	100.00	100.00	100	96.43
20%	96.55	100.00	100	100.00	100.00	100.00	100.00	100.00	100.00	100	96.55
30%	100.00	96.55	100	96.55	96.55	100.00	100.00	100.00	93.10	100	82.76
40%	100.00	100.00	100	100.00	100.00	100.00	100.00	100.00	100.00	100	100.00
Imputation using predictive mean matching method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	100.00	96.43	100	100.00	100.00	100.00	100.00	100.00	100.00	100	96.43
20%	96.55	100.00	100	100.00	100.00	100.00	100.00	100.00	100.00	100	96.55
30%	100.00	100.00	100	96.55	100.00	100.00	100.00	100.00	93.10	100	89.66
40%	100.00	96.43	100	100.00	100.00	100.00	100.00	100.00	100.00	100	96.43
Imputation using propensity−score method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	100.00	96.43	100	100.00	100.00	96.43	96.43	100.00	100.00	100	92.86
20%	96.55	100.00	100	100.00	96.55	100.00	100.00	100.00	100.00	100	96.55
30%	100.00	100.00	100	100.00	96.55	100.00	100.00	96.55	93.10	100	86.21
40%	100.00	100.00	100	100.00	100.00	100.00	100.00	100.00	100.00	100	100.00
Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu10=6.0	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
0%	100.00	100.00	100	100.00	100.00	100.00	100.00	100.00	100.00	100	100.00
Imputation using linear regression	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA

SampleSize_n200_10Treatments/table2_nonshrink_10_treatments_MAR_n200

MR	T1vs.T0_many	T2vs.T0_many	T3vs.T0_many	T4vs.T0_many	T5vs.T0_many	T6vs.T0_many	T7vs.T0_many	T8vs.T0_many	T9vs.T0_many	T10vs.T0_many	Joint_many
10%	100.00	100.00	100	100.00	100.00	100.00	100	100.00	100.00	100	100.00
20%	96.55	100.00	100	93.10	100.00	96.55	100	96.55	100.00	100	93.10
30%	100.00	100.00	100	100.00	100.00	100.00	100	96.43	100.00	100	96.43
40%	100.00	96.43	100	100.00	100.00	100.00	100	100.00	96.43	100	96.43
Imputation using predictive mean matching method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	100.00	100.00	100	100.00	100.00	100.00	100	100.00	100.00	100	100.00
20%	96.55	100.00	100	93.10	100.00	96.55	100	100.00	100.00	100	89.66
30%	100.00	100.00	100	100.00	100.00	100.00	100	96.43	100.00	100	96.43
40%	100.00	96.43	100	100.00	96.43	100.00	100	100.00	96.43	100	96.43
Imputation using propensity−score method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	100.00	100.00	100	100.00	100.00	100.00	100	100.00	100.00	100	100.00
20%	96.55	100.00	100	89.66	100.00	96.55	100	96.55	100.00	100	89.66
30%	100.00	100.00	100	100.00	100.00	100.00	100	96.43	100.00	100	96.43
40%	100.00	100.00	100	100.00	96.43	100.00	100	100.00	96.43	100	96.43

SampleSize_n200_10Treatments/table2_shrink_10_treatments_MAR_n200

MR	T1vs.T0_none	T2vs.T0_none	T3vs.T0_none	T4vs.T0_none	T5vs.T0_none	T6vs.T0_none	T7vs.T0_none	T8vs.T0_none	T9vs.T0_none	T10vs.T0_none	Joint_none
Scenario I: all means are 0 for T0,T1,...,T10	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
0%	96.43	96.43	100.00	85.71	100.00	100.00	82.14	96.43	96.43	92.86	67.86
Imputation using linear regression	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	92.86	96.43	92.86	96.43	96.43	96.43	89.29	100.00	100.00	89.29	64.29
20%	89.66	82.76	96.55	96.55	93.10	100.00	93.10	100.00	96.55	93.10	65.52
30%	86.21	93.10	93.10	96.55	89.66	96.55	96.55	93.10	86.21	100.00	62.07
40%	96.43	96.43	96.43	96.43	96.43	92.86	92.86	92.86	96.43	96.43	75.00
Imputation using predictive mean matching method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	92.86	96.43	92.86	96.43	96.43	96.43	89.29	100.00	100.00	89.29	64.29
20%	86.21	82.76	96.55	96.55	93.10	100.00	93.10	100.00	96.55	93.10	62.07
30%	89.66	93.10	86.21	89.66	89.66	96.55	96.55	89.66	93.10	96.55	62.07
40%	96.43	92.86	85.71	100.00	92.86	85.71	85.71	96.43	96.43	96.43	60.71
Imputation using propensity−score method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	92.86	96.43	92.86	100.00	96.43	96.43	89.29	100.00	100.00	89.29	67.86
20%	89.66	79.31	96.55	96.55	93.10	100.00	93.10	100.00	96.55	93.10	65.52
30%	89.66	93.10	89.66	89.66	89.66	96.55	96.55	93.10	86.21	96.55	62.07
40%	92.86	96.43	96.43	100.00	96.43	92.86	92.86	96.43	100.00	96.43	67.86
Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu10=6.0	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
0%	100.00	100.00	96.43	96.43	96.43	92.86	92.86	100.00	96.43	100.00	82.14
Imputation using linear regression	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA

SampleSize_n200_10Treatments/table2_shrink_10_treatments_MAR_n200

MR	T1vs.T0_none	T2vs.T0_none	T3vs.T0_none	T4vs.T0_none	T5vs.T0_none	T6vs.T0_none	T7vs.T0_none	T8vs.T0_none	T9vs.T0_none	T10vs.T0_none	Joint_none
10%	100.00	100.00	96.55	100.00	100.00	100.00	93.10	93.10	100.00	100.00	82.76
20%	93.10	93.10	93.10	86.21	96.55	79.31	100.00	93.10	96.55	93.10	68.97
30%	96.43	100.00	96.43	100.00	96.43	92.86	100.00	92.86	96.43	96.43	78.57
40%	92.86	82.14	96.43	100.00	92.86	92.86	100.00	89.29	92.86	100.00	60.71
Imputation using predictive mean matching method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	100.00	100.00	96.55	100.00	100.00	100.00	93.10	96.55	96.55	100.00	82.76
20%	93.10	96.55	96.55	86.21	93.10	82.76	96.55	93.10	96.55	93.10	68.97
30%	96.43	100.00	100.00	100.00	100.00	92.86	96.43	96.43	92.86	96.43	82.14
40%	89.29	82.14	96.43	100.00	92.86	92.86	100.00	96.43	92.86	96.43	71.43
Imputation using propensity−score method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	96.55	100.00	93.10	100.00	100.00	100.00	96.55	93.10	100.00	100.00	82.76
20%	93.10	93.10	93.10	86.21	96.55	86.21	100.00	93.10	96.55	93.10	72.41
30%	96.43	100.00	100.00	100.00	96.43	92.86	96.43	92.86	92.86	96.43	78.57
40%	92.86	85.71	96.43	100.00	89.29	96.43	96.43	96.43	96.43	96.43	75.00

SampleSize_n200_10Treatments/table2_shrink_10_treatments_MAR_n200

MR	T1vs.T0_many	T2vs.T0_many	T3vs.T0_many	T4vs.T0_many	T5vs.T0_many	T6vs.T0_many	T7vs.T0_many	T8vs.T0_many	T9vs.T0_many	T10vs.T0_many	Joint_many
Scenario I: all means are 0 for T0,T1,...,T10	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
0%	100.00	100.00	100	100.00	100.00	100.00	100.00	100.00	96.43	100	96.43
Imputation using linear regression	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	100.00	96.43	100	100.00	100.00	96.43	96.43	100.00	100.00	100	92.86
20%	96.55	100.00	100	100.00	100.00	100.00	100.00	100.00	100.00	100	96.55
30%	100.00	96.55	100	96.55	96.55	100.00	100.00	100.00	93.10	100	82.76
40%	100.00	100.00	100	100.00	100.00	100.00	100.00	100.00	100.00	100	100.00
Imputation using predictive mean matching method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	100.00	96.43	100	100.00	100.00	96.43	96.43	100.00	100.00	100	89.29
20%	96.55	100.00	100	100.00	100.00	100.00	100.00	100.00	100.00	100	96.55
30%	100.00	100.00	100	96.55	100.00	100.00	100.00	100.00	93.10	100	89.66
40%	100.00	96.43	100	100.00	100.00	100.00	100.00	100.00	100.00	100	96.43
Imputation using propensity−score method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	100.00	96.43	100	100.00	100.00	96.43	92.86	100.00	100.00	100	89.29
20%	96.55	100.00	100	100.00	96.55	100.00	100.00	100.00	100.00	100	96.55
30%	100.00	100.00	100	100.00	96.55	100.00	100.00	96.55	93.10	100	86.21
40%	100.00	100.00	100	100.00	100.00	100.00	100.00	100.00	100.00	100	100.00
Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu10=6.0	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
0%	100.00	100.00	100	100.00	100.00	100.00	100.00	100.00	100.00	100	100.00
Imputation using linear regression	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA

SampleSize_n200_10Treatments/table2_shrink_10_treatments_MAR_n200

MR	T1vs.T0_many	T2vs.T0_many	T3vs.T0_many	T4vs.T0_many	T5vs.T0_many	T6vs.T0_many	T7vs.T0_many	T8vs.T0_many	T9vs.T0_many	T10vs.T0_many	Joint_many
10%	100.00	100.00	100	100.00	100.00	100.00	100	100.00	100.00	100.00	100.00
20%	96.55	100.00	100	93.10	100.00	96.55	100	93.10	100.00	100.00	89.66
30%	100.00	100.00	100	100.00	100.00	100.00	100	96.43	100.00	100.00	96.43
40%	100.00	96.43	100	100.00	100.00	100.00	100	100.00	96.43	100.00	96.43
Imputation using predictive mean matching method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	100.00	100.00	100	100.00	100.00	100.00	100	100.00	100.00	100.00	100.00
20%	96.55	100.00	100	96.55	100.00	96.55	100	100.00	100.00	96.55	93.10
30%	100.00	100.00	100	100.00	100.00	100.00	100	96.43	100.00	100.00	96.43
40%	100.00	96.43	100	100.00	96.43	100.00	100	100.00	96.43	100.00	96.43
Imputation using propensity−score method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	100.00	100.00	100	100.00	100.00	100.00	100	100.00	100.00	100.00	100.00
20%	96.55	100.00	100	89.66	100.00	96.55	100	96.55	100.00	96.55	89.66
30%	100.00	100.00	100	100.00	100.00	100.00	100	96.43	100.00	100.00	96.43
40%	100.00	100.00	100	100.00	96.43	100.00	100	100.00	96.43	100.00	96.43

SampleSize_n200_10Treatments/table3_nonshrink_10_treatments_MAR_n200

MR	None_T1vs.T0	None_T2vs.T0	None_T3vs.T0	None_T4vs.T0	None_T5vs.T0	None_T6vs.T0	None_T7vs.T0	None_T8vs.T0	None_T9vs.T0	None_T10vs.T0	Many_T1vs.T0
Scenario I: all means are 0 for T0,T1,...,T10											
0%	1.255	1.248	1.256	1.267	1.263	1.271	1.279	1.294	1.294	1.301	1.728
Imputation using linear regression											
10%	1.310 (4.38)	1.308 (4.76)	1.322 (5.31)	1.326 (4.61)	1.324 (4.89)	1.333 (4.92)	1.336 (4.46)	1.333 (2.96)	1.358 (4.96)	1.367 (5.09)	1.805 (4.47)
20%	1.361 (8.48)	1.375 (10.12)	1.364 (8.61)	1.381 (8.96)	1.383 (9.52)	1.394 (9.69)	1.402 (9.58)	1.395 (7.75)	1.411 (9.08)	1.421 (9.22)	1.882 (8.92)
30%	1.463 (16.58)	1.472 (17.92)	1.483 (18.09)	1.484 (17.08)	1.501 (18.86)	1.517 (19.36)	1.505 (17.69)	1.504 (16.19)	1.524 (17.81)	1.524 (17.10)	2.021 (16.95)
40%	1.601 (27.61)	1.628 (30.37)	1.622 (29.21)	1.665 (31.35)	1.649 (30.58)	1.662 (30.77)	1.659 (29.74)	1.658 (28.10)	1.645 (27.14)	1.690 (29.89)	2.208 (27.76)
Imputation using predictive mean matching method											
10%	1.302 (3.79)	1.301 (4.19)	1.313 (4.60)	1.324 (4.47)	1.320 (4.54)	1.328 (4.50)	1.329 (3.89)	1.327 (2.53)	1.350 (4.33)	1.355 (4.12)	1.795 (3.88)
20%	1.346 (7.24)	1.351 (8.23)	1.343 (6.99)	1.355 (6.96)	1.372 (8.64)	1.369 (7.72)	1.387 (8.48)	1.386 (7.10)	1.391 (7.52)	1.401 (7.66)	1.860 (7.62)
30%	1.445 (15.16)	1.465 (17.35)	1.457 (16.01)	1.429 (12.74)	1.474 (16.77)	1.489 (17.16)	1.485 (16.12)	1.480 (14.34)	1.519 (17.41)	1.493 (14.71)	1.992 (15.24)
40%	1.541 (22.80)	1.570 (25.78)	1.573 (25.26)	1.604 (26.55)	1.614 (27.81)	1.594 (25.39)	1.592 (24.45)	1.591 (22.95)	1.579 (22.04)	1.593 (22.45)	2.123 (22.84)
Imputation using propensity−score method											
10%	1.307 (4.20)	1.308 (4.80)	1.319 (5.06)	1.325 (4.59)	1.323 (4.79)	1.332 (4.82)	1.333 (4.24)	1.339 (3.48)	1.361 (5.16)	1.361 (4.60)	1.802 (4.26)
20%	1.346 (7.24)	1.353 (8.39)	1.356 (7.99)	1.355 (6.95)	1.378 (9.16)	1.379 (8.50)	1.395 (9.07)	1.383 (6.88)	1.394 (7.71)	1.411 (8.43)	1.862 (7.72)
30%	1.458 (16.17)	1.444 (15.67)	1.470 (17.05)	1.438 (13.44)	1.461 (15.71)	1.471 (15.75)	1.471 (15.03)	1.479 (14.29)	1.500 (15.92)	1.493 (14.71)	2.014 (16.53)
40%	1.564 (24.68)	1.616 (29.44)	1.592 (26.76)	1.626 (28.28)	1.630 (29.12)	1.620 (27.42)	1.626 (27.11)	1.632 (26.07)	1.620 (25.24)	1.645 (26.41)	2.159 (24.95)
Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu10=6.0											
0%	1.244	1.257	1.256	1.263	1.277	1.279	1.292	1.277	1.299	1.295	1.712
Imputation using linear regression											

SampleSize_n200_10Treatments/table3_nonshrink_10_treatments_MAR_n200

MR	None_T1vs.T0	None_T2vs.T0	None_T3vs.T0	None_T4vs.T0	None_T5vs.T0	None_T6vs.T0	None_T7vs.T0	None_T8vs.T0	None_T9vs.T0	None_T10vs.T0	Many_T1vs.T0
10%	1.291 (3.81)	1.294 (2.90)	1.307 (4.06)	1.305 (3.36)	1.309 (2.50)	1.319 (3.06)	1.330 (2.96)	1.338 (4.75)	1.342 (3.32)	1.348 (4.14)	1.783 (4.16)
20%	1.372 (10.25)	1.382 (9.91)	1.382 (10.00)	1.388 (9.88)	1.395 (9.17)	1.406 (9.87)	1.407 (8.89)	1.419 (11.09)	1.409 (8.51)	1.430 (10.49)	1.894 (10.63)
30%	1.445 (16.16)	1.478 (17.54)	1.475 (17.47)	1.481 (17.27)	1.481 (15.93)	1.489 (16.34)	1.505 (16.51)	1.484 (16.19)	1.524 (17.32)	1.499 (15.80)	1.999 (16.75)
40%	1.595 (28.23)	1.603 (27.50)	1.615 (28.60)	1.604 (27.03)	1.629 (27.51)	1.611 (25.93)	1.614 (24.90)	1.608 (25.94)	1.668 (28.44)	1.678 (29.64)	2.201 (28.60)
Imputation using predictive mean matching method											
10%	1.279 (2.79)	1.285 (2.21)	1.296 (3.15)	1.298 (2.76)	1.303 (2.01)	1.309 (2.34)	1.313 (1.62)	1.325 (3.74)	1.327 (2.22)	1.336 (3.17)	1.766 (3.17)
20%	1.350 (8.50)	1.366 (8.65)	1.374 (9.36)	1.370 (8.49)	1.386 (8.52)	1.389 (8.57)	1.392 (7.75)	1.404 (9.96)	1.386 (6.75)	1.408 (8.77)	1.862 (8.79)
30%	1.431 (15.06)	1.434 (14.07)	1.443 (14.88)	1.424 (12.77)	1.430 (11.95)	1.445 (12.91)	1.462 (13.17)	1.452 (13.70)	1.472 (13.31)	1.475 (13.97)	1.980 (15.69)
40%	1.546 (24.29)	1.562 (24.22)	1.560 (24.17)	1.570 (24.32)	1.567 (22.66)	1.592 (24.45)	1.609 (24.59)	1.596 (24.98)	1.605 (23.61)	1.602 (23.75)	2.124 (24.07)
Imputation using propensity−score method											
10%	1.286 (3.33)	1.286 (2.29)	1.301 (3.60)	1.301 (2.99)	1.306 (2.22)	1.308 (2.20)	1.321 (2.25)	1.331 (4.21)	1.332 (2.60)	1.339 (3.45)	1.775 (3.68)
20%	1.360 (9.33)	1.373 (9.20)	1.365 (8.68)	1.374 (8.84)	1.389 (8.76)	1.394 (8.94)	1.410 (9.14)	1.409 (10.35)	1.398 (7.67)	1.419 (9.60)	1.878 (9.70)
30%	1.443 (15.96)	1.455 (15.72)	1.441 (14.74)	1.458 (15.46)	1.466 (14.81)	1.458 (13.99)	1.479 (14.47)	1.477 (15.63)	1.491 (14.79)	1.486 (14.79)	1.996 (16.58)
40%	1.539 (23.72)	1.556 (23.80)	1.603 (27.60)	1.593 (26.16)	1.571 (23.01)	1.607 (25.63)	1.603 (24.06)	1.597 (25.05)	1.625 (25.11)	1.634 (26.21)	2.125 (24.15)

SampleSize_n200_10Treatments/table3_nonshrink_10_treatments_MAR_n200

MR	Many_T2vs.T0	Many_T3vs.T0	Many_T4vs.T0	Many_T5vs.T0	Many_T6vs.T0	Many_T7vs.T0	Many_T8vs.T0	Many_T9vs.T0	Many_T10vs.T0
Scenario I: all means are 0 for T0,T1,...,T10									
0%	1.719	1.729	1.745	1.739	1.751	1.762	1.783	1.782	1.792
Imputation using linear regression									
10%	1.803 (4.86)	1.823 (5.41)	1.828 (4.71)	1.826 (4.99)	1.838 (5.01)	1.842 (4.56)	1.837 (3.05)	1.872 (5.06)	1.885 (5.18)
20%	1.901 (10.55)	1.886 (9.05)	1.909 (9.37)	1.912 (9.96)	1.928 (10.12)	1.938 (10.02)	1.929 (8.19)	1.951 (9.51)	1.965 (9.64)
30%	2.034 (18.30)	2.049 (18.45)	2.050 (17.44)	2.073 (19.23)	2.096 (19.71)	2.080 (18.06)	2.078 (16.58)	2.106 (18.18)	2.105 (17.45)
40%	2.245 (30.54)	2.238 (29.41)	2.295 (31.47)	2.274 (30.76)	2.292 (30.90)	2.288 (29.90)	2.287 (28.27)	2.269 (27.31)	2.330 (30.01)
Imputation using predictive mean matching method									
10%	1.793 (4.28)	1.811 (4.70)	1.825 (4.55)	1.820 (4.63)	1.831 (4.59)	1.832 (3.99)	1.829 (2.62)	1.861 (4.42)	1.868 (4.21)
20%	1.867 (8.61)	1.857 (7.39)	1.873 (7.33)	1.896 (9.04)	1.892 (8.10)	1.918 (8.86)	1.916 (7.47)	1.923 (7.90)	1.937 (8.04)
30%	2.019 (17.42)	2.008 (16.11)	1.970 (12.87)	2.032 (16.86)	2.052 (17.24)	2.047 (16.21)	2.040 (14.44)	2.094 (17.48)	2.058 (14.81)
40%	2.163 (25.80)	2.167 (25.32)	2.208 (26.52)	2.222 (27.78)	2.196 (25.44)	2.193 (24.50)	2.193 (23.00)	2.177 (22.14)	2.196 (22.50)
Imputation using propensity−score method									
10%	1.803 (4.86)	1.818 (5.12)	1.827 (4.65)	1.823 (4.86)	1.836 (4.88)	1.837 (4.30)	1.846 (3.53)	1.875 (5.22)	1.876 (4.66)
20%	1.872 (8.87)	1.876 (8.47)	1.875 (7.41)	1.907 (9.65)	1.907 (8.96)	1.930 (9.54)	1.914 (7.35)	1.928 (8.18)	1.952 (8.89)
30%	1.996 (16.08)	2.031 (17.42)	1.987 (13.83)	2.020 (16.13)	2.033 (16.15)	2.034 (15.43)	2.045 (14.68)	2.073 (16.30)	2.063 (15.09)
40%	2.229 (29.66)	2.197 (27.04)	2.243 (28.49)	2.249 (29.35)	2.235 (27.70)	2.244 (27.36)	2.252 (26.32)	2.236 (25.50)	2.270 (26.65)
Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu10=6.0									
0%	1.730	1.728	1.738	1.758	1.761	1.778	1.757	1.787	1.782
Imputation using linear regression									

SampleSize_n200_10Treatments/table3_nonshrink_10_treatments_MAR_n200

MR	Many_T2vs.T0	Many_T3vs.T0	Many_T4vs.T0	Many_T5vs.T0	Many_T6vs.T0	Many_T7vs.T0	Many_T8vs.T0	Many_T9vs.T0	Many_T10vs.T0
10%	1.786 (3.24)	1.804 (4.40)	1.802 (3.70)	1.808 (2.84)	1.820 (3.40)	1.836 (3.29)	1.847 (5.10)	1.852 (3.65)	1.861 (4.48)
20%	1.908 (10.28)	1.908 (10.38)	1.916 (10.26)	1.925 (9.53)	1.941 (10.24)	1.942 (9.26)	1.959 (11.47)	1.946 (8.89)	1.975 (10.86)
30%	2.043 (18.12)	2.041 (18.06)	2.048 (17.87)	2.048 (16.51)	2.059 (16.93)	2.081 (17.09)	2.053 (16.81)	2.107 (17.91)	2.074 (16.41)
40%	2.212 (27.89)	2.229 (28.97)	2.215 (27.47)	2.247 (27.85)	2.224 (26.33)	2.228 (25.34)	2.220 (26.36)	2.302 (28.81)	2.316 (30.00)
Imputation using predictive mean matching method									
10%	1.774 (2.58)	1.789 (3.53)	1.792 (3.13)	1.800 (2.38)	1.808 (2.71)	1.813 (1.99)	1.830 (4.13)	1.833 (2.59)	1.845 (3.54)
20%	1.884 (8.93)	1.895 (9.65)	1.891 (8.79)	1.912 (8.79)	1.917 (8.86)	1.921 (8.04)	1.937 (10.25)	1.913 (7.04)	1.943 (9.06)
30%	1.985 (14.73)	1.997 (15.53)	1.972 (13.45)	1.980 (12.62)	1.999 (13.57)	2.023 (13.81)	2.010 (14.36)	2.037 (13.98)	2.042 (14.63)
40%	2.145 (24.01)	2.142 (23.96)	2.157 (24.15)	2.152 (22.45)	2.186 (24.16)	2.211 (24.36)	2.192 (24.77)	2.206 (23.43)	2.201 (23.56)
Imputation using propensity−score method									
10%	1.775 (2.63)	1.797 (3.94)	1.796 (3.33)	1.803 (2.56)	1.805 (2.55)	1.824 (2.59)	1.837 (4.56)	1.839 (2.94)	1.849 (3.79)
20%	1.895 (9.57)	1.885 (9.07)	1.898 (9.22)	1.918 (9.13)	1.925 (9.31)	1.947 (9.50)	1.946 (10.74)	1.931 (8.04)	1.959 (9.97)
30%	2.012 (16.33)	1.994 (15.39)	2.017 (16.09)	2.029 (15.44)	2.018 (14.62)	2.046 (15.10)	2.043 (16.25)	2.063 (15.44)	2.057 (15.44)
40%	2.149 (24.23)	2.212 (27.96)	2.200 (26.58)	2.169 (23.40)	2.218 (26.00)	2.213 (24.49)	2.204 (25.45)	2.243 (25.52)	2.256 (26.61)

SampleSize_n200_10Treatments/table3_shrink_10_treatments_MAR_n200

MR	None_T1vs.T0	None_T2vs.T0	None_T3vs.T0	None_T4vs.T0	None_T5vs.T0	None_T6vs.T0	None_T7vs.T0	None_T8vs.T0	None_T9vs.T0	None_T10vs.T0	Many_T1vs.T0
Scenario I: all means are 0 for T0,T1,...,T10											
0%	1.255	1.248	1.256	1.267	1.263	1.271	1.279	1.294	1.294	1.301	1.724
Imputation using linear regression											
10%	1.310 (4.38)	1.308 (4.76)	1.322 (5.31)	1.326 (4.61)	1.324 (4.89)	1.333 (4.92)	1.336 (4.46)	1.333 (2.96)	1.358 (4.96)	1.367 (5.09)	1.804 (4.64)
20%	1.361 (8.48)	1.375 (10.12)	1.364 (8.61)	1.381 (8.96)	1.383 (9.52)	1.394 (9.69)	1.402 (9.58)	1.395 (7.75)	1.411 (9.08)	1.421 (9.22)	1.887 (9.45)
30%	1.463 (16.58)	1.472 (17.92)	1.483 (18.09)	1.484 (17.08)	1.501 (18.86)	1.517 (19.36)	1.505 (17.69)	1.504 (16.19)	1.524 (17.81)	1.524 (17.10)	2.023 (17.33)
40%	1.601 (27.61)	1.628 (30.37)	1.622 (29.21)	1.665 (31.35)	1.649 (30.58)	1.662 (30.77)	1.659 (29.74)	1.658 (28.10)	1.645 (27.14)	1.690 (29.89)	2.209 (28.10)
Imputation using predictive mean matching method											
10%	1.302 (3.79)	1.301 (4.19)	1.313 (4.60)	1.324 (4.47)	1.320 (4.54)	1.328 (4.50)	1.329 (3.89)	1.327 (2.53)	1.350 (4.33)	1.355 (4.12)	1.794 (4.04)
20%	1.346 (7.24)	1.351 (8.23)	1.343 (6.99)	1.355 (6.96)	1.372 (8.64)	1.369 (7.72)	1.387 (8.48)	1.386 (7.10)	1.391 (7.52)	1.401 (7.66)	1.864 (8.09)
30%	1.445 (15.16)	1.465 (17.35)	1.457 (16.01)	1.429 (12.74)	1.474 (16.77)	1.489 (17.16)	1.485 (16.12)	1.480 (14.34)	1.519 (17.41)	1.493 (14.71)	1.995 (15.67)
40%	1.541 (22.80)	1.570 (25.78)	1.573 (25.26)	1.604 (26.55)	1.614 (27.81)	1.594 (25.39)	1.592 (24.45)	1.591 (22.95)	1.579 (22.04)	1.593 (22.45)	2.124 (23.17)
Imputation using propensity−score method											
10%	1.307 (4.20)	1.308 (4.80)	1.319 (5.06)	1.325 (4.59)	1.323 (4.79)	1.332 (4.82)	1.333 (4.24)	1.339 (3.48)	1.361 (5.16)	1.361 (4.60)	1.801 (4.42)
20%	1.346 (7.24)	1.353 (8.39)	1.356 (7.99)	1.355 (6.95)	1.378 (9.16)	1.379 (8.50)	1.395 (9.07)	1.383 (6.88)	1.394 (7.71)	1.411 (8.43)	1.866 (8.19)
30%	1.458 (16.17)	1.444 (15.67)	1.470 (17.05)	1.438 (13.44)	1.461 (15.71)	1.471 (15.75)	1.471 (15.03)	1.479 (14.29)	1.500 (15.92)	1.493 (14.71)	2.018 (17.01)
40%	1.564 (24.68)	1.616 (29.44)	1.592 (26.76)	1.626 (28.28)	1.630 (29.12)	1.620 (27.42)	1.626 (27.11)	1.632 (26.07)	1.620 (25.24)	1.645 (26.41)	2.160 (25.28)
Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu10=6.0											
0%	1.244	1.257	1.256	1.263	1.277	1.279	1.292	1.277	1.299	1.295	1.717
Imputation using linear regression											

SampleSize_n200_10Treatments/table3_shrink_10_treatments_MAR_n200

MR	None_T1vs.T0	None_T2vs.T0	None_T3vs.T0	None_T4vs.T0	None_T5vs.T0	None_T6vs.T0	None_T7vs.T0	None_T8vs.T0	None_T9vs.T0	None_T10vs.T0	Many_T1vs.T0
10%	1.291 (3.81)	1.294 (2.90)	1.307 (4.06)	1.305 (3.36)	1.309 (2.50)	1.319 (3.06)	1.330 (2.96)	1.338 (4.75)	1.342 (3.32)	1.348 (4.14)	1.788 (4.13)
20%	1.372 (10.25)	1.382 (9.91)	1.382 (10.00)	1.388 (9.88)	1.395 (9.17)	1.406 (9.87)	1.407 (8.89)	1.419 (11.09)	1.409 (8.51)	1.430 (10.49)	1.901 (10.72)
30%	1.445 (16.16)	1.478 (17.54)	1.475 (17.47)	1.481 (17.27)	1.481 (15.93)	1.489 (16.34)	1.505 (16.51)	1.484 (16.19)	1.524 (17.32)	1.499 (15.80)	2.010 (17.12)
40%	1.595 (28.23)	1.603 (27.50)	1.615 (28.60)	1.604 (27.03)	1.629 (27.51)	1.611 (25.93)	1.614 (24.90)	1.608 (25.94)	1.668 (28.44)	1.678 (29.64)	2.202 (28.29)
Imputation using predictive mean matching method											
10%	1.279 (2.79)	1.285 (2.21)	1.296 (3.15)	1.298 (2.76)	1.303 (2.01)	1.309 (2.34)	1.313 (1.62)	1.325 (3.74)	1.327 (2.22)	1.336 (3.17)	1.772 (3.22)
20%	1.350 (8.50)	1.366 (8.65)	1.374 (9.36)	1.370 (8.49)	1.386 (8.52)	1.389 (8.57)	1.392 (7.75)	1.404 (9.96)	1.386 (6.75)	1.408 (8.77)	1.871 (9.00)
30%	1.431 (15.06)	1.434 (14.07)	1.443 (14.88)	1.424 (12.77)	1.430 (11.95)	1.445 (12.91)	1.462 (13.17)	1.452 (13.70)	1.472 (13.31)	1.475 (13.97)	1.991 (15.98)
40%	1.546 (24.29)	1.562 (24.22)	1.560 (24.17)	1.570 (24.32)	1.567 (22.66)	1.592 (24.45)	1.609 (24.59)	1.596 (24.98)	1.605 (23.61)	1.602 (23.75)	2.124 (23.75)
Imputation using propensity−score method											
10%	1.286 (3.33)	1.286 (2.29)	1.301 (3.60)	1.301 (2.99)	1.306 (2.22)	1.308 (2.20)	1.321 (2.25)	1.331 (4.21)	1.332 (2.60)	1.339 (3.45)	1.780 (3.68)
20%	1.360 (9.33)	1.373 (9.20)	1.365 (8.68)	1.374 (8.84)	1.389 (8.76)	1.394 (8.94)	1.410 (9.14)	1.409 (10.35)	1.398 (7.67)	1.419 (9.60)	1.884 (9.76)
30%	1.443 (15.96)	1.455 (15.72)	1.441 (14.74)	1.458 (15.46)	1.466 (14.81)	1.458 (13.99)	1.479 (14.47)	1.477 (15.63)	1.491 (14.79)	1.486 (14.79)	2.006 (16.85)
40%	1.539 (23.72)	1.556 (23.80)	1.603 (27.60)	1.593 (26.16)	1.571 (23.01)	1.607 (25.63)	1.603 (24.06)	1.597 (25.05)	1.625 (25.11)	1.634 (26.21)	2.127 (23.89)

MR	Many_T2vs.T0	Many_T3vs.T0	Many_T4vs.T0	Many_T5vs.T0	Many_T6vs.T0	Many_T7vs.T0	Many_T8vs.T0	Many_T9vs.T0	Many_T10vs.T0
Scenario I: all means are 0 for T0,T1,...,T10									
0%	1.722	1.727	1.731	1.728	1.731	1.738	1.743	1.746	1.748
Imputation using linear regression									
10%	1.799 (4.51)	1.809 (4.77)	1.811 (4.62)	1.811 (4.75)	1.817 (4.94)	1.816 (4.48)	1.810 (3.83)	1.832 (4.91)	1.839 (5.23)
20%	1.899 (10.32)	1.887 (9.27)	1.911 (10.36)	1.901 (9.96)	1.914 (10.53)	1.916 (10.25)	1.914 (9.81)	1.930 (10.51)	1.940 (10.98)
30%	2.034 (18.15)	2.046 (18.44)	2.049 (18.38)	2.064 (19.45)	2.083 (20.33)	2.067 (18.92)	2.062 (18.34)	2.086 (19.46)	2.086 (19.37)
40%	2.235 (29.82)	2.227 (28.94)	2.283 (31.85)	2.261 (30.80)	2.275 (31.38)	2.275 (30.89)	2.267 (30.07)	2.253 (28.99)	2.310 (32.17)
Imputation using predictive mean matching method									
10%	1.789 (3.94)	1.797 (4.06)	1.807 (4.37)	1.804 (4.36)	1.808 (4.43)	1.805 (3.87)	1.801 (3.36)	1.821 (4.26)	1.824 (4.35)
20%	1.866 (8.36)	1.859 (7.64)	1.876 (8.35)	1.883 (8.93)	1.879 (8.54)	1.895 (9.02)	1.901 (9.06)	1.899 (8.76)	1.909 (9.24)
30%	2.020 (17.30)	2.005 (16.10)	1.971 (13.82)	2.025 (17.17)	2.041 (17.91)	2.033 (16.96)	2.026 (16.24)	2.072 (18.64)	2.038 (16.63)
40%	2.154 (25.12)	2.156 (24.82)	2.195 (26.80)	2.207 (27.70)	2.176 (25.69)	2.175 (25.12)	2.172 (24.62)	2.159 (23.64)	2.173 (24.36)
Imputation using propensity−score method									
10%	1.800 (4.57)	1.805 (4.51)	1.810 (4.52)	1.806 (4.52)	1.813 (4.73)	1.811 (4.22)	1.818 (4.32)	1.834 (5.04)	1.831 (4.75)
20%	1.870 (8.59)	1.877 (8.68)	1.878 (8.49)	1.894 (9.58)	1.895 (9.48)	1.908 (9.76)	1.898 (8.90)	1.904 (9.06)	1.925 (10.15)
30%	1.997 (16.02)	2.029 (17.50)	1.991 (14.97)	2.014 (16.53)	2.023 (16.87)	2.022 (16.32)	2.032 (16.58)	2.055 (17.66)	2.046 (17.07)
40%	2.221 (28.99)	2.186 (26.57)	2.231 (28.88)	2.236 (29.37)	2.219 (28.13)	2.229 (28.25)	2.232 (28.04)	2.222 (27.24)	2.251 (28.81)
Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu10=6.0									
0%	1.723	1.722	1.725	1.729	1.725	1.735	1.728	1.740	1.737
Imputation using linear regression									

MR	Many_T2vs.T0	Many_T3vs.T0	Many_T4vs.T0	Many_T5vs.T0	Many_T6vs.T0	Many_T7vs.T0	Many_T8vs.T0	Many_T9vs.T0	Many_T10vs.T0
10%	1.789 (3.80)	1.797 (4.35)	1.797 (4.16)	1.798 (4.00)	1.802 (4.51)	1.814 (4.51)	1.816 (5.07)	1.821 (4.67)	1.827 (5.22)
20%	1.906 (10.63)	1.905 (10.59)	1.906 (10.49)	1.919 (11.01)	1.922 (11.46)	1.922 (10.78)	1.932 (11.80)	1.926 (10.70)	1.946 (12.05)
30%	2.042 (18.54)	2.036 (18.21)	2.044 (18.49)	2.043 (18.20)	2.050 (18.85)	2.067 (19.13)	2.040 (18.05)	2.083 (19.73)	2.054 (18.26)
40%	2.207 (28.08)	2.223 (29.08)	2.204 (27.77)	2.241 (29.63)	2.216 (28.46)	2.211 (27.43)	2.207 (27.73)	2.282 (31.17)	2.294 (32.08)
Imputation using predictive mean matching method									
10%	1.779 (3.27)	1.783 (3.52)	1.786 (3.57)	1.789 (3.49)	1.791 (3.86)	1.791 (3.24)	1.798 (4.06)	1.803 (3.60)	1.810 (4.24)
20%	1.885 (9.38)	1.892 (9.86)	1.881 (9.04)	1.905 (10.18)	1.897 (9.96)	1.899 (9.44)	1.910 (10.52)	1.894 (8.84)	1.913 (10.17)
30%	1.987 (15.31)	1.995 (15.85)	1.967 (14.06)	1.973 (14.14)	1.991 (15.44)	2.009 (15.77)	1.997 (15.59)	2.010 (15.53)	2.021 (16.39)
40%	2.141 (24.26)	2.136 (24.03)	2.145 (24.34)	2.147 (24.20)	2.177 (26.20)	2.192 (26.35)	2.177 (26.00)	2.185 (25.61)	2.177 (25.38)
Imputation using propensity−score method									
10%	1.779 (3.23)	1.790 (3.93)	1.790 (3.78)	1.792 (3.68)	1.788 (3.66)	1.800 (3.76)	1.805 (4.46)	1.808 (3.94)	1.814 (4.49)
20%	1.895 (10.01)	1.883 (9.33)	1.889 (9.53)	1.911 (10.55)	1.907 (10.54)	1.927 (11.05)	1.918 (11.00)	1.911 (9.83)	1.931 (11.19)
30%	2.013 (16.85)	1.991 (15.63)	2.014 (16.76)	2.020 (16.86)	2.012 (16.63)	2.031 (17.06)	2.031 (17.56)	2.038 (17.14)	2.035 (17.18)
40%	2.145 (24.50)	2.205 (28.05)	2.189 (26.89)	2.165 (25.22)	2.210 (28.12)	2.197 (26.59)	2.192 (26.85)	2.225 (27.86)	2.234 (28.63)

SampleSize_n250_10Treatments/table2_nonshrink_10_treatments_MAR_n250

MR	T1vs.T0_none	T2vs.T0_none	T3vs.T0_none	T4vs.T0_none	T5vs.T0_none	T6vs.T0_none	T7vs.T0_none	T8vs.T0_none	T9vs.T0_none	T10vs.T0_none	Joint_none
Scenario I: all means are 0 for T0,T1,...,T10	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
0%	89.29	100.00	100.00	100.00	96.43	96.43	100.00	89.29	96.43	96.43	67.86
Imputation using linear regression	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	93.10	96.55	96.55	100.00	96.55	96.55	93.10	100.00	93.10	93.10	79.31
20%	93.10	96.55	96.55	100.00	96.55	96.55	100.00	96.55	93.10	93.10	68.97
30%	100.00	89.29	89.29	92.86	96.43	96.43	100.00	89.29	89.29	92.86	67.86
40%	96.43	100.00	89.29	100.00	100.00	92.86	96.43	100.00	85.71	96.43	75.00
Imputation using predictive mean matching method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	93.10	96.55	93.10	100.00	96.55	96.55	93.10	100.00	93.10	93.10	79.31
20%	89.66	89.66	96.55	100.00	96.55	96.55	96.55	96.55	93.10	89.66	58.62
30%	96.43	89.29	89.29	89.29	92.86	89.29	96.43	89.29	89.29	92.86	64.29
40%	96.43	96.43	96.43	100.00	100.00	89.29	96.43	100.00	85.71	96.43	78.57
Imputation using propensity−score method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	93.10	96.55	93.10	100.00	96.55	100.00	96.55	100.00	96.55	93.10	82.76
20%	89.66	89.66	93.10	100.00	93.10	96.55	100.00	96.55	93.10	89.66	62.07
30%	100.00	85.71	89.29	89.29	96.43	96.43	100.00	96.43	92.86	96.43	67.86
40%	96.43	96.43	92.86	100.00	100.00	89.29	100.00	100.00	85.71	96.43	75.00
Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu10=6.0	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
0%	96.43	92.86	92.86	96.43	100.00	96.43	96.43	96.43	96.43	100.00	71.43
Imputation using linear regression	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA

SampleSize_n250_10Treatments/table2_nonshrink_10_treatments_MAR_n250

MR	T1vs.T0_none	T2vs.T0_none	T3vs.T0_none	T4vs.T0_none	T5vs.T0_none	T6vs.T0_none	T7vs.T0_none	T8vs.T0_none	T9vs.T0_none	T10vs.T0_none	Joint_none
10%	96.55	93.10	89.66	93.10	96.55	86.21	96.55	100.00	96.55	96.55	65.52
20%	96.55	86.21	100.00	89.66	86.21	86.21	93.10	89.66	100.00	86.21	65.52
30%	92.86	92.86	92.86	85.71	82.14	92.86	82.14	89.29	85.71	92.86	46.43
40%	96.55	89.66	93.10	96.55	96.55	93.10	96.55	100.00	93.10	89.66	65.52
Imputation using predictive mean matching method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	96.55	93.10	93.10	93.10	93.10	89.66	100.00	96.55	100.00	96.55	65.52
20%	96.55	96.55	93.10	86.21	89.66	86.21	96.55	89.66	100.00	89.66	65.52
30%	92.86	89.29	89.29	85.71	82.14	89.29	82.14	85.71	92.86	89.29	42.86
40%	96.55	89.66	89.66	93.10	96.55	93.10	96.55	96.55	93.10	89.66	65.52
Imputation using propensity−score method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	93.10	93.10	89.66	89.66	93.10	86.21	100.00	96.55	96.55	96.55	58.62
20%	89.66	93.10	96.55	86.21	93.10	86.21	93.10	93.10	100.00	89.66	68.97
30%	89.29	92.86	89.29	85.71	85.71	82.14	85.71	85.71	82.14	92.86	46.43
40%	96.55	89.66	96.55	96.55	89.66	96.55	93.10	100.00	93.10	93.10	65.52

SampleSize_n250_10Treatments/table2_nonshrink_10_treatments_MAR_n250

MR	T1vs.T0_many	T2vs.T0_many	T3vs.T0_many	T4vs.T0_many	T5vs.T0_many	T6vs.T0_many	T7vs.T0_many	T8vs.T0_many	T9vs.T0_many	T10vs.T0_many	Joint_many
Scenario I: all means are 0 for T0,T1,...,T10	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
0%	100	100	100.00	100	96.43	100	100	100.00	100.00	100	96.43
Imputation using linear regression	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	100	100	100.00	100	100.00	100	100	100.00	100.00	100	100.00
20%	100	100	100.00	100	100.00	100	100	100.00	96.55	100	96.55
30%	100	100	100.00	100	100.00	100	100	100.00	100.00	100	100.00
40%	100	100	100.00	100	100.00	100	100	100.00	96.43	100	96.43
Imputation using predictive mean matching method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	100	100	100.00	100	100.00	100	100	100.00	100.00	100	100.00
20%	100	100	100.00	100	96.55	100	100	100.00	96.55	100	93.10
30%	100	100	96.43	100	96.43	100	100	100.00	92.86	100	85.71
40%	100	100	100.00	100	100.00	100	100	100.00	96.43	100	96.43
Imputation using propensity−score method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	100	100	100.00	100	100.00	100	100	100.00	100.00	100	100.00
20%	100	100	100.00	100	100.00	100	100	100.00	100.00	100	100.00
30%	100	100	100.00	100	100.00	100	100	100.00	96.43	100	96.43
40%	100	100	100.00	100	100.00	100	100	100.00	100.00	100	100.00
Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu10=6.0	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
0%	100	100	100.00	100	100.00	100	100	96.43	100.00	100	96.43
Imputation using linear regression	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA

SampleSize_n250_10Treatments/table2_nonshrink_10_treatments_MAR_n250

MR	T1vs.T0_many	T2vs.T0_many	T3vs.T0_many	T4vs.T0_many	T5vs.T0_many	T6vs.T0_many	T7vs.T0_many	T8vs.T0_many	T9vs.T0_many	T10vs.T0_many	Joint_many
10%	96.55	96.55	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	96.55
20%	96.55	100.00	100.00	96.55	100.00	100.00	100.00	100.00	100.00	100.00	93.10
30%	100.00	100.00	100.00	96.43	96.43	96.43	96.43	100.00	100.00	96.43	82.14
40%	100.00	100.00	96.55	100.00	96.55	100.00	100.00	100.00	96.55	96.55	96.55
Imputation using predictive mean matching method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	96.55	96.55	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	96.55
20%	100.00	100.00	100.00	96.55	100.00	100.00	100.00	100.00	100.00	100.00	96.55
30%	100.00	100.00	100.00	100.00	96.43	96.43	100.00	92.86	92.86	100.00	85.71
40%	100.00	100.00	96.55	100.00	96.55	100.00	100.00	100.00	96.55	96.55	96.55
Imputation using propensity−score method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	96.55	96.55	100.00	96.55	100.00	100.00	100.00	100.00	100.00	100.00	93.10
20%	100.00	100.00	100.00	96.55	100.00	100.00	100.00	100.00	100.00	100.00	96.55
30%	100.00	100.00	100.00	100.00	96.43	96.43	96.43	89.29	96.43	96.43	82.14
40%	100.00	100.00	96.55	100.00	96.55	100.00	100.00	100.00	96.55	93.10	93.10

SampleSize_n250_10Treatments/table2_shrink_10_treatments_MAR_n250

MR	T1vs.T0_none	T2vs.T0_none	T3vs.T0_none	T4vs.T0_none	T5vs.T0_none	T6vs.T0_none	T7vs.T0_none	T8vs.T0_none	T9vs.T0_none	T10vs.T0_none	Joint_none
Scenario I: all means are 0 for T0,T1,...,T10	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
0%	89.29	100.00	100.00	100.00	96.43	96.43	100.00	89.29	96.43	96.43	67.86
Imputation using linear regression	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	93.10	96.55	96.55	100.00	96.55	96.55	93.10	100.00	93.10	93.10	79.31
20%	93.10	96.55	96.55	100.00	96.55	96.55	100.00	96.55	93.10	93.10	68.97
30%	100.00	89.29	89.29	92.86	96.43	96.43	100.00	89.29	89.29	92.86	67.86
40%	96.43	100.00	89.29	100.00	100.00	92.86	96.43	100.00	85.71	96.43	75.00
Imputation using predictive mean matching method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	93.10	96.55	93.10	100.00	96.55	96.55	93.10	100.00	93.10	93.10	79.31
20%	89.66	89.66	96.55	100.00	96.55	96.55	96.55	96.55	93.10	89.66	58.62
30%	96.43	89.29	89.29	89.29	92.86	89.29	96.43	89.29	89.29	92.86	64.29
40%	96.43	96.43	96.43	100.00	100.00	89.29	96.43	100.00	85.71	96.43	78.57
Imputation using propensity−score method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	93.10	96.55	93.10	100.00	96.55	100.00	96.55	100.00	96.55	93.10	82.76
20%	89.66	89.66	93.10	100.00	93.10	96.55	100.00	96.55	93.10	89.66	62.07
30%	100.00	85.71	89.29	89.29	96.43	96.43	100.00	96.43	92.86	96.43	67.86
40%	96.43	96.43	92.86	100.00	100.00	89.29	100.00	100.00	85.71	96.43	75.00
Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu10=6.0	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
0%	96.43	92.86	92.86	96.43	100.00	96.43	96.43	96.43	96.43	100.00	71.43
Imputation using linear regression	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA

SampleSize_n250_10Treatments/table2_shrink_10_treatments_MAR_n250

MR	T1vs.T0_none	T2vs.T0_none	T3vs.T0_none	T4vs.T0_none	T5vs.T0_none	T6vs.T0_none	T7vs.T0_none	T8vs.T0_none	T9vs.T0_none	T10vs.T0_none	Joint_none
10%	96.55	93.10	89.66	93.10	96.55	86.21	96.55	100.00	96.55	96.55	65.52
20%	96.55	86.21	100.00	89.66	86.21	86.21	93.10	89.66	100.00	86.21	65.52
30%	92.86	92.86	92.86	85.71	82.14	92.86	82.14	89.29	85.71	92.86	46.43
40%	96.55	89.66	93.10	96.55	96.55	93.10	96.55	100.00	93.10	89.66	65.52
Imputation using predictive mean matching method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	96.55	93.10	93.10	93.10	93.10	89.66	100.00	96.55	100.00	96.55	65.52
20%	96.55	96.55	93.10	86.21	89.66	86.21	96.55	89.66	100.00	89.66	65.52
30%	92.86	89.29	89.29	85.71	82.14	89.29	82.14	85.71	92.86	89.29	42.86
40%	96.55	89.66	89.66	93.10	96.55	93.10	96.55	96.55	93.10	89.66	65.52
Imputation using propensity−score method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	93.10	93.10	89.66	89.66	93.10	86.21	100.00	96.55	96.55	96.55	58.62
20%	89.66	93.10	96.55	86.21	93.10	86.21	93.10	93.10	100.00	89.66	68.97
30%	89.29	92.86	89.29	85.71	85.71	82.14	85.71	85.71	82.14	92.86	46.43
40%	96.55	89.66	96.55	96.55	89.66	96.55	93.10	100.00	93.10	93.10	65.52

SampleSize_n250_10Treatments/table2_shrink_10_treatments_MAR_n250

MR	T1vs.T0_many	T2vs.T0_many	T3vs.T0_many	T4vs.T0_many	T5vs.T0_many	T6vs.T0_many	T7vs.T0_many	T8vs.T0_many	T9vs.T0_many	T10vs.T0_many	Joint_many
Scenario I: all means are 0 for T0,T1,...,T10	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
0%	100	100	100.00	100	96.43	100	100	100.00	100.00	100	96.43
Imputation using linear regression	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	100	100	100.00	100	100.00	100	100	100.00	100.00	100	100.00
20%	100	100	100.00	100	100.00	100	100	100.00	96.55	100	96.55
30%	100	100	100.00	100	100.00	100	100	100.00	100.00	100	100.00
40%	100	100	100.00	100	100.00	100	100	100.00	96.43	100	96.43
Imputation using predictive mean matching method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	100	100	100.00	100	100.00	100	100	100.00	100.00	100	100.00
20%	100	100	100.00	100	100.00	100	100	96.55	96.55	100	93.10
30%	100	100	96.43	100	96.43	100	100	100.00	92.86	100	85.71
40%	100	100	100.00	100	100.00	100	100	100.00	96.43	100	96.43
Imputation using propensity−score method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	100	100	100.00	100	100.00	100	100	100.00	100.00	100	100.00
20%	100	100	100.00	100	100.00	100	100	100.00	100.00	100	100.00
30%	100	100	100.00	100	100.00	100	100	100.00	96.43	100	96.43
40%	100	100	100.00	100	100.00	100	100	100.00	100.00	100	100.00
Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu10=6.0	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
0%	100	100	100.00	100	100.00	100	100	96.43	100.00	100	96.43
Imputation using linear regression	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA

SampleSize_n250_10Treatments/table2_shrink_10_treatments_MAR_n250

MR	T1vs.T0_many	T2vs.T0_many	T3vs.T0_many	T4vs.T0_many	T5vs.T0_many	T6vs.T0_many	T7vs.T0_many	T8vs.T0_many	T9vs.T0_many	T10vs.T0_many	Joint_many
10%	96.55	96.55	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	96.55
20%	96.55	100.00	100.00	96.55	100.00	100.00	100.00	100.00	100.00	100.00	93.10
30%	100.00	100.00	100.00	96.43	96.43	100.00	96.43	100.00	100.00	96.43	85.71
40%	100.00	100.00	96.55	100.00	96.55	100.00	100.00	100.00	96.55	96.55	96.55
Imputation using predictive mean matching method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	96.55	96.55	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	96.55
20%	100.00	100.00	100.00	96.55	100.00	100.00	100.00	100.00	100.00	100.00	96.55
30%	100.00	100.00	100.00	100.00	96.43	96.43	100.00	92.86	92.86	100.00	85.71
40%	100.00	100.00	96.55	100.00	96.55	100.00	100.00	100.00	96.55	96.55	96.55
Imputation using propensity−score method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	96.55	96.55	100.00	96.55	100.00	100.00	100.00	100.00	100.00	100.00	93.10
20%	100.00	100.00	100.00	96.55	100.00	100.00	100.00	100.00	100.00	100.00	96.55
30%	100.00	100.00	100.00	100.00	96.43	96.43	96.43	92.86	96.43	96.43	82.14
40%	100.00	100.00	96.55	100.00	96.55	100.00	100.00	100.00	96.55	96.55	96.55

SampleSize_n250_10Treatments/table3_nonshrink_10_treatments_MAR_n250

MR	None_T1vs.T0	None_T2vs.T0	None_T3vs.T0	None_T4vs.T0	None_T5vs.T0	None_T6vs.T0	None_T7vs.T0	None_T8vs.T0	None_T9vs.T0	None_T10vs.T0	Many_T1vs.T0
Scenario I: all means are 0 for T0,T1,...,T10											
0%	1.124	1.111	1.126	1.131	1.131	1.135	1.140	1.147	1.162	1.154	1.549
Imputation using linear regression											
10%	1.177 (4.68)	1.179 (6.14)	1.179 (4.72)	1.176 (3.96)	1.199 (6.03)	1.195 (5.29)	1.208 (6.03)	1.209 (5.33)	1.216 (4.65)	1.221 (5.83)	1.622 (4.71)
20%	1.216 (8.19)	1.220 (9.85)	1.244 (10.50)	1.233 (8.95)	1.233 (9.02)	1.252 (10.34)	1.247 (9.42)	1.260 (9.79)	1.265 (8.87)	1.265 (9.61)	1.681 (8.51)
30%	1.328 (18.11)	1.341 (20.71)	1.343 (19.28)	1.351 (19.37)	1.357 (20.02)	1.363 (20.09)	1.366 (19.89)	1.372 (19.54)	1.371 (18.00)	1.389 (20.40)	1.831 (18.17)
40%	1.417 (26.08)	1.414 (27.26)	1.439 (27.79)	1.417 (25.22)	1.465 (29.53)	1.432 (26.17)	1.440 (26.39)	1.473 (28.37)	1.454 (25.16)	1.483 (28.53)	1.961 (26.55)
Imputation using predictive mean matching method											
10%	1.172 (4.24)	1.173 (5.65)	1.174 (4.23)	1.174 (3.74)	1.193 (5.49)	1.194 (5.18)	1.204 (5.62)	1.201 (4.66)	1.211 (4.23)	1.213 (5.09)	1.615 (4.25)
20%	1.199 (6.65)	1.211 (9.07)	1.230 (9.19)	1.226 (8.34)	1.209 (6.90)	1.240 (9.21)	1.236 (8.48)	1.239 (7.94)	1.245 (7.21)	1.250 (8.36)	1.657 (6.94)
30%	1.289 (14.67)	1.293 (16.37)	1.309 (16.24)	1.294 (14.39)	1.295 (14.46)	1.313 (15.66)	1.314 (15.29)	1.333 (16.16)	1.334 (14.84)	1.333 (15.56)	1.781 (14.95)
40%	1.373 (22.10)	1.378 (24.06)	1.406 (24.90)	1.421 (25.57)	1.410 (24.69)	1.397 (23.07)	1.409 (23.61)	1.439 (25.42)	1.429 (22.97)	1.431 (24.00)	1.894 (22.28)
Imputation using propensity−score method											
10%	1.172 (4.21)	1.174 (5.65)	1.178 (4.58)	1.174 (3.77)	1.197 (5.87)	1.196 (5.40)	1.202 (5.47)	1.205 (4.99)	1.212 (4.30)	1.216 (5.37)	1.614 (4.20)
20%	1.212 (7.78)	1.217 (9.52)	1.228 (9.03)	1.240 (9.61)	1.228 (8.60)	1.245 (9.67)	1.252 (9.87)	1.253 (9.22)	1.262 (8.59)	1.261 (9.30)	1.674 (8.07)
30%	1.297 (15.41)	1.313 (18.22)	1.314 (16.67)	1.320 (16.69)	1.330 (17.61)	1.331 (17.26)	1.349 (18.40)	1.359 (18.41)	1.351 (16.28)	1.351 (17.08)	1.792 (15.64)
40%	1.397 (24.30)	1.431 (28.80)	1.436 (27.55)	1.420 (25.47)	1.452 (28.39)	1.430 (26.01)	1.449 (27.16)	1.482 (29.14)	1.461 (25.77)	1.489 (29.02)	1.931 (24.61)
Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu10=6.0											
0%	1.097	1.109	1.116	1.114	1.123	1.131	1.134	1.146	1.149	1.155	1.513
Imputation using linear regression											

SampleSize_n250_10Treatments/table3_nonshrink_10_treatments_MAR_n250

MR	None_T1vs.T0	None_T2vs.T0	None_T3vs.T0	None_T4vs.T0	None_T5vs.T0	None_T6vs.T0	None_T7vs.T0	None_T8vs.T0	None_T9vs.T0	None_T10vs.T0	Many_T1vs.T0
10%	1.165 (6.18)	1.161 (4.66)	1.170 (4.89)	1.174 (5.31)	1.180 (5.16)	1.184 (4.70)	1.197 (5.60)	1.206 (5.25)	1.207 (5.02)	1.209 (4.71)	1.607 (6.21)
20%	1.224 (11.61)	1.237 (11.47)	1.231 (10.34)	1.246 (11.77)	1.247 (11.10)	1.248 (10.42)	1.276 (12.57)	1.260 (9.95)	1.275 (10.95)	1.277 (10.58)	1.691 (11.71)
30%	1.321 (20.41)	1.305 (17.61)	1.326 (18.89)	1.326 (18.96)	1.335 (18.91)	1.330 (17.62)	1.350 (19.09)	1.350 (17.87)	1.362 (18.55)	1.356 (17.44)	1.824 (20.50)
40%	1.431 (30.41)	1.425 (28.46)	1.432 (28.36)	1.454 (30.46)	1.427 (27.09)	1.447 (28.00)	1.449 (27.84)	1.454 (26.94)	1.480 (28.77)	1.486 (28.68)	1.977 (30.61)
Imputation using predictive mean matching method											
10%	1.158 (5.58)	1.153 (3.91)	1.160 (3.99)	1.166 (4.66)	1.170 (4.20)	1.172 (3.71)	1.184 (4.47)	1.195 (4.28)	1.198 (4.26)	1.196 (3.55)	1.599 (5.65)
20%	1.213 (10.58)	1.219 (9.89)	1.225 (9.78)	1.234 (10.71)	1.224 (9.02)	1.243 (9.93)	1.263 (11.39)	1.253 (9.34)	1.253 (9.03)	1.260 (9.08)	1.674 (10.63)
30%	1.285 (17.11)	1.282 (15.55)	1.290 (15.60)	1.298 (16.44)	1.299 (15.73)	1.305 (15.41)	1.315 (15.99)	1.334 (16.41)	1.330 (15.71)	1.318 (14.09)	1.775 (17.26)
40%	1.385 (26.24)	1.356 (22.26)	1.368 (22.63)	1.413 (26.84)	1.386 (23.47)	1.395 (23.36)	1.404 (23.84)	1.396 (21.89)	1.431 (24.51)	1.423 (23.18)	1.914 (26.45)
Imputation using propensity−score method											
10%	1.159 (5.65)	1.157 (4.28)	1.164 (4.34)	1.171 (5.05)	1.175 (4.68)	1.175 (3.90)	1.187 (4.69)	1.197 (4.46)	1.199 (4.37)	1.204 (4.20)	1.599 (5.67)
20%	1.208 (10.12)	1.225 (10.41)	1.229 (10.18)	1.231 (10.50)	1.228 (9.42)	1.246 (10.22)	1.259 (11.07)	1.245 (8.69)	1.259 (9.55)	1.266 (9.64)	1.668 (10.20)
30%	1.301 (18.59)	1.305 (17.62)	1.319 (18.19)	1.315 (17.97)	1.319 (17.54)	1.301 (15.10)	1.329 (17.24)	1.336 (16.61)	1.353 (17.71)	1.332 (15.37)	1.799 (18.85)
40%	1.409 (28.43)	1.403 (26.50)	1.399 (25.37)	1.432 (28.54)	1.427 (27.09)	1.449 (28.14)	1.457 (28.48)	1.438 (25.53)	1.485 (29.28)	1.476 (27.81)	1.946 (28.58)

MR	Many_T2vs.T0	Many_T3vs.T0	Many_T4vs.T0	Many_T5vs.T0	Many_T6vs.T0	Many_T7vs.T0	Many_T8vs.T0	Many_T9vs.T0	Many_T10vs.T0
Scenario I: all means are 0 for T0,T1,...,T10									
0%	1.531	1.552	1.559	1.559	1.564	1.571	1.581	1.601	1.590
Imputation using linear regression									
10%	1.625 (6.18)	1.626 (4.76)	1.621 (4.00)	1.653 (6.07)	1.648 (5.33)	1.666 (6.06)	1.666 (5.36)	1.676 (4.69)	1.683 (5.87)
20%	1.687 (10.19)	1.719 (10.80)	1.704 (9.28)	1.704 (9.34)	1.731 (10.67)	1.724 (9.75)	1.741 (10.11)	1.748 (9.19)	1.748 (9.94)
30%	1.848 (20.75)	1.852 (19.33)	1.862 (19.41)	1.871 (20.06)	1.879 (20.13)	1.884 (19.96)	1.892 (19.62)	1.890 (18.08)	1.915 (20.45)
40%	1.956 (27.78)	1.991 (28.31)	1.961 (25.75)	2.026 (29.97)	1.981 (26.64)	1.993 (26.89)	2.038 (28.88)	2.012 (25.67)	2.051 (29.00)
Imputation using predictive mean matching method									
10%	1.617 (5.67)	1.618 (4.25)	1.618 (3.76)	1.645 (5.52)	1.646 (5.20)	1.659 (5.64)	1.655 (4.69)	1.669 (4.25)	1.671 (5.11)
20%	1.674 (9.36)	1.699 (9.46)	1.694 (8.63)	1.671 (7.19)	1.713 (9.49)	1.708 (8.76)	1.712 (8.24)	1.721 (7.50)	1.728 (8.66)
30%	1.786 (16.67)	1.808 (16.52)	1.788 (14.68)	1.789 (14.78)	1.814 (15.95)	1.816 (15.61)	1.842 (16.46)	1.843 (15.15)	1.842 (15.86)
40%	1.902 (24.27)	1.941 (25.08)	1.960 (25.70)	1.946 (24.86)	1.927 (23.23)	1.944 (23.77)	1.986 (25.62)	1.971 (23.12)	1.975 (24.20)
Imputation using propensity−score method									
10%	1.617 (5.66)	1.623 (4.57)	1.618 (3.77)	1.650 (5.87)	1.649 (5.40)	1.656 (5.46)	1.660 (4.99)	1.670 (4.30)	1.675 (5.36)
20%	1.681 (9.83)	1.696 (9.32)	1.713 (9.89)	1.697 (8.89)	1.720 (9.96)	1.730 (10.15)	1.732 (9.51)	1.743 (8.88)	1.743 (9.61)
30%	1.813 (18.44)	1.814 (16.91)	1.823 (16.90)	1.836 (17.82)	1.837 (17.47)	1.863 (18.62)	1.876 (18.63)	1.865 (16.51)	1.866 (17.32)
40%	1.976 (29.09)	1.984 (27.86)	1.961 (25.78)	2.005 (28.65)	1.975 (26.30)	2.001 (27.42)	2.046 (29.39)	2.018 (26.07)	2.056 (29.29)
Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu10=6.0									
0%	1.530	1.539	1.537	1.549	1.560	1.564	1.580	1.585	1.593
Imputation using linear regression									

SampleSize_n250_10Treatments/table3_nonshrink_10_treatments_MAR_n250

MR	Many_T2vs.T0	Many_T3vs.T0	Many_T4vs.T0	Many_T5vs.T0	Many_T6vs.T0	Many_T7vs.T0	Many_T8vs.T0	Many_T9vs.T0	Many_T10vs.T0
10%	1.602 (4.70)	1.615 (4.93)	1.619 (5.35)	1.629 (5.19)	1.633 (4.74)	1.652 (5.64)	1.664 (5.28)	1.665 (5.06)	1.669 (4.74)
20%	1.707 (11.57)	1.700 (10.45)	1.720 (11.87)	1.722 (11.19)	1.724 (10.53)	1.762 (12.66)	1.739 (10.05)	1.760 (11.05)	1.763 (10.67)
30%	1.802 (17.72)	1.832 (19.00)	1.830 (19.06)	1.843 (19.00)	1.836 (17.72)	1.864 (19.18)	1.864 (17.96)	1.880 (18.63)	1.873 (17.54)
40%	1.969 (28.65)	1.979 (28.58)	2.008 (30.64)	1.972 (27.34)	2.000 (28.23)	2.004 (28.10)	2.010 (27.18)	2.044 (28.98)	2.053 (28.86)
Imputation using predictive mean matching method									
10%	1.591 (3.98)	1.602 (4.06)	1.610 (4.73)	1.615 (4.26)	1.618 (3.77)	1.635 (4.53)	1.649 (4.35)	1.654 (4.32)	1.651 (3.62)
20%	1.683 (9.95)	1.690 (9.83)	1.703 (10.76)	1.689 (9.08)	1.715 (9.99)	1.743 (11.43)	1.729 (9.39)	1.729 (9.10)	1.739 (9.15)
30%	1.771 (15.70)	1.782 (15.76)	1.792 (16.58)	1.795 (15.89)	1.802 (15.56)	1.816 (16.15)	1.841 (16.51)	1.836 (15.86)	1.820 (14.23)
40%	1.874 (22.48)	1.891 (22.88)	1.953 (27.02)	1.915 (23.69)	1.928 (23.61)	1.941 (24.08)	1.931 (22.16)	1.977 (24.73)	1.966 (23.39)
Imputation using propensity−score method									
10%	1.596 (4.30)	1.606 (4.37)	1.615 (5.07)	1.621 (4.70)	1.621 (3.93)	1.638 (4.72)	1.651 (4.49)	1.655 (4.40)	1.661 (4.22)
20%	1.691 (10.48)	1.697 (10.24)	1.700 (10.57)	1.696 (9.50)	1.720 (10.28)	1.738 (11.13)	1.719 (8.76)	1.738 (9.62)	1.748 (9.71)
30%	1.804 (17.86)	1.823 (18.45)	1.817 (18.22)	1.824 (17.80)	1.800 (15.40)	1.838 (17.50)	1.847 (16.85)	1.870 (17.95)	1.842 (15.62)
40%	1.938 (26.65)	1.933 (25.58)	1.978 (28.66)	1.970 (27.23)	2.001 (28.29)	2.012 (28.62)	1.987 (25.71)	2.051 (29.37)	2.038 (27.91)

SampleSize_n250_10Treatments/table3_shrink_10_treatments_MAR_n250

MR	None_T1vs.T0	None_T2vs.T0	None_T3vs.T0	None_T4vs.T0	None_T5vs.T0	None_T6vs.T0	None_T7vs.T0	None_T8vs.T0	None_T9vs.T0	None_T10vs.T0	Many_T1vs.T0
Scenario I: all means are 0 for T0,T1,...,T10											
0%	1.124	1.111	1.126	1.131	1.131	1.135	1.140	1.147	1.162	1.154	1.543
Imputation using linear regression											
10%	1.177 (4.68)	1.179 (6.14)	1.179 (4.72)	1.176 (3.96)	1.199 (6.03)	1.195 (5.29)	1.208 (6.03)	1.209 (5.33)	1.216 (4.65)	1.221 (5.83)	1.625 (5.29)
20%	1.216 (8.19)	1.220 (9.85)	1.244 (10.50)	1.233 (8.95)	1.233 (9.02)	1.252 (10.34)	1.247 (9.42)	1.260 (9.79)	1.265 (8.87)	1.265 (9.61)	1.689 (9.45)
30%	1.328 (18.11)	1.341 (20.71)	1.343 (19.28)	1.351 (19.37)	1.357 (20.02)	1.363 (20.09)	1.366 (19.89)	1.372 (19.54)	1.371 (18.00)	1.389 (20.40)	1.830 (18.61)
40%	1.417 (26.08)	1.414 (27.26)	1.439 (27.79)	1.417 (25.22)	1.465 (29.53)	1.432 (26.17)	1.440 (26.39)	1.473 (28.37)	1.454 (25.16)	1.483 (28.53)	1.967 (27.51)
Imputation using predictive mean matching method											
10%	1.172 (4.24)	1.173 (5.65)	1.174 (4.23)	1.174 (3.74)	1.193 (5.49)	1.194 (5.18)	1.204 (5.62)	1.201 (4.66)	1.211 (4.23)	1.213 (5.09)	1.616 (4.77)
20%	1.199 (6.65)	1.211 (9.07)	1.230 (9.19)	1.226 (8.34)	1.209 (6.90)	1.240 (9.21)	1.236 (8.48)	1.239 (7.94)	1.245 (7.21)	1.250 (8.36)	1.663 (7.81)
30%	1.289 (14.67)	1.293 (16.37)	1.309 (16.24)	1.294 (14.39)	1.295 (14.46)	1.313 (15.66)	1.314 (15.29)	1.333 (16.16)	1.334 (14.84)	1.333 (15.56)	1.780 (15.39)
40%	1.373 (22.10)	1.378 (24.06)	1.406 (24.90)	1.421 (25.57)	1.410 (24.69)	1.397 (23.07)	1.409 (23.61)	1.439 (25.42)	1.429 (22.97)	1.431 (24.00)	1.901 (23.18)
Imputation using propensity−score method											
10%	1.172 (4.21)	1.174 (5.65)	1.178 (4.58)	1.174 (3.77)	1.197 (5.87)	1.196 (5.40)	1.202 (5.47)	1.205 (4.99)	1.212 (4.30)	1.216 (5.37)	1.616 (4.73)
20%	1.212 (7.78)	1.217 (9.52)	1.228 (9.03)	1.240 (9.61)	1.228 (8.60)	1.245 (9.67)	1.252 (9.87)	1.253 (9.22)	1.262 (8.59)	1.261 (9.30)	1.680 (8.91)
30%	1.297 (15.41)	1.313 (18.22)	1.314 (16.67)	1.320 (16.69)	1.330 (17.61)	1.331 (17.26)	1.349 (18.40)	1.359 (18.41)	1.351 (16.28)	1.351 (17.08)	1.791 (16.07)
40%	1.397 (24.30)	1.431 (28.80)	1.436 (27.55)	1.420 (25.47)	1.452 (28.39)	1.430 (26.01)	1.449 (27.16)	1.482 (29.14)	1.461 (25.77)	1.489 (29.02)	1.936 (25.49)
Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu10=6.0											
0%	1.097	1.109	1.116	1.114	1.123	1.131	1.134	1.146	1.149	1.155	1.530
Imputation using linear regression											

MR	None_T1vs.T0	None_T2vs.T0	None_T3vs.T0	None_T4vs.T0	None_T5vs.T0	None_T6vs.T0	None_T7vs.T0	None_T8vs.T0	None_T9vs.T0	None_T10vs.T0	Many_T1vs.T0
10%	1.165 (6.18)	1.161 (4.66)	1.170 (4.89)	1.174 (5.31)	1.180 (5.16)	1.184 (4.70)	1.197 (5.60)	1.206 (5.25)	1.207 (5.02)	1.209 (4.71)	1.607 (5.00)
20%	1.224 (11.61)	1.237 (11.47)	1.231 (10.34)	1.246 (11.77)	1.247 (11.10)	1.248 (10.42)	1.276 (12.57)	1.260 (9.95)	1.275 (10.95)	1.277 (10.58)	1.696 (10.84)
30%	1.321 (20.41)	1.305 (17.61)	1.326 (18.89)	1.326 (18.96)	1.335 (18.91)	1.330 (17.62)	1.350 (19.09)	1.350 (17.87)	1.362 (18.55)	1.356 (17.44)	1.821 (19.02)
40%	1.431 (30.41)	1.425 (28.46)	1.432 (28.36)	1.454 (30.46)	1.427 (27.09)	1.447 (28.00)	1.449 (27.84)	1.454 (26.94)	1.480 (28.77)	1.486 (28.68)	1.978 (29.27)
Imputation using predictive mean matching method											
10%	1.158 (5.58)	1.153 (3.91)	1.160 (3.99)	1.166 (4.66)	1.170 (4.20)	1.172 (3.71)	1.184 (4.47)	1.195 (4.28)	1.198 (4.26)	1.196 (3.55)	1.598 (4.43)
20%	1.213 (10.58)	1.219 (9.89)	1.225 (9.78)	1.234 (10.71)	1.224 (9.02)	1.243 (9.93)	1.263 (11.39)	1.253 (9.34)	1.253 (9.03)	1.260 (9.08)	1.679 (9.74)
30%	1.285 (17.11)	1.282 (15.55)	1.290 (15.60)	1.298 (16.44)	1.299 (15.73)	1.305 (15.41)	1.315 (15.99)	1.334 (16.41)	1.330 (15.71)	1.318 (14.09)	1.773 (15.88)
40%	1.385 (26.24)	1.356 (22.26)	1.368 (22.63)	1.413 (26.84)	1.386 (23.47)	1.395 (23.36)	1.404 (23.84)	1.396 (21.89)	1.431 (24.51)	1.423 (23.18)	1.915 (25.11)
Imputation using propensity−score method											
10%	1.159 (5.65)	1.157 (4.28)	1.164 (4.34)	1.171 (5.05)	1.175 (4.68)	1.175 (3.90)	1.187 (4.69)	1.197 (4.46)	1.199 (4.37)	1.204 (4.20)	1.599 (4.46)
20%	1.208 (10.12)	1.225 (10.41)	1.229 (10.18)	1.231 (10.50)	1.228 (9.42)	1.246 (10.22)	1.259 (11.07)	1.245 (8.69)	1.259 (9.55)	1.266 (9.64)	1.673 (9.32)
30%	1.301 (18.59)	1.305 (17.62)	1.319 (18.19)	1.315 (17.97)	1.319 (17.54)	1.301 (15.10)	1.329 (17.24)	1.336 (16.61)	1.353 (17.71)	1.332 (15.37)	1.797 (17.44)
40%	1.409 (28.43)	1.403 (26.50)	1.399 (25.37)	1.432 (28.54)	1.427 (27.09)	1.449 (28.14)	1.457 (28.48)	1.438 (25.53)	1.485 (29.28)	1.476 (27.81)	1.949 (27.33)

MR	Many_T2vs.T0	Many_T3vs.T0	Many_T4vs.T0	Many_T5vs.T0	Many_T6vs.T0	Many_T7vs.T0	Many_T8vs.T0	Many_T9vs.T0	Many_T10vs.T0
Scenario I: all means are 0 for T0,T1,...,T10									
0%	1.540	1.548	1.547	1.548	1.550	1.547	1.553	1.558	1.557
Imputation using linear regression									
10%	1.621 (5.27)	1.622 (4.82)	1.615 (4.42)	1.637 (5.78)	1.632 (5.28)	1.643 (6.20)	1.642 (5.75)	1.646 (5.65)	1.654 (6.21)
20%	1.687 (9.56)	1.716 (10.88)	1.695 (9.53)	1.702 (9.96)	1.715 (10.67)	1.709 (10.51)	1.724 (10.99)	1.725 (10.69)	1.722 (10.62)
30%	1.846 (19.85)	1.844 (19.11)	1.855 (19.92)	1.863 (20.33)	1.868 (20.55)	1.866 (20.65)	1.869 (20.38)	1.867 (19.82)	1.895 (21.71)
40%	1.958 (27.14)	1.987 (28.39)	1.958 (26.56)	2.022 (30.63)	1.981 (27.80)	1.988 (28.54)	2.024 (30.31)	2.002 (28.46)	2.037 (30.84)
Imputation using predictive mean matching method									
10%	1.613 (4.73)	1.613 (4.25)	1.611 (4.15)	1.628 (5.16)	1.628 (5.04)	1.635 (5.70)	1.631 (5.01)	1.638 (5.12)	1.640 (5.37)
20%	1.674 (8.69)	1.695 (9.51)	1.684 (8.84)	1.670 (7.87)	1.697 (9.53)	1.695 (9.59)	1.692 (8.99)	1.698 (8.95)	1.701 (9.26)
30%	1.783 (15.75)	1.801 (16.39)	1.782 (15.19)	1.780 (15.01)	1.802 (16.24)	1.798 (16.26)	1.819 (17.16)	1.820 (16.79)	1.823 (17.09)
40%	1.906 (23.75)	1.936 (25.08)	1.956 (26.43)	1.941 (25.41)	1.926 (24.26)	1.939 (25.33)	1.969 (26.81)	1.960 (25.77)	1.960 (25.87)
Imputation using propensity−score method									
10%	1.613 (4.70)	1.619 (4.63)	1.611 (4.13)	1.634 (5.55)	1.632 (5.27)	1.634 (5.60)	1.637 (5.40)	1.639 (5.18)	1.644 (5.63)
20%	1.680 (9.10)	1.693 (9.36)	1.703 (10.09)	1.696 (9.55)	1.704 (9.96)	1.715 (10.87)	1.714 (10.36)	1.720 (10.39)	1.716 (10.24)
30%	1.810 (17.55)	1.808 (16.80)	1.819 (17.56)	1.829 (18.14)	1.828 (17.95)	1.847 (19.38)	1.855 (19.44)	1.843 (18.27)	1.846 (18.59)
40%	1.977 (28.37)	1.979 (27.86)	1.959 (26.59)	2.000 (29.21)	1.974 (27.36)	1.997 (29.11)	2.032 (30.82)	2.007 (28.77)	2.041 (31.08)
Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu10=6.0									
0%	1.536	1.539	1.539	1.540	1.545	1.547	1.551	1.553	1.555
Imputation using linear regression									

MR	Many_T2vs.T0	Many_T3vs.T0	Many_T4vs.T0	Many_T5vs.T0	Many_T6vs.T0	Many_T7vs.T0	Many_T8vs.T0	Many_T9vs.T0	Many_T10vs.T0
10%	1.604 (4.43)	1.609 (4.57)	1.613 (4.81)	1.618 (5.06)	1.618 (4.74)	1.631 (5.41)	1.634 (5.37)	1.635 (5.23)	1.638 (5.33)
20%	1.706 (11.08)	1.698 (10.30)	1.712 (11.25)	1.715 (11.35)	1.710 (10.67)	1.744 (12.72)	1.723 (11.09)	1.736 (11.77)	1.740 (11.90)
30%	1.802 (17.31)	1.822 (18.40)	1.823 (18.47)	1.832 (18.96)	1.826 (18.19)	1.848 (19.43)	1.847 (19.13)	1.858 (19.62)	1.856 (19.36)
40%	1.971 (28.32)	1.976 (28.38)	2.004 (30.21)	1.967 (27.70)	1.991 (28.90)	1.989 (28.56)	1.998 (28.83)	2.028 (30.55)	2.038 (31.09)
Imputation using predictive mean matching method									
10%	1.593 (3.75)	1.597 (3.78)	1.604 (4.20)	1.604 (4.17)	1.603 (3.73)	1.614 (4.32)	1.619 (4.38)	1.622 (4.44)	1.619 (4.14)
20%	1.683 (9.59)	1.689 (9.76)	1.696 (10.24)	1.684 (9.35)	1.702 (10.14)	1.726 (11.53)	1.712 (10.39)	1.706 (9.83)	1.714 (10.24)
30%	1.771 (15.32)	1.775 (15.33)	1.786 (16.09)	1.783 (15.77)	1.793 (16.05)	1.800 (16.35)	1.824 (17.61)	1.814 (16.76)	1.805 (16.06)
40%	1.879 (22.32)	1.889 (22.77)	1.948 (26.59)	1.912 (24.14)	1.920 (24.25)	1.928 (24.60)	1.920 (23.79)	1.959 (26.12)	1.952 (25.57)
Imputation using propensity−score method									
10%	1.598 (4.07)	1.602 (4.10)	1.609 (4.56)	1.610 (4.56)	1.606 (3.95)	1.618 (4.58)	1.622 (4.62)	1.624 (4.57)	1.630 (4.80)
20%	1.689 (10.00)	1.694 (10.10)	1.692 (9.97)	1.689 (9.70)	1.706 (10.45)	1.720 (11.18)	1.702 (9.76)	1.714 (10.36)	1.724 (10.86)
30%	1.803 (17.40)	1.815 (17.95)	1.811 (17.69)	1.814 (17.78)	1.790 (15.87)	1.822 (17.75)	1.830 (17.99)	1.848 (18.99)	1.827 (17.48)
40%	1.941 (26.39)	1.931 (25.48)	1.974 (28.31)	1.966 (27.66)	1.993 (28.99)	2.000 (29.25)	1.975 (27.38)	2.035 (31.01)	2.025 (30.26)

SampleSize_n300_10Treatments/table2_nonshrink_10_treatments_MAR_n300

MR	T1vs.T0_none	T2vs.T0_none	T3vs.T0_none	T4vs.T0_none	T5vs.T0_none	T6vs.T0_none	T7vs.T0_none	T8vs.T0_none	T9vs.T0_none	T10vs.T0_none	Joint_none
Scenario I: all means are 0 for T0,T1,...,T10	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
0%	85.71	89.29	89.29	92.86	96.43	89.29	96.43	89.29	100.00	96.43	64.29
Imputation using linear regression	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	96.55	96.55	96.55	96.55	89.66	100.00	100.00	96.55	93.10	93.10	72.41
20%	96.55	100.00	93.10	93.10	96.55	96.55	96.55	93.10	93.10	96.55	82.76
30%	92.86	100.00	92.86	96.43	89.29	92.86	96.43	89.29	92.86	96.43	64.29
40%	89.66	96.55	100.00	93.10	96.55	96.55	96.55	96.55	89.66	100.00	72.41
Imputation using predictive mean matching method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	96.55	96.55	96.55	100.00	86.21	100.00	100.00	96.55	89.66	89.66	65.52
20%	100.00	96.55	96.55	96.55	96.55	93.10	96.55	96.55	93.10	100.00	86.21
30%	96.43	100.00	92.86	96.43	89.29	96.43	96.43	92.86	89.29	96.43	71.43
40%	89.66	93.10	96.55	93.10	96.55	89.66	93.10	96.55	96.55	96.55	65.52
Imputation using propensity−score method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	96.55	100.00	96.55	100.00	89.66	100.00	100.00	96.55	93.10	93.10	68.97
20%	96.55	100.00	93.10	89.66	96.55	96.55	100.00	96.55	93.10	100.00	79.31
30%	92.86	100.00	92.86	92.86	96.43	100.00	100.00	85.71	92.86	96.43	67.86
40%	89.66	96.55	93.10	93.10	96.55	93.10	96.55	96.55	89.66	96.55	65.52
Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu10=6.0	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
0%	100.00	100.00	93.10	93.10	93.10	96.55	100.00	93.10	93.10	100.00	75.86
Imputation using linear regression	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA

SampleSize_n300_10Treatments/table2_nonshrink_10_treatments_MAR_n300

MR	T1vs.T0_none	T2vs.T0_none	T3vs.T0_none	T4vs.T0_none	T5vs.T0_none	T6vs.T0_none	T7vs.T0_none	T8vs.T0_none	T9vs.T0_none	T10vs.T0_none	Joint_none
10%	96.55	100.00	100.00	96.55	96.55	100.00	96.55	100.00	100.00	100.00	86.21
20%	92.86	100.00	78.57	96.43	85.71	89.29	85.71	89.29	96.43	92.86	64.29
30%	100.00	96.43	82.14	96.43	85.71	92.86	96.43	96.43	96.43	92.86	75.00
40%	89.66	93.10	96.55	100.00	93.10	100.00	93.10	93.10	100.00	100.00	62.07
Imputation using predictive mean matching method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	96.55	100.00	100.00	96.55	96.55	100.00	96.55	100.00	100.00	96.55	82.76
20%	92.86	100.00	78.57	96.43	89.29	92.86	85.71	89.29	96.43	89.29	64.29
30%	96.43	96.43	85.71	96.43	85.71	89.29	92.86	92.86	96.43	92.86	71.43
40%	96.55	82.76	96.55	100.00	93.10	96.55	93.10	86.21	100.00	100.00	55.17
Imputation using propensity−score method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	100.00	100.00	100.00	96.55	96.55	100.00	96.55	100.00	100.00	100.00	89.66
20%	92.86	100.00	85.71	96.43	89.29	89.29	82.14	89.29	89.29	85.71	67.86
30%	96.43	96.43	85.71	96.43	89.29	92.86	100.00	100.00	96.43	92.86	75.00
40%	93.10	89.66	96.55	100.00	93.10	100.00	93.10	96.55	100.00	100.00	68.97

SampleSize_n300_10Treatments/table2_nonshrink_10_treatments_MAR_n300

MR	T1vs.T0_many	T2vs.T0_many	T3vs.T0_many	T4vs.T0_many	T5vs.T0_many	T6vs.T0_many	T7vs.T0_many	T8vs.T0_many	T9vs.T0_many	T10vs.T0_many	Joint_many
Scenario I: all means are 0 for T0,T1,...,T10	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
0%	96.43	100	100	100.00	100.00	96.43	100	96.43	100.00	100	89.29
Imputation using linear regression	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	100.00	100	100	100.00	100.00	100.00	100	100.00	100.00	100	100.00
20%	100.00	100	100	100.00	100.00	96.55	100	100.00	100.00	100	96.55
30%	96.43	100	100	100.00	100.00	100.00	100	96.43	100.00	100	92.86
40%	100.00	100	100	100.00	100.00	100.00	100	100.00	100.00	100	100.00
Imputation using predictive mean matching method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	100.00	100	100	100.00	100.00	100.00	100	100.00	96.55	100	96.55
20%	100.00	100	100	100.00	100.00	96.55	100	96.55	100.00	100	93.10
30%	96.43	100	100	100.00	100.00	100.00	100	96.43	100.00	100	92.86
40%	100.00	100	100	100.00	100.00	100.00	100	100.00	100.00	100	100.00
Imputation using propensity−score method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	100.00	100	100	100.00	100.00	100.00	100	100.00	100.00	100	100.00
20%	100.00	100	100	100.00	100.00	100.00	100	100.00	100.00	100	100.00
30%	96.43	100	100	100.00	100.00	100.00	100	96.43	96.43	100	89.29
40%	100.00	100	100	100.00	100.00	100.00	100	100.00	100.00	100	100.00
Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu10=6.0	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
0%	100.00	100	100	96.55	96.55	100.00	100	100.00	100.00	100	93.10
Imputation using linear regression	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA

SampleSize_n300_10Treatments/table2_nonshrink_10_treatments_MAR_n300

MR	T1vs.T0_many	T2vs.T0_many	T3vs.T0_many	T4vs.T0_many	T5vs.T0_many	T6vs.T0_many	T7vs.T0_many	T8vs.T0_many	T9vs.T0_many	T10vs.T0_many	Joint_many
10%	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100	100.00	100.00
20%	100.00	100.00	96.43	100.00	100.00	96.43	100.00	100.00	100	100.00	96.43
30%	100.00	96.43	96.43	96.43	100.00	96.43	100.00	100.00	100	96.43	96.43
40%	100.00	100.00	100.00	100.00	96.55	100.00	100.00	100.00	100	100.00	96.55
Imputation using predictive mean matching method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100	100.00	100.00
20%	96.43	100.00	96.43	100.00	96.43	96.43	96.43	96.43	100	100.00	92.86
30%	100.00	96.43	92.86	100.00	96.43	96.43	100.00	100.00	100	96.43	89.29
40%	100.00	100.00	96.55	100.00	93.10	100.00	100.00	100.00	100	100.00	89.66
Imputation using propensity−score method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100	100.00	100.00
20%	100.00	100.00	96.43	100.00	96.43	96.43	100.00	96.43	100	100.00	92.86
30%	100.00	96.43	96.43	96.43	100.00	96.43	100.00	100.00	100	96.43	96.43
40%	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100	100.00	100.00

SampleSize_n300_10Treatments/table2_shrink_10_treatments_MAR_n300

MR	T1vs.T0_none	T2vs.T0_none	T3vs.T0_none	T4vs.T0_none	T5vs.T0_none	T6vs.T0_none	T7vs.T0_none	T8vs.T0_none	T9vs.T0_none	T10vs.T0_none	Joint_none
Scenario I: all means are 0 for T0,T1,...,T10	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
0%	85.71	89.29	89.29	92.86	96.43	89.29	96.43	89.29	100.00	96.43	64.29
Imputation using linear regression	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	96.55	96.55	96.55	96.55	89.66	100.00	100.00	96.55	93.10	93.10	72.41
20%	96.55	100.00	93.10	93.10	96.55	96.55	96.55	93.10	93.10	96.55	82.76
30%	92.86	100.00	92.86	96.43	89.29	92.86	96.43	89.29	92.86	96.43	64.29
40%	89.66	96.55	100.00	93.10	96.55	96.55	96.55	96.55	89.66	100.00	72.41
Imputation using predictive mean matching method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	96.55	96.55	96.55	100.00	86.21	100.00	100.00	96.55	89.66	89.66	65.52
20%	100.00	96.55	96.55	96.55	96.55	93.10	96.55	96.55	93.10	100.00	86.21
30%	96.43	100.00	92.86	96.43	89.29	96.43	96.43	92.86	89.29	96.43	71.43
40%	89.66	93.10	96.55	93.10	96.55	89.66	93.10	96.55	96.55	96.55	65.52
Imputation using propensity−score method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	96.55	100.00	96.55	100.00	89.66	100.00	100.00	96.55	93.10	93.10	68.97
20%	96.55	100.00	93.10	89.66	96.55	96.55	100.00	96.55	93.10	100.00	79.31
30%	92.86	100.00	92.86	92.86	96.43	100.00	100.00	85.71	92.86	96.43	67.86
40%	89.66	96.55	93.10	93.10	96.55	93.10	96.55	96.55	89.66	96.55	65.52
Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu10=6.0	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
0%	100.00	100.00	93.10	93.10	93.10	96.55	100.00	93.10	93.10	100.00	75.86
Imputation using linear regression	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA

SampleSize_n300_10Treatments/table2_shrink_10_treatments_MAR_n300

MR	T1vs.T0_none	T2vs.T0_none	T3vs.T0_none	T4vs.T0_none	T5vs.T0_none	T6vs.T0_none	T7vs.T0_none	T8vs.T0_none	T9vs.T0_none	T10vs.T0_none	Joint_none
10%	96.55	100.00	100.00	96.55	96.55	100.00	96.55	100.00	100.00	100.00	86.21
20%	92.86	100.00	78.57	96.43	85.71	89.29	85.71	89.29	96.43	92.86	64.29
30%	100.00	96.43	82.14	96.43	85.71	92.86	96.43	96.43	96.43	92.86	75.00
40%	89.66	93.10	96.55	100.00	93.10	100.00	93.10	93.10	100.00	100.00	62.07
Imputation using predictive mean matching method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	96.55	100.00	100.00	96.55	96.55	100.00	96.55	100.00	100.00	96.55	82.76
20%	92.86	100.00	78.57	96.43	89.29	92.86	85.71	89.29	96.43	89.29	64.29
30%	96.43	96.43	85.71	96.43	85.71	89.29	92.86	92.86	96.43	92.86	71.43
40%	96.55	82.76	96.55	100.00	93.10	96.55	93.10	86.21	100.00	100.00	55.17
Imputation using propensity−score method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	100.00	100.00	100.00	96.55	96.55	100.00	96.55	100.00	100.00	100.00	89.66
20%	92.86	100.00	85.71	96.43	89.29	89.29	82.14	89.29	89.29	85.71	67.86
30%	96.43	96.43	85.71	96.43	89.29	92.86	100.00	100.00	96.43	92.86	75.00
40%	93.10	89.66	96.55	100.00	93.10	100.00	93.10	96.55	100.00	100.00	68.97

SampleSize_n300_10Treatments/table2_shrink_10_treatments_MAR_n300

MR	T1vs.T0_many	T2vs.T0_many	T3vs.T0_many	T4vs.T0_many	T5vs.T0_many	T6vs.T0_many	T7vs.T0_many	T8vs.T0_many	T9vs.T0_many	T10vs.T0_many	Joint_many
Scenario I: all means are 0 for T0,T1,...,T10	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
0%	96.43	100	100	100.00	100.00	96.43	96.43	96.43	100.00	100	85.71
Imputation using linear regression	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	100.00	100	100	100.00	100.00	100.00	100.00	100.00	100.00	100	100.00
20%	100.00	100	100	100.00	100.00	96.55	100.00	96.55	100.00	100	93.10
30%	96.43	100	100	100.00	100.00	100.00	100.00	96.43	100.00	100	92.86
40%	100.00	100	100	100.00	100.00	100.00	100.00	100.00	100.00	100	100.00
Imputation using predictive mean matching method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	100.00	100	100	100.00	100.00	100.00	100.00	100.00	96.55	100	96.55
20%	100.00	100	100	100.00	100.00	96.55	100.00	96.55	100.00	100	93.10
30%	96.43	100	100	100.00	100.00	100.00	100.00	96.43	100.00	100	92.86
40%	100.00	100	100	100.00	100.00	100.00	100.00	100.00	100.00	100	100.00
Imputation using propensity−score method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	100.00	100	100	100.00	100.00	100.00	100.00	100.00	100.00	100	100.00
20%	100.00	100	100	100.00	100.00	100.00	100.00	100.00	100.00	100	100.00
30%	96.43	100	100	100.00	100.00	100.00	100.00	96.43	96.43	100	89.29
40%	96.55	100	100	100.00	100.00	100.00	100.00	100.00	100.00	100	96.55
Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu10=6.0	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
0%	100.00	100	100	96.55	96.55	100.00	100.00	100.00	100.00	100	93.10
Imputation using linear regression	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA

SampleSize_n300_10Treatments/table2_shrink_10_treatments_MAR_n300

MR	T1vs.T0_many	T2vs.T0_many	T3vs.T0_many	T4vs.T0_many	T5vs.T0_many	T6vs.T0_many	T7vs.T0_many	T8vs.T0_many	T9vs.T0_many	T10vs.T0_many	Joint_many
10%	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100	100.00	100.00
20%	100.00	100.00	96.43	100.00	100.00	96.43	100.00	100.00	100	100.00	96.43
30%	100.00	96.43	96.43	96.43	96.43	96.43	100.00	100.00	100	96.43	92.86
40%	100.00	100.00	100.00	100.00	96.55	100.00	100.00	100.00	100	100.00	96.55
Imputation using predictive mean matching method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100	100.00	100.00
20%	96.43	100.00	96.43	100.00	96.43	96.43	96.43	96.43	100	100.00	92.86
30%	100.00	96.43	92.86	100.00	96.43	96.43	100.00	100.00	100	96.43	89.29
40%	100.00	100.00	96.55	100.00	93.10	100.00	100.00	100.00	100	100.00	89.66
Imputation using propensity−score method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100	100.00	100.00
20%	100.00	100.00	96.43	100.00	96.43	96.43	100.00	96.43	100	100.00	92.86
30%	100.00	96.43	96.43	96.43	100.00	96.43	100.00	100.00	100	96.43	96.43
40%	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100	100.00	100.00

SampleSize_n300_10Treatments/table3_nonshrink_10_treatments_MAR_n300

MR	None_T1vs.T0	None_T2vs.T0	None_T3vs.T0	None_T4vs.T0	None_T5vs.T0	None_T6vs.T0	None_T7vs.T0	None_T8vs.T0	None_T9vs.T0	None_T10vs.T0	Many_T1vs.T0
Scenario I: all means are 0 for T0,T1,...,T10											
0%	1.016	1.022	1.020	1.028	1.033	1.043	1.040	1.043	1.061	1.060	1.400
Imputation using linear regression											
10%	1.061 (4.43)	1.061 (3.80)	1.077 (5.62)	1.077 (4.72)	1.081 (4.60)	1.093 (4.89)	1.098 (5.52)	1.100 (5.45)	1.106 (4.28)	1.109 (4.67)	1.463 (4.52)
20%	1.124 (10.65)	1.124 (9.91)	1.130 (10.84)	1.138 (10.64)	1.145 (10.78)	1.149 (10.23)	1.149 (10.42)	1.161 (11.36)	1.164 (9.70)	1.169 (10.36)	1.551 (10.80)
30%	1.190 (17.10)	1.220 (19.30)	1.214 (19.03)	1.238 (20.36)	1.232 (19.20)	1.241 (19.09)	1.242 (19.42)	1.241 (18.98)	1.247 (17.56)	1.266 (19.44)	1.642 (17.32)
40%	1.300 (27.94)	1.304 (27.53)	1.308 (28.27)	1.329 (29.23)	1.321 (27.86)	1.324 (27.03)	1.350 (29.76)	1.360 (30.38)	1.341 (26.41)	1.365 (28.78)	1.795 (28.23)
Imputation using predictive mean matching method											
10%	1.054 (3.76)	1.062 (3.90)	1.072 (5.06)	1.071 (4.11)	1.075 (4.02)	1.084 (3.99)	1.091 (4.88)	1.092 (4.67)	1.099 (3.59)	1.100 (3.83)	1.454 (3.86)
20%	1.120 (10.21)	1.112 (8.77)	1.123 (10.11)	1.124 (9.28)	1.130 (9.38)	1.133 (8.70)	1.143 (9.91)	1.148 (10.05)	1.148 (8.19)	1.149 (8.42)	1.545 (10.39)
30%	1.177 (15.89)	1.183 (15.73)	1.189 (16.59)	1.188 (15.49)	1.184 (14.57)	1.215 (16.57)	1.212 (16.50)	1.218 (16.81)	1.219 (14.90)	1.227 (15.75)	1.624 (16.00)
40%	1.261 (24.17)	1.252 (22.48)	1.256 (23.11)	1.300 (26.47)	1.294 (25.26)	1.296 (24.32)	1.295 (24.50)	1.320 (26.54)	1.295 (22.08)	1.323 (24.81)	1.744 (24.57)
Imputation using propensity−score method											
10%	1.060 (4.35)	1.066 (4.30)	1.079 (5.76)	1.076 (4.67)	1.079 (4.42)	1.093 (4.87)	1.098 (5.60)	1.094 (4.88)	1.101 (3.77)	1.106 (4.36)	1.462 (4.41)
20%	1.125 (10.69)	1.129 (10.45)	1.133 (11.10)	1.140 (10.90)	1.143 (10.64)	1.154 (10.69)	1.151 (10.64)	1.161 (11.34)	1.166 (9.89)	1.172 (10.63)	1.551 (10.79)
30%	1.176 (15.74)	1.184 (15.83)	1.190 (16.68)	1.201 (16.76)	1.207 (16.78)	1.232 (18.15)	1.227 (17.93)	1.222 (17.23)	1.227 (15.68)	1.246 (17.56)	1.624 (16.00)
40%	1.301 (28.06)	1.301 (27.24)	1.300 (27.43)	1.336 (29.89)	1.318 (27.59)	1.319 (26.51)	1.333 (28.14)	1.351 (29.52)	1.323 (24.72)	1.363 (28.62)	1.796 (28.28)
Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu10=6.0											
0%	1.014	1.021	1.025	1.022	1.036	1.035	1.045	1.045	1.050	1.057	1.398
Imputation using linear regression											

SampleSize_n300_10Treatments/table3_nonshrink_10_treatments_MAR_n300

MR	None_T1vs.T0	None_T2vs.T0	None_T3vs.T0	None_T4vs.T0	None_T5vs.T0	None_T6vs.T0	None_T7vs.T0	None_T8vs.T0	None_T9vs.T0	None_T10vs.T0	Many_T1vs.T0
10%	1.055 (4.05)	1.062 (4.04)	1.062 (3.56)	1.074 (5.14)	1.080 (4.30)	1.086 (4.97)	1.084 (3.70)	1.095 (4.75)	1.101 (4.90)	1.095 (3.60)	1.458 (4.27)
20%	1.112 (9.64)	1.117 (9.38)	1.126 (9.78)	1.132 (10.78)	1.130 (9.13)	1.141 (10.30)	1.150 (10.07)	1.164 (11.38)	1.149 (9.42)	1.160 (9.84)	1.535 (9.79)
30%	1.198 (18.13)	1.194 (17.01)	1.207 (17.73)	1.205 (17.92)	1.198 (15.72)	1.222 (18.04)	1.224 (17.18)	1.238 (18.52)	1.251 (19.20)	1.253 (18.56)	1.655 (18.37)
40%	1.294 (27.59)	1.292 (26.60)	1.292 (26.01)	1.289 (26.13)	1.320 (27.47)	1.328 (28.29)	1.329 (27.16)	1.356 (29.76)	1.361 (29.63)	1.352 (27.93)	1.790 (28.02)
Imputation using predictive mean matching method											
10%	1.053 (3.83)	1.058 (3.62)	1.060 (3.37)	1.069 (4.63)	1.070 (3.34)	1.080 (4.34)	1.077 (3.09)	1.090 (4.33)	1.095 (4.35)	1.091 (3.27)	1.455 (4.04)
20%	1.115 (9.89)	1.106 (8.32)	1.119 (9.15)	1.122 (9.84)	1.136 (9.71)	1.140 (10.16)	1.139 (9.04)	1.154 (10.44)	1.141 (8.72)	1.150 (8.85)	1.537 (9.95)
30%	1.194 (17.72)	1.169 (14.52)	1.198 (16.89)	1.204 (17.83)	1.199 (15.74)	1.220 (17.90)	1.212 (16.00)	1.224 (17.18)	1.239 (18.05)	1.236 (16.95)	1.646 (17.76)
40%	1.256 (23.81)	1.247 (22.13)	1.261 (22.98)	1.244 (21.78)	1.253 (21.01)	1.274 (23.11)	1.284 (22.92)	1.293 (23.77)	1.303 (24.14)	1.305 (23.49)	1.738 (24.28)
Imputation using propensity−score method											
10%	1.054 (3.87)	1.059 (3.71)	1.059 (3.27)	1.069 (4.62)	1.072 (3.54)	1.081 (4.41)	1.082 (3.50)	1.094 (4.67)	1.098 (4.62)	1.094 (3.53)	1.455 (4.08)
20%	1.111 (9.51)	1.110 (8.72)	1.129 (10.09)	1.125 (10.08)	1.130 (9.15)	1.137 (9.88)	1.145 (9.59)	1.150 (10.03)	1.154 (9.96)	1.161 (9.92)	1.532 (9.58)
30%	1.182 (16.51)	1.177 (15.28)	1.202 (17.26)	1.207 (18.14)	1.210 (16.84)	1.207 (16.63)	1.221 (16.88)	1.226 (17.29)	1.229 (17.08)	1.232 (16.61)	1.633 (16.78)
40%	1.276 (25.82)	1.276 (24.99)	1.281 (24.95)	1.292 (26.50)	1.309 (26.44)	1.327 (28.27)	1.317 (26.02)	1.334 (27.65)	1.361 (29.67)	1.337 (26.54)	1.764 (26.15)

MR	Many_T2vs.T0	Many_T3vs.T0	Many_T4vs.T0	Many_T5vs.T0	Many_T6vs.T0	Many_T7vs.T0	Many_T8vs.T0	Many_T9vs.T0	Many_T10vs.T0
Scenario I: all means are 0 for T0,T1,...,T10									
0%	1.409	1.406	1.417	1.424	1.437	1.433	1.437	1.462	1.460
Imputation using linear regression									
10%	1.464 (3.90)	1.486 (5.72)	1.485 (4.81)	1.491 (4.69)	1.508 (4.98)	1.514 (5.61)	1.517 (5.54)	1.526 (4.36)	1.530 (4.76)
20%	1.551 (10.05)	1.560 (10.99)	1.570 (10.79)	1.579 (10.92)	1.586 (10.37)	1.585 (10.57)	1.602 (11.50)	1.606 (9.84)	1.614 (10.49)
30%	1.683 (19.46)	1.676 (19.22)	1.708 (20.51)	1.700 (19.38)	1.714 (19.28)	1.715 (19.62)	1.712 (19.15)	1.722 (17.74)	1.747 (19.64)
40%	1.801 (27.83)	1.807 (28.59)	1.835 (29.51)	1.825 (28.16)	1.829 (27.32)	1.864 (30.02)	1.877 (30.64)	1.853 (26.71)	1.885 (29.07)
Imputation using predictive mean matching method									
10%	1.465 (4.01)	1.478 (5.17)	1.477 (4.22)	1.483 (4.13)	1.496 (4.10)	1.505 (4.99)	1.506 (4.78)	1.516 (3.69)	1.518 (3.94)
20%	1.535 (8.95)	1.550 (10.30)	1.551 (9.47)	1.560 (9.56)	1.564 (8.89)	1.578 (10.08)	1.584 (10.22)	1.585 (8.37)	1.586 (8.61)
30%	1.632 (15.84)	1.641 (16.71)	1.639 (15.63)	1.634 (14.73)	1.677 (16.69)	1.672 (16.63)	1.680 (16.90)	1.682 (15.02)	1.692 (15.89)
40%	1.732 (22.92)	1.737 (23.57)	1.797 (26.84)	1.789 (25.65)	1.792 (24.70)	1.791 (24.92)	1.824 (26.92)	1.791 (22.50)	1.829 (25.22)
Imputation using propensity−score method									
10%	1.470 (4.36)	1.487 (5.81)	1.484 (4.73)	1.488 (4.48)	1.507 (4.92)	1.514 (5.65)	1.508 (4.93)	1.518 (3.82)	1.525 (4.42)
20%	1.557 (10.54)	1.563 (11.19)	1.573 (10.99)	1.577 (10.73)	1.592 (10.77)	1.587 (10.73)	1.601 (11.42)	1.608 (9.98)	1.617 (10.72)
30%	1.636 (16.10)	1.644 (16.95)	1.658 (17.03)	1.667 (17.05)	1.701 (18.39)	1.694 (18.19)	1.688 (17.47)	1.695 (15.93)	1.721 (17.82)
40%	1.796 (27.49)	1.795 (27.70)	1.844 (30.09)	1.820 (27.83)	1.821 (26.75)	1.840 (28.39)	1.865 (29.76)	1.828 (25.00)	1.882 (28.85)
Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu10=6.0									
0%	1.407	1.413	1.408	1.427	1.426	1.440	1.440	1.447	1.456
Imputation using linear regression									

SampleSize_n300_10Treatments/table3_nonshrink_10_treatments_MAR_n300

MR	Many_T2vs.T0	Many_T3vs.T0	Many_T4vs.T0	Many_T5vs.T0	Many_T6vs.T0	Many_T7vs.T0	Many_T8vs.T0	Many_T9vs.T0	Many_T10vs.T0
10%	1.467 (4.26)	1.467 (3.78)	1.484 (5.36)	1.492 (4.51)	1.500 (5.18)	1.497 (3.91)	1.512 (4.96)	1.521 (5.11)	1.512 (3.82)
20%	1.541 (9.53)	1.554 (9.94)	1.562 (10.93)	1.560 (9.28)	1.575 (10.44)	1.587 (10.22)	1.606 (11.53)	1.585 (9.58)	1.602 (9.98)
30%	1.650 (17.25)	1.667 (17.98)	1.664 (18.18)	1.656 (15.99)	1.687 (18.30)	1.691 (17.41)	1.711 (18.78)	1.728 (19.44)	1.730 (18.80)
40%	1.788 (27.07)	1.787 (26.45)	1.783 (26.59)	1.826 (27.90)	1.836 (28.72)	1.838 (27.59)	1.874 (30.14)	1.882 (30.06)	1.869 (28.34)
Imputation using predictive mean matching method									
10%	1.461 (3.83)	1.464 (3.58)	1.476 (4.85)	1.478 (3.55)	1.491 (4.55)	1.488 (3.29)	1.506 (4.53)	1.513 (4.56)	1.507 (3.48)
20%	1.525 (8.41)	1.544 (9.24)	1.548 (9.92)	1.567 (9.77)	1.572 (10.22)	1.572 (9.11)	1.592 (10.51)	1.574 (8.80)	1.586 (8.92)
30%	1.613 (14.61)	1.653 (16.96)	1.660 (17.88)	1.653 (15.82)	1.683 (17.96)	1.672 (16.07)	1.689 (17.27)	1.709 (18.12)	1.704 (17.03)
40%	1.726 (22.65)	1.745 (23.45)	1.722 (22.31)	1.735 (21.55)	1.763 (23.61)	1.777 (23.39)	1.790 (24.26)	1.803 (24.64)	1.805 (23.97)
Imputation using propensity−score method									
10%	1.462 (3.91)	1.462 (3.48)	1.476 (4.83)	1.481 (3.75)	1.492 (4.62)	1.494 (3.70)	1.511 (4.88)	1.517 (4.83)	1.511 (3.73)
20%	1.531 (8.80)	1.557 (10.16)	1.551 (10.16)	1.559 (9.22)	1.568 (9.96)	1.579 (9.66)	1.586 (10.11)	1.592 (10.03)	1.602 (9.99)
30%	1.626 (15.55)	1.661 (17.51)	1.667 (18.39)	1.671 (17.08)	1.668 (16.91)	1.687 (17.12)	1.693 (17.56)	1.698 (17.36)	1.702 (16.88)
40%	1.763 (25.33)	1.770 (25.27)	1.786 (26.80)	1.809 (26.75)	1.834 (28.54)	1.819 (26.31)	1.843 (27.94)	1.880 (29.95)	1.847 (26.85)

SampleSize_n300_10Treatments/table3_shrink_10_treatments_MAR_n300

MR	None_T1vs.T0	None_T2vs.T0	None_T3vs.T0	None_T4vs.T0	None_T5vs.T0	None_T6vs.T0	None_T7vs.T0	None_T8vs.T0	None_T9vs.T0	None_T10vs.T0	Many_T1vs.T0
Scenario I: all means are 0 for T0,T1,...,T10											
0%	1.016	1.022	1.020	1.028	1.033	1.043	1.040	1.043	1.061	1.060	1.404
Imputation using linear regression											
10%	1.061 (4.43)	1.061 (3.80)	1.077 (5.62)	1.077 (4.72)	1.081 (4.60)	1.093 (4.89)	1.098 (5.52)	1.100 (5.45)	1.106 (4.28)	1.109 (4.67)	1.467 (4.44)
20%	1.124 (10.65)	1.124 (9.91)	1.130 (10.84)	1.138 (10.64)	1.145 (10.78)	1.149 (10.23)	1.149 (10.42)	1.161 (11.36)	1.164 (9.70)	1.169 (10.36)	1.550 (10.33)
30%	1.190 (17.10)	1.220 (19.30)	1.214 (19.03)	1.238 (20.36)	1.232 (19.20)	1.241 (19.09)	1.242 (19.42)	1.241 (18.98)	1.247 (17.56)	1.266 (19.44)	1.647 (17.29)
40%	1.300 (27.94)	1.304 (27.53)	1.308 (28.27)	1.329 (29.23)	1.321 (27.86)	1.324 (27.03)	1.350 (29.76)	1.360 (30.38)	1.341 (26.41)	1.365 (28.78)	1.796 (27.89)
Imputation using predictive mean matching method											
10%	1.054 (3.76)	1.062 (3.90)	1.072 (5.06)	1.071 (4.11)	1.075 (4.02)	1.084 (3.99)	1.091 (4.88)	1.092 (4.67)	1.099 (3.59)	1.100 (3.83)	1.459 (3.85)
20%	1.120 (10.21)	1.112 (8.77)	1.123 (10.11)	1.124 (9.28)	1.130 (9.38)	1.133 (8.70)	1.143 (9.91)	1.148 (10.05)	1.148 (8.19)	1.149 (8.42)	1.542 (9.82)
30%	1.177 (15.89)	1.183 (15.73)	1.189 (16.59)	1.188 (15.49)	1.184 (14.57)	1.215 (16.57)	1.212 (16.50)	1.218 (16.81)	1.219 (14.90)	1.227 (15.75)	1.629 (15.98)
40%	1.261 (24.17)	1.252 (22.48)	1.256 (23.11)	1.300 (26.47)	1.294 (25.26)	1.296 (24.32)	1.295 (24.50)	1.320 (26.54)	1.295 (22.08)	1.323 (24.81)	1.745 (24.23)
Imputation using propensity−score method											
10%	1.060 (4.35)	1.066 (4.30)	1.079 (5.76)	1.076 (4.67)	1.079 (4.42)	1.093 (4.87)	1.098 (5.60)	1.094 (4.88)	1.101 (3.77)	1.106 (4.36)	1.465 (4.33)
20%	1.125 (10.69)	1.129 (10.45)	1.133 (11.10)	1.140 (10.90)	1.143 (10.64)	1.154 (10.69)	1.151 (10.64)	1.161 (11.34)	1.166 (9.89)	1.172 (10.63)	1.549 (10.30)
30%	1.176 (15.74)	1.184 (15.83)	1.190 (16.68)	1.201 (16.76)	1.207 (16.78)	1.232 (18.15)	1.227 (17.93)	1.222 (17.23)	1.227 (15.68)	1.246 (17.56)	1.629 (15.99)
40%	1.301 (28.06)	1.301 (27.24)	1.300 (27.43)	1.336 (29.89)	1.318 (27.59)	1.319 (26.51)	1.333 (28.14)	1.351 (29.52)	1.323 (24.72)	1.363 (28.62)	1.797 (27.93)
Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu10=6.0											
0%	1.014	1.021	1.025	1.022	1.036	1.035	1.045	1.045	1.050	1.057	1.399
Imputation using linear regression											

SampleSize_n300_10Treatments/table3_shrink_10_treatments_MAR_n300

MR	None_T1vs.T0	None_T2vs.T0	None_T3vs.T0	None_T4vs.T0	None_T5vs.T0	None_T6vs.T0	None_T7vs.T0	None_T8vs.T0	None_T9vs.T0	None_T10vs.T0	Many_T1vs.T0
10%	1.055 (4.05)	1.062 (4.04)	1.062 (3.56)	1.074 (5.14)	1.080 (4.30)	1.086 (4.97)	1.084 (3.70)	1.095 (4.75)	1.101 (4.90)	1.095 (3.60)	1.464 (4.65)
20%	1.112 (9.64)	1.117 (9.38)	1.126 (9.78)	1.132 (10.78)	1.130 (9.13)	1.141 (10.30)	1.150 (10.07)	1.164 (11.38)	1.149 (9.42)	1.160 (9.84)	1.537 (9.85)
30%	1.198 (18.13)	1.194 (17.01)	1.207 (17.73)	1.205 (17.92)	1.198 (15.72)	1.222 (18.04)	1.224 (17.18)	1.238 (18.52)	1.251 (19.20)	1.253 (18.56)	1.657 (18.39)
40%	1.294 (27.59)	1.292 (26.60)	1.292 (26.01)	1.289 (26.13)	1.320 (27.47)	1.328 (28.29)	1.329 (27.16)	1.356 (29.76)	1.361 (29.63)	1.352 (27.93)	1.791 (28.01)
Imputation using predictive mean matching method											
10%	1.053 (3.83)	1.058 (3.62)	1.060 (3.37)	1.069 (4.63)	1.070 (3.34)	1.080 (4.34)	1.077 (3.09)	1.090 (4.33)	1.095 (4.35)	1.091 (3.27)	1.461 (4.39)
20%	1.115 (9.89)	1.106 (8.32)	1.119 (9.15)	1.122 (9.84)	1.136 (9.71)	1.140 (10.16)	1.139 (9.04)	1.154 (10.44)	1.141 (8.72)	1.150 (8.85)	1.539 (9.98)
30%	1.194 (17.72)	1.169 (14.52)	1.198 (16.89)	1.204 (17.83)	1.199 (15.74)	1.220 (17.90)	1.212 (16.00)	1.224 (17.18)	1.239 (18.05)	1.236 (16.95)	1.647 (17.72)
40%	1.256 (23.81)	1.247 (22.13)	1.261 (22.98)	1.244 (21.78)	1.253 (21.01)	1.274 (23.11)	1.284 (22.92)	1.293 (23.77)	1.303 (24.14)	1.305 (23.49)	1.740 (24.33)
Imputation using propensity−score method											
10%	1.054 (3.87)	1.059 (3.71)	1.059 (3.27)	1.069 (4.62)	1.072 (3.54)	1.081 (4.41)	1.082 (3.50)	1.094 (4.67)	1.098 (4.62)	1.094 (3.53)	1.461 (4.41)
20%	1.111 (9.51)	1.110 (8.72)	1.129 (10.09)	1.125 (10.08)	1.130 (9.15)	1.137 (9.88)	1.145 (9.59)	1.150 (10.03)	1.154 (9.96)	1.161 (9.92)	1.534 (9.64)
30%	1.182 (16.51)	1.177 (15.28)	1.202 (17.26)	1.207 (18.14)	1.210 (16.84)	1.207 (16.63)	1.221 (16.88)	1.226 (17.29)	1.229 (17.08)	1.232 (16.61)	1.634 (16.78)
40%	1.276 (25.82)	1.276 (24.99)	1.281 (24.95)	1.292 (26.50)	1.309 (26.44)	1.327 (28.27)	1.317 (26.02)	1.334 (27.65)	1.361 (29.67)	1.337 (26.54)	1.766 (26.24)

MR	Many_T2vs.T0	Many_T3vs.T0	Many_T4vs.T0	Many_T5vs.T0	Many_T6vs.T0	Many_T7vs.T0	Many_T8vs.T0	Many_T9vs.T0	Many_T10vs.T0
Scenario I: all means are 0 for T0,T1,...,T10									
0%	1.409	1.406	1.410	1.414	1.418	1.418	1.417	1.429	1.429
Imputation using linear regression									
10%	1.464 (3.93)	1.477 (5.03)	1.477 (4.74)	1.479 (4.58)	1.492 (5.20)	1.491 (5.18)	1.496 (5.62)	1.503 (5.15)	1.501 (5.07)
20%	1.549 (9.95)	1.554 (10.52)	1.559 (10.59)	1.568 (10.86)	1.571 (10.78)	1.569 (10.68)	1.581 (11.63)	1.585 (10.89)	1.592 (11.38)
30%	1.685 (19.60)	1.672 (18.93)	1.703 (20.81)	1.692 (19.64)	1.699 (19.85)	1.700 (19.91)	1.702 (20.13)	1.708 (19.48)	1.727 (20.85)
40%	1.801 (27.82)	1.804 (28.34)	1.829 (29.74)	1.818 (28.58)	1.821 (28.42)	1.854 (30.80)	1.865 (31.61)	1.841 (28.81)	1.869 (30.79)
Imputation using predictive mean matching method									
10%	1.466 (4.06)	1.469 (4.51)	1.468 (4.17)	1.471 (4.02)	1.479 (4.30)	1.483 (4.59)	1.485 (4.84)	1.493 (4.49)	1.489 (4.23)
20%	1.532 (8.73)	1.543 (9.73)	1.541 (9.30)	1.547 (9.41)	1.548 (9.18)	1.560 (10.04)	1.562 (10.29)	1.563 (9.35)	1.563 (9.38)
30%	1.633 (15.92)	1.637 (16.40)	1.634 (15.88)	1.626 (14.95)	1.663 (17.29)	1.656 (16.84)	1.669 (17.78)	1.666 (16.56)	1.672 (17.05)
40%	1.733 (22.96)	1.735 (23.39)	1.791 (27.07)	1.783 (26.07)	1.784 (25.80)	1.781 (25.61)	1.812 (27.89)	1.779 (24.50)	1.813 (26.88)
Imputation using propensity−score method									
10%	1.471 (4.40)	1.478 (5.09)	1.475 (4.61)	1.475 (4.30)	1.491 (5.18)	1.492 (5.24)	1.488 (5.00)	1.495 (4.57)	1.496 (4.72)
20%	1.556 (10.39)	1.556 (10.70)	1.563 (10.87)	1.565 (10.69)	1.577 (11.23)	1.571 (10.84)	1.581 (11.62)	1.588 (11.08)	1.595 (11.60)
30%	1.637 (16.17)	1.640 (16.67)	1.653 (17.26)	1.659 (17.33)	1.688 (19.04)	1.680 (18.49)	1.678 (18.45)	1.681 (17.61)	1.701 (19.07)
40%	1.796 (27.49)	1.793 (27.53)	1.838 (30.36)	1.815 (28.32)	1.814 (27.92)	1.831 (29.19)	1.853 (30.79)	1.817 (27.14)	1.868 (30.70)
Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu10=6.0									
0%	1.403	1.403	1.404	1.407	1.408	1.412	1.414	1.417	1.417
Imputation using linear regression									

MR	Many_T2vs.T0	Many_T3vs.T0	Many_T4vs.T0	Many_T5vs.T0	Many_T6vs.T0	Many_T7vs.T0	Many_T8vs.T0	Many_T9vs.T0	Many_T10vs.T0
10%	1.468 (4.62)	1.468 (4.65)	1.476 (5.17)	1.480 (5.24)	1.484 (5.45)	1.483 (5.00)	1.491 (5.43)	1.495 (5.53)	1.489 (5.09)
20%	1.541 (9.81)	1.548 (10.38)	1.557 (10.91)	1.552 (10.36)	1.564 (11.13)	1.573 (11.37)	1.586 (12.11)	1.568 (10.67)	1.581 (11.57)
30%	1.653 (17.77)	1.665 (18.68)	1.659 (18.22)	1.651 (17.37)	1.676 (19.06)	1.682 (19.09)	1.693 (19.69)	1.710 (20.69)	1.709 (20.61)
40%	1.786 (27.26)	1.785 (27.25)	1.781 (26.88)	1.818 (29.26)	1.828 (29.83)	1.829 (29.56)	1.864 (31.78)	1.864 (31.57)	1.853 (30.81)
Imputation using predictive mean matching method									
10%	1.462 (4.21)	1.464 (4.42)	1.469 (4.66)	1.467 (4.26)	1.476 (4.83)	1.475 (4.43)	1.484 (4.96)	1.487 (4.94)	1.483 (4.69)
20%	1.526 (8.74)	1.537 (9.60)	1.542 (9.87)	1.559 (10.83)	1.560 (10.80)	1.557 (10.29)	1.571 (11.08)	1.556 (9.83)	1.565 (10.46)
30%	1.616 (15.13)	1.649 (17.61)	1.655 (17.95)	1.647 (17.07)	1.671 (18.69)	1.661 (17.61)	1.671 (18.15)	1.691 (19.33)	1.682 (18.70)
40%	1.724 (22.88)	1.744 (24.33)	1.721 (22.63)	1.728 (22.85)	1.756 (24.72)	1.769 (25.30)	1.778 (25.69)	1.786 (26.06)	1.789 (26.29)
Imputation using propensity−score method									
10%	1.463 (4.27)	1.464 (4.36)	1.469 (4.65)	1.470 (4.51)	1.477 (4.93)	1.480 (4.78)	1.489 (5.30)	1.490 (5.17)	1.488 (5.01)
20%	1.530 (9.05)	1.551 (10.55)	1.545 (10.09)	1.551 (10.23)	1.556 (10.51)	1.566 (10.87)	1.566 (10.71)	1.574 (11.06)	1.581 (11.57)
30%	1.629 (16.10)	1.658 (18.20)	1.662 (18.42)	1.666 (18.45)	1.656 (17.63)	1.677 (18.74)	1.675 (18.46)	1.680 (18.56)	1.681 (18.63)
40%	1.762 (25.55)	1.770 (26.22)	1.785 (27.18)	1.803 (28.16)	1.825 (29.67)	1.812 (28.30)	1.833 (29.58)	1.864 (31.59)	1.833 (29.36)

SampleSize_n50_10Treatments/table2_nonshrink_10_treatments_MAR_n50

MR	T1vs.T0_none	T2vs.T0_none	T3vs.T0_none	T4vs.T0_none	T5vs.T0_none	T6vs.T0_none	T7vs.T0_none	T8vs.T0_none	T9vs.T0_none	T10vs.T0_none	Joint_none
Scenario I: all means are 0 for T0,T1,...,T10	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
0%	100.00	89.66	89.66	89.66	96.55	93.10	96.55	96.55	93.10	100.00	72.41
Imputation using linear regression	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	86.21	82.76	89.66	93.10	86.21	100.00	93.10	93.10	93.10	96.55	72.41
20%	92.86	96.43	96.43	92.86	96.43	92.86	100.00	100.00	92.86	100.00	78.57
30%	100.00	89.66	96.55	96.55	93.10	93.10	93.10	89.66	89.66	89.66	62.07
40%	89.66	96.55	96.55	96.55	89.66	96.55	100.00	100.00	96.55	96.55	79.31
Imputation using predictive mean matching method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	86.21	82.76	89.66	89.66	86.21	100.00	93.10	93.10	93.10	96.55	72.41
20%	92.86	92.86	92.86	92.86	96.43	92.86	100.00	96.43	89.29	96.43	67.86
30%	96.55	89.66	96.55	96.55	89.66	96.55	89.66	89.66	89.66	89.66	58.62
40%	86.21	93.10	89.66	86.21	86.21	89.66	96.55	96.55	93.10	96.55	58.62
Imputation using propensity−score method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	86.21	82.76	89.66	86.21	86.21	96.55	93.10	93.10	93.10	93.10	68.97
20%	92.86	96.43	96.43	92.86	96.43	96.43	100.00	96.43	92.86	96.43	75.00
30%	93.10	79.31	89.66	96.55	93.10	96.55	96.55	93.10	93.10	89.66	55.17
40%	93.10	96.55	96.55	89.66	86.21	89.66	89.66	96.55	93.10	96.55	65.52
Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu10=6.0	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
0%	100.00	96.55	100.00	96.55	100.00	100.00	96.55	100.00	93.10	100.00	82.76
Imputation using linear regression	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA

SampleSize_n50_10Treatments/table2_nonshrink_10_treatments_MAR_n50

MR	T1vs.T0_none	T2vs.T0_none	T3vs.T0_none	T4vs.T0_none	T5vs.T0_none	T6vs.T0_none	T7vs.T0_none	T8vs.T0_none	T9vs.T0_none	T10vs.T0_none	Joint_none
10%	92.86	100.00	89.29	92.86	100.00	96.43	96.43	96.43	96.43	96.43	78.57
20%	92.86	100.00	89.29	100.00	100.00	100.00	96.43	100.00	89.29	96.43	75.00
30%	96.55	100.00	96.55	96.55	96.55	100.00	89.66	100.00	100.00	96.55	86.21
40%	93.10	96.55	89.66	93.10	93.10	96.55	100.00	93.10	100.00	96.55	65.52
Imputation using predictive mean matching method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	96.43	100.00	89.29	96.43	100.00	96.43	96.43	96.43	96.43	96.43	78.57
20%	89.29	92.86	85.71	96.43	96.43	100.00	96.43	100.00	89.29	96.43	64.29
30%	93.10	100.00	100.00	89.66	96.55	100.00	93.10	100.00	96.55	96.55	79.31
40%	96.55	100.00	82.76	89.66	89.66	96.55	93.10	89.66	96.55	96.55	51.72
Imputation using propensity−score method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	92.86	100.00	92.86	92.86	100.00	96.43	96.43	100.00	96.43	96.43	82.14
20%	89.29	96.43	89.29	100.00	96.43	96.43	96.43	100.00	92.86	96.43	71.43
30%	96.55	100.00	96.55	93.10	96.55	100.00	93.10	100.00	100.00	96.55	86.21
40%	93.10	100.00	86.21	86.21	89.66	96.55	96.55	82.76	96.55	93.10	65.52

SampleSize_n50_10Treatments/table2_nonshrink_10_treatments_MAR_n50

MR	T1vs.T0_many	T2vs.T0_many	T3vs.T0_many	T4vs.T0_many	T5vs.T0_many	T6vs.T0_many	T7vs.T0_many	T8vs.T0_many	T9vs.T0_many	T10vs.T0_many	Joint_many
Scenario I: all means are 0 for T0,T1,...,T10	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
0%	100.00	100.00	100	100.00	100.00	100.00	96.55	100.00	96.55	100.00	96.55
Imputation using linear regression	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	100.00	100.00	100	100.00	96.55	100.00	100.00	100.00	96.55	100.00	93.10
20%	100.00	100.00	100	96.43	100.00	100.00	100.00	100.00	100.00	100.00	96.43
30%	100.00	100.00	100	100.00	100.00	96.55	96.55	100.00	100.00	100.00	93.10
40%	100.00	100.00	100	96.55	100.00	100.00	100.00	100.00	100.00	100.00	96.55
Imputation using predictive mean matching method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	100.00	100.00	100	100.00	96.55	100.00	100.00	100.00	96.55	100.00	93.10
20%	100.00	96.43	100	96.43	100.00	100.00	100.00	100.00	100.00	100.00	96.43
30%	100.00	93.10	100	100.00	96.55	96.55	96.55	96.55	100.00	96.55	82.76
40%	96.55	100.00	100	96.55	96.55	100.00	100.00	100.00	100.00	100.00	96.55
Imputation using propensity−score method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	96.55	100.00	100	100.00	96.55	100.00	100.00	96.55	96.55	100.00	93.10
20%	100.00	96.43	100	100.00	96.43	100.00	100.00	100.00	100.00	100.00	92.86
30%	100.00	100.00	100	100.00	100.00	96.55	96.55	100.00	100.00	100.00	93.10
40%	96.55	100.00	100	96.55	100.00	100.00	100.00	100.00	100.00	100.00	96.55
Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu10=6.0	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
0%	100.00	100.00	100	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00
Imputation using linear regression	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA

SampleSize_n50_10Treatments/table2_nonshrink_10_treatments_MAR_n50

MR	T1vs.T0_many	T2vs.T0_many	T3vs.T0_many	T4vs.T0_many	T5vs.T0_many	T6vs.T0_many	T7vs.T0_many	T8vs.T0_many	T9vs.T0_many	T10vs.T0_many	Joint_many
10%	100.00	100	100.00	100.00	100.00	100.00	100	100.00	100.00	100.00	100.00
20%	100.00	100	96.43	100.00	100.00	100.00	100	100.00	96.43	100.00	92.86
30%	100.00	100	100.00	100.00	100.00	100.00	100	100.00	100.00	100.00	100.00
40%	100.00	100	100.00	100.00	100.00	100.00	100	100.00	100.00	96.55	96.55
Imputation using predictive mean matching method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	100.00	100	100.00	100.00	100.00	100.00	100	100.00	100.00	100.00	100.00
20%	96.43	100	92.86	100.00	100.00	100.00	100	100.00	96.43	100.00	85.71
30%	100.00	100	100.00	100.00	96.55	100.00	100	100.00	100.00	100.00	96.55
40%	100.00	100	100.00	96.55	96.55	100.00	100	96.55	100.00	96.55	86.21
Imputation using propensity−score method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	100.00	100	100.00	96.43	100.00	100.00	100	100.00	100.00	100.00	96.43
20%	96.43	100	96.43	100.00	100.00	100.00	100	100.00	96.43	100.00	89.29
30%	100.00	100	100.00	100.00	100.00	100.00	100	100.00	100.00	100.00	100.00
40%	100.00	100	100.00	100.00	93.10	96.55	100	93.10	100.00	96.55	79.31

SampleSize_n50_10Treatments/table2_shrink_10_treatments_MAR_n50

MR	T1vs.T0_none	T2vs.T0_none	T3vs.T0_none	T4vs.T0_none	T5vs.T0_none	T6vs.T0_none	T7vs.T0_none	T8vs.T0_none	T9vs.T0_none	T10vs.T0_none	Joint_none
Scenario I: all means are 0 for T0,T1,...,T10	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
0%	100.00	89.66	89.66	89.66	96.55	93.10	96.55	96.55	93.10	100.00	72.41
Imputation using linear regression	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	86.21	82.76	89.66	93.10	86.21	100.00	93.10	93.10	93.10	96.55	72.41
20%	92.86	96.43	96.43	92.86	96.43	92.86	100.00	100.00	92.86	100.00	78.57
30%	100.00	89.66	96.55	96.55	93.10	93.10	93.10	89.66	89.66	89.66	62.07
40%	89.66	96.55	96.55	96.55	89.66	96.55	100.00	100.00	96.55	96.55	79.31
Imputation using predictive mean matching method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	86.21	82.76	89.66	89.66	86.21	100.00	93.10	93.10	93.10	96.55	72.41
20%	92.86	92.86	92.86	92.86	96.43	92.86	100.00	96.43	89.29	96.43	67.86
30%	96.55	89.66	96.55	96.55	89.66	96.55	89.66	89.66	89.66	89.66	58.62
40%	86.21	93.10	89.66	86.21	86.21	89.66	96.55	96.55	93.10	96.55	58.62
Imputation using propensity−score method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	86.21	82.76	89.66	86.21	86.21	96.55	93.10	93.10	93.10	93.10	68.97
20%	92.86	96.43	96.43	92.86	96.43	96.43	100.00	96.43	92.86	96.43	75.00
30%	93.10	79.31	89.66	96.55	93.10	96.55	96.55	93.10	93.10	89.66	55.17
40%	93.10	96.55	96.55	89.66	86.21	89.66	89.66	96.55	93.10	96.55	65.52
Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu10=6.0	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
0%	100.00	96.55	100.00	96.55	100.00	100.00	96.55	100.00	93.10	100.00	82.76
Imputation using linear regression	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA

SampleSize_n50_10Treatments/table2_shrink_10_treatments_MAR_n50

MR	T1vs.T0_none	T2vs.T0_none	T3vs.T0_none	T4vs.T0_none	T5vs.T0_none	T6vs.T0_none	T7vs.T0_none	T8vs.T0_none	T9vs.T0_none	T10vs.T0_none	Joint_none
10%	92.86	100.00	89.29	92.86	100.00	96.43	96.43	96.43	96.43	96.43	78.57
20%	92.86	100.00	89.29	100.00	100.00	100.00	96.43	100.00	89.29	96.43	75.00
30%	96.55	100.00	96.55	96.55	96.55	100.00	89.66	100.00	100.00	96.55	86.21
40%	93.10	96.55	89.66	93.10	93.10	96.55	100.00	93.10	100.00	96.55	65.52
Imputation using predictive mean matching method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	96.43	100.00	89.29	96.43	100.00	96.43	96.43	96.43	96.43	96.43	78.57
20%	89.29	92.86	85.71	96.43	96.43	100.00	96.43	100.00	89.29	96.43	64.29
30%	93.10	100.00	100.00	89.66	96.55	100.00	93.10	100.00	96.55	96.55	79.31
40%	96.55	100.00	82.76	89.66	89.66	96.55	93.10	89.66	96.55	96.55	51.72
Imputation using propensity−score method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	92.86	100.00	92.86	92.86	100.00	96.43	96.43	100.00	96.43	96.43	82.14
20%	89.29	96.43	89.29	100.00	96.43	96.43	96.43	100.00	92.86	96.43	71.43
30%	96.55	100.00	96.55	93.10	96.55	100.00	93.10	100.00	100.00	96.55	86.21
40%	93.10	100.00	86.21	86.21	89.66	96.55	96.55	82.76	96.55	93.10	65.52

SampleSize_n50_10Treatments/table2_shrink_10_treatments_MAR_n50

MR	T1vs.T0_many	T2vs.T0_many	T3vs.T0_many	T4vs.T0_many	T5vs.T0_many	T6vs.T0_many	T7vs.T0_many	T8vs.T0_many	T9vs.T0_many	T10vs.T0_many	Joint_many
Scenario I: all means are 0 for T0,T1,...,T10	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
0%	100.00	100.00	100	100.00	100.00	100.00	96.55	100.00	96.55	100.00	96.55
Imputation using linear regression	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	100.00	100.00	100	100.00	96.55	100.00	100.00	100.00	96.55	100.00	93.10
20%	100.00	100.00	100	96.43	100.00	100.00	100.00	100.00	100.00	100.00	96.43
30%	100.00	100.00	100	100.00	100.00	96.55	96.55	100.00	100.00	100.00	93.10
40%	100.00	100.00	100	96.55	100.00	100.00	100.00	100.00	100.00	100.00	96.55
Imputation using predictive mean matching method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	100.00	100.00	100	100.00	96.55	100.00	100.00	100.00	96.55	100.00	93.10
20%	100.00	96.43	100	96.43	96.43	100.00	100.00	100.00	100.00	100.00	92.86
30%	100.00	93.10	100	100.00	100.00	96.55	96.55	96.55	96.55	96.55	82.76
40%	96.55	100.00	100	96.55	96.55	100.00	100.00	96.55	100.00	100.00	96.55
Imputation using propensity−score method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	100.00	100.00	100	100.00	96.55	100.00	100.00	100.00	96.55	100.00	93.10
20%	100.00	96.43	100	100.00	96.43	100.00	100.00	100.00	100.00	100.00	92.86
30%	100.00	100.00	100	100.00	100.00	96.55	96.55	100.00	100.00	100.00	93.10
40%	96.55	100.00	100	96.55	100.00	100.00	100.00	100.00	100.00	100.00	96.55
Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu10=6.0	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
0%	100.00	100.00	100	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00
Imputation using linear regression	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA

SampleSize_n50_10Treatments/table2_shrink_10_treatments_MAR_n50

MR	T1vs.T0_many	T2vs.T0_many	T3vs.T0_many	T4vs.T0_many	T5vs.T0_many	T6vs.T0_many	T7vs.T0_many	T8vs.T0_many	T9vs.T0_many	T10vs.T0_many	Joint_many
10%	100.00	100	100	100.00	100.00	100.00	100	100.00	100.00	100.00	100.00
20%	96.43	100	100	100.00	100.00	100.00	100	100.00	96.43	100.00	92.86
30%	100.00	100	100	100.00	100.00	100.00	100	100.00	100.00	100.00	100.00
40%	100.00	100	100	100.00	100.00	100.00	100	100.00	100.00	96.55	96.55
Imputation using predictive mean matching method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	100.00	100	100	100.00	100.00	100.00	100	100.00	100.00	100.00	100.00
20%	96.43	100	100	100.00	100.00	100.00	100	100.00	96.43	100.00	92.86
30%	100.00	100	100	100.00	100.00	100.00	100	100.00	100.00	100.00	100.00
40%	100.00	100	100	100.00	96.55	96.55	100	96.55	100.00	96.55	86.21
Imputation using propensity−score method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	100.00	100	100	96.43	100.00	100.00	100	100.00	100.00	100.00	96.43
20%	92.86	100	100	100.00	100.00	100.00	100	100.00	96.43	100.00	89.29
30%	100.00	100	100	100.00	100.00	100.00	100	100.00	100.00	100.00	100.00
40%	100.00	100	100	100.00	93.10	96.55	100	93.10	100.00	96.55	79.31

SampleSize_n50_10Treatments/table3_nonshrink_10_treatments_MAR_n50

MR	None_T1vs.T0	None_T2vs.T0	None_T3vs.T0	None_T4vs.T0	None_T5vs.T0	None_T6vs.T0	None_T7vs.T0	None_T8vs.T0	None_T9vs.T0	None_T10vs.T0	Many_T1vs.T0
Scenario I: all means are 0 for T0,T1,...,T10											
0%	2.467	2.560	2.584	2.574	2.542	2.621	2.640	2.627	2.618	2.620	3.367
Imputation using linear regression											
10%	2.589 (4.96)	2.604 (1.71)	2.673 (3.45)	2.696 (4.75)	2.664 (4.77)	2.658 (1.42)	2.694 (2.04)	2.734 (4.06)	2.712 (3.59)	2.724 (3.98)	3.572 (6.10)
20%	2.764 (12.04)	2.772 (8.26)	2.798 (8.27)	2.769 (7.57)	2.852 (12.19)	2.803 (6.94)	2.859 (8.29)	2.880 (9.62)	2.855 (9.05)	2.793 (6.63)	3.811 (13.19)
30%	2.980 (20.79)	3.030 (18.36)	3.057 (18.30)	3.066 (19.11)	3.051 (20.00)	3.062 (16.83)	3.044 (15.33)	3.092 (17.68)	3.096 (18.27)	3.073 (17.30)	4.107 (21.97)
40%	3.400 (37.83)	3.318 (29.58)	3.330 (28.87)	3.425 (33.08)	3.413 (34.25)	3.447 (31.53)	3.404 (28.94)	3.430 (30.58)	3.400 (29.87)	3.484 (32.97)	4.684 (39.10)
Imputation using predictive mean matching method											
10%	2.530 (2.56)	2.558 (−0.08)	2.625 (1.59)	2.643 (2.68)	2.620 (3.06)	2.606 (−0.56)	2.647 (0.28)	2.679 (1.98)	2.669 (1.95)	2.676 (2.14)	3.489 (3.62)
20%	2.639 (6.97)	2.667 (4.17)	2.697 (4.36)	2.660 (3.35)	2.735 (7.56)	2.694 (2.79)	2.742 (3.89)	2.778 (5.74)	2.765 (5.63)	2.740 (4.58)	3.637 (8.01)
30%	2.804 (13.65)	2.835 (10.74)	2.838 (9.82)	2.874 (11.65)	2.856 (12.33)	2.830 (8.00)	2.851 (7.99)	2.905 (10.59)	2.922 (11.60)	2.843 (8.51)	3.862 (14.71)
40%	2.969 (20.35)	3.028 (18.26)	3.002 (16.15)	3.030 (17.70)	2.974 (16.98)	3.033 (15.73)	3.010 (14.03)	3.044 (15.87)	3.047 (16.39)	3.123 (19.23)	4.093 (21.57)
Imputation using propensity−score method											
10%	2.542 (3.06)	2.570 (0.39)	2.635 (1.96)	2.639 (2.52)	2.620 (3.03)	2.616 (−0.18)	2.641 (0.05)	2.677 (1.90)	2.668 (1.90)	2.681 (2.35)	3.508 (4.18)
20%	2.653 (7.54)	2.688 (5.00)	2.709 (4.83)	2.685 (4.31)	2.760 (8.57)	2.731 (4.22)	2.755 (4.35)	2.788 (6.13)	2.768 (5.74)	2.753 (5.10)	3.658 (8.66)
30%	2.780 (12.71)	2.812 (9.85)	2.886 (11.69)	2.843 (10.45)	2.901 (14.12)	2.855 (8.95)	2.896 (9.71)	2.934 (11.69)	2.905 (10.98)	2.868 (9.47)	3.844 (14.18)
40%	3.143 (27.39)	3.117 (21.73)	3.101 (20.00)	3.226 (25.31)	3.090 (21.53)	3.143 (19.93)	3.167 (19.98)	3.175 (20.87)	3.166 (20.93)	3.256 (24.27)	4.325 (28.47)
Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu10=6.0											
0%	2.484	2.503	2.536	2.501	2.540	2.562	2.556	2.565	2.601	2.581	3.396
Imputation using linear regression											

MR	None_T1vs.T0	None_T2vs.T0	None_T3vs.T0	None_T4vs.T0	None_T5vs.T0	None_T6vs.T0	None_T7vs.T0	None_T8vs.T0	None_T9vs.T0	None_T10vs.T0	Many_T1vs.T0
10%	2.662 (7.17)	2.572 (2.75)	2.639 (4.06)	2.654 (6.13)	2.638 (3.87)	2.643 (3.16)	2.664 (4.22)	2.653 (3.41)	2.702 (3.91)	2.689 (4.20)	3.669 (8.03)
20%	2.687 (8.19)	2.769 (10.60)	2.763 (8.94)	2.739 (9.50)	2.790 (9.85)	2.805 (9.47)	2.807 (9.80)	2.833 (10.41)	2.854 (9.75)	2.853 (10.56)	3.711 (9.26)
30%	2.992 (20.47)	3.060 (22.23)	2.979 (17.44)	3.074 (22.92)	3.050 (20.11)	3.074 (19.99)	3.152 (23.31)	3.171 (23.61)	3.162 (21.58)	3.084 (19.51)	4.121 (21.35)
40%	3.248 (30.78)	3.316 (32.46)	3.289 (29.70)	3.305 (32.14)	3.350 (31.91)	3.399 (32.64)	3.384 (32.38)	3.375 (31.55)	3.469 (33.41)	3.381 (31.00)	4.489 (32.17)
Imputation using predictive mean matching method											
10%	2.601 (4.73)	2.511 (0.30)	2.579 (1.67)	2.594 (3.71)	2.583 (1.69)	2.581 (0.74)	2.606 (1.95)	2.596 (1.17)	2.638 (1.45)	2.637 (2.17)	3.586 (5.60)
20%	2.573 (3.62)	2.640 (5.44)	2.660 (4.89)	2.654 (6.13)	2.678 (5.44)	2.673 (4.32)	2.713 (6.12)	2.720 (6.01)	2.741 (5.39)	2.703 (4.73)	3.557 (4.74)
30%	2.838 (14.27)	2.836 (13.30)	2.790 (9.99)	2.906 (16.17)	2.875 (13.22)	2.901 (13.21)	2.947 (15.30)	3.018 (17.64)	2.946 (13.27)	2.855 (10.65)	3.903 (14.92)
40%	2.971 (19.63)	2.948 (17.75)	2.924 (15.27)	2.968 (18.68)	2.983 (17.46)	3.003 (17.20)	3.038 (18.86)	3.012 (17.39)	3.027 (16.40)	3.023 (17.12)	4.104 (20.85)
Imputation using propensity−score method											
10%	2.615 (5.28)	2.534 (1.24)	2.591 (2.14)	2.607 (4.24)	2.592 (2.07)	2.611 (1.89)	2.634 (3.03)	2.612 (1.83)	2.637 (1.40)	2.658 (3.00)	3.603 (6.10)
20%	2.626 (5.75)	2.683 (7.19)	2.706 (6.70)	2.693 (7.67)	2.731 (7.52)	2.719 (6.13)	2.749 (7.54)	2.765 (7.79)	2.767 (6.40)	2.758 (6.86)	3.623 (6.69)
30%	2.861 (15.18)	2.905 (16.05)	2.854 (12.54)	2.971 (18.78)	2.949 (16.13)	2.947 (15.03)	2.986 (16.81)	3.058 (19.19)	3.005 (15.57)	2.932 (13.63)	3.941 (16.04)
40%	2.997 (20.68)	3.063 (22.38)	2.976 (17.32)	3.032 (21.24)	3.100 (22.07)	3.038 (18.56)	3.123 (22.19)	3.072 (19.73)	3.183 (22.41)	3.115 (20.69)	4.140 (21.89)

MR	Many_T2vs.T0	Many_T3vs.T0	Many_T4vs.T0	Many_T5vs.T0	Many_T6vs.T0	Many_T7vs.T0	Many_T8vs.T0	Many_T9vs.T0	Many_T10vs.T0
Scenario I: all means are 0 for T0,T1,...,T10									
0%	3.495	3.528	3.514	3.471	3.577	3.604	3.587	3.574	3.576
Imputation using linear regression									
10%	3.593 (2.80)	3.690 (4.58)	3.720 (5.86)	3.676 (5.92)	3.668 (2.52)	3.717 (3.14)	3.773 (5.19)	3.743 (4.71)	3.760 (5.13)
20%	3.823 (9.37)	3.858 (9.36)	3.818 (8.65)	3.934 (13.35)	3.864 (8.01)	3.941 (9.35)	3.972 (10.74)	3.938 (10.16)	3.853 (7.73)
30%	4.177 (19.50)	4.213 (19.42)	4.227 (20.28)	4.203 (21.10)	4.219 (17.93)	4.198 (16.47)	4.262 (18.81)	4.269 (19.42)	4.235 (18.42)
40%	4.575 (30.89)	4.592 (30.16)	4.723 (34.40)	4.702 (35.47)	4.750 (32.78)	4.693 (30.22)	4.729 (31.84)	4.687 (31.14)	4.803 (34.29)
Imputation using predictive mean matching method									
10%	3.528 (0.95)	3.622 (2.66)	3.646 (3.74)	3.614 (4.14)	3.595 (0.48)	3.651 (1.30)	3.696 (3.04)	3.682 (3.00)	3.691 (3.22)
20%	3.676 (5.17)	3.717 (5.35)	3.666 (4.32)	3.769 (8.61)	3.712 (3.76)	3.779 (4.86)	3.829 (6.74)	3.811 (6.62)	3.776 (5.58)
30%	3.907 (11.77)	3.912 (10.89)	3.961 (12.70)	3.935 (13.38)	3.900 (9.01)	3.929 (9.02)	4.004 (11.62)	4.027 (12.66)	3.918 (9.55)
40%	4.173 (19.39)	4.138 (17.30)	4.179 (18.91)	4.101 (18.17)	4.181 (16.86)	4.151 (15.18)	4.197 (17.01)	4.202 (17.55)	4.307 (20.42)
Imputation using propensity−score method									
10%	3.547 (1.47)	3.637 (3.08)	3.642 (3.63)	3.616 (4.17)	3.610 (0.91)	3.645 (1.13)	3.695 (3.01)	3.682 (3.01)	3.701 (3.48)
20%	3.707 (6.07)	3.736 (5.88)	3.702 (5.35)	3.807 (9.69)	3.765 (5.26)	3.799 (5.40)	3.846 (7.21)	3.818 (6.80)	3.797 (6.16)
30%	3.888 (11.24)	3.989 (13.08)	3.931 (11.87)	4.009 (15.52)	3.946 (10.30)	4.004 (11.11)	4.056 (13.08)	4.017 (12.38)	3.965 (10.87)
40%	4.291 (22.77)	4.270 (21.04)	4.442 (26.39)	4.255 (22.59)	4.327 (20.97)	4.362 (21.02)	4.373 (21.92)	4.359 (21.96)	4.482 (25.33)
Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu10=6.0									
0%	3.423	3.468	3.420	3.473	3.504	3.496	3.508	3.557	3.529
Imputation using linear regression									

SampleSize_n50_10Treatments/table3_nonshrink_10_treatments_MAR_n50

MR	Many_T2vs.T0	Many_T3vs.T0	Many_T4vs.T0	Many_T5vs.T0	Many_T6vs.T0	Many_T7vs.T0	Many_T8vs.T0	Many_T9vs.T0	Many_T10vs.T0
10%	3.546 (3.60)	3.639 (4.91)	3.660 (6.99)	3.637 (4.73)	3.644 (4.00)	3.673 (5.06)	3.658 (4.25)	3.725 (4.73)	3.708 (5.06)
20%	3.825 (11.74)	3.817 (10.05)	3.783 (10.60)	3.852 (10.93)	3.874 (10.57)	3.877 (10.90)	3.912 (11.49)	3.943 (10.86)	3.940 (11.65)
30%	4.213 (23.07)	4.105 (18.34)	4.234 (23.79)	4.203 (21.02)	4.233 (20.81)	4.341 (24.18)	4.368 (24.51)	4.354 (22.42)	4.247 (20.35)
40%	4.580 (33.79)	4.542 (30.96)	4.567 (33.52)	4.627 (33.23)	4.693 (33.92)	4.674 (33.71)	4.662 (32.90)	4.792 (34.74)	4.671 (32.35)
Imputation using predictive mean matching method									
10%	3.462 (1.14)	3.556 (2.51)	3.577 (4.57)	3.561 (2.53)	3.559 (1.56)	3.594 (2.80)	3.579 (2.01)	3.637 (2.27)	3.636 (3.02)
20%	3.650 (6.63)	3.678 (6.04)	3.669 (7.27)	3.701 (6.57)	3.696 (5.46)	3.749 (7.25)	3.759 (7.14)	3.789 (6.54)	3.736 (5.87)
30%	3.900 (13.94)	3.838 (10.66)	3.997 (16.85)	3.955 (13.88)	3.988 (13.80)	4.052 (15.92)	4.150 (18.29)	4.052 (13.92)	3.927 (11.28)
40%	4.071 (18.94)	4.037 (16.39)	4.101 (19.89)	4.122 (18.68)	4.150 (18.43)	4.197 (20.05)	4.160 (18.58)	4.183 (17.61)	4.176 (18.33)
Imputation using propensity−score method									
10%	3.493 (2.05)	3.571 (2.95)	3.593 (5.05)	3.573 (2.87)	3.598 (2.68)	3.630 (3.84)	3.601 (2.63)	3.634 (2.19)	3.663 (3.81)
20%	3.703 (8.17)	3.735 (7.67)	3.716 (8.63)	3.767 (8.47)	3.752 (7.07)	3.792 (8.48)	3.815 (8.73)	3.819 (7.38)	3.804 (7.80)
30%	4.000 (16.85)	3.933 (13.40)	4.092 (19.62)	4.063 (16.99)	4.060 (15.85)	4.114 (17.68)	4.212 (20.06)	4.141 (16.42)	4.038 (14.42)
40%	4.230 (23.58)	4.110 (18.51)	4.188 (22.44)	4.281 (23.26)	4.198 (19.81)	4.314 (23.41)	4.243 (20.95)	4.395 (23.59)	4.302 (21.91)

SampleSize_n50_10Treatments/table3_shrink_10_treatments_MAR_n50

MR	None_T1vs.T0	None_T2vs.T0	None_T3vs.T0	None_T4vs.T0	None_T5vs.T0	None_T6vs.T0	None_T7vs.T0	None_T8vs.T0	None_T9vs.T0	None_T10vs.T0	Many_T1vs.T0
Scenario I: all means are 0 for T0,T1,...,T10											
0%	2.467	2.560	2.584	2.574	2.542	2.621	2.640	2.627	2.618	2.620	3.436
Imputation using linear regression											
10%	2.589 (4.96)	2.604 (1.71)	2.673 (3.45)	2.696 (4.75)	2.664 (4.77)	2.658 (1.42)	2.694 (2.04)	2.734 (4.06)	2.712 (3.59)	2.724 (3.98)	3.630 (5.67)
20%	2.764 (12.04)	2.772 (8.26)	2.798 (8.27)	2.769 (7.57)	2.852 (12.19)	2.803 (6.94)	2.859 (8.29)	2.880 (9.62)	2.855 (9.05)	2.793 (6.63)	3.819 (11.17)
30%	2.980 (20.79)	3.030 (18.36)	3.057 (18.30)	3.066 (19.11)	3.051 (20.00)	3.062 (16.83)	3.044 (15.33)	3.092 (17.68)	3.096 (18.27)	3.073 (17.30)	4.125 (20.07)
40%	3.400 (37.83)	3.318 (29.58)	3.330 (28.87)	3.425 (33.08)	3.413 (34.25)	3.447 (31.53)	3.404 (28.94)	3.430 (30.58)	3.400 (29.87)	3.484 (32.97)	4.696 (36.68)
Imputation using predictive mean matching method											
10%	2.530 (2.56)	2.558 (−0.08)	2.625 (1.59)	2.643 (2.68)	2.620 (3.06)	2.606 (−0.56)	2.647 (0.28)	2.679 (1.98)	2.669 (1.95)	2.676 (2.14)	3.550 (3.34)
20%	2.639 (6.97)	2.667 (4.17)	2.697 (4.36)	2.660 (3.35)	2.735 (7.56)	2.694 (2.79)	2.742 (3.89)	2.778 (5.74)	2.765 (5.63)	2.740 (4.58)	3.649 (6.20)
30%	2.804 (13.65)	2.835 (10.74)	2.838 (9.82)	2.874 (11.65)	2.856 (12.33)	2.830 (8.00)	2.851 (7.99)	2.905 (10.59)	2.922 (11.60)	2.843 (8.51)	3.895 (13.36)
40%	2.969 (20.35)	3.028 (18.26)	3.002 (16.15)	3.030 (17.70)	2.974 (16.98)	3.033 (15.73)	3.010 (14.03)	3.044 (15.87)	3.047 (16.39)	3.123 (19.23)	4.113 (19.72)
Imputation using propensity−score method											
10%	2.542 (3.06)	2.570 (0.39)	2.635 (1.96)	2.639 (2.52)	2.620 (3.03)	2.616 (−0.18)	2.641 (0.05)	2.677 (1.90)	2.668 (1.90)	2.681 (2.35)	3.571 (3.93)
20%	2.653 (7.54)	2.688 (5.00)	2.709 (4.83)	2.685 (4.31)	2.760 (8.57)	2.731 (4.22)	2.755 (4.35)	2.788 (6.13)	2.768 (5.74)	2.753 (5.10)	3.667 (6.74)
30%	2.780 (12.71)	2.812 (9.85)	2.886 (11.69)	2.843 (10.45)	2.901 (14.12)	2.855 (8.95)	2.896 (9.71)	2.934 (11.69)	2.905 (10.98)	2.868 (9.47)	3.871 (12.67)
40%	3.143 (27.39)	3.117 (21.73)	3.101 (20.00)	3.226 (25.31)	3.090 (21.53)	3.143 (19.93)	3.167 (19.98)	3.175 (20.87)	3.166 (20.93)	3.256 (24.27)	4.343 (26.42)
Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu10=6.0											
0%	2.484	2.503	2.536	2.501	2.540	2.562	2.556	2.565	2.601	2.581	3.444
Imputation using linear regression											

MR	None_T1vs.T0	None_T2vs.T0	None_T3vs.T0	None_T4vs.T0	None_T5vs.T0	None_T6vs.T0	None_T7vs.T0	None_T8vs.T0	None_T9vs.T0	None_T10vs.T0	Many_T1vs.T0
10%	2.662 (7.17)	2.572 (2.75)	2.639 (4.06)	2.654 (6.13)	2.638 (3.87)	2.643 (3.16)	2.664 (4.22)	2.653 (3.41)	2.702 (3.91)	2.689 (4.20)	3.636 (5.57)
20%	2.687 (8.19)	2.769 (10.60)	2.763 (8.94)	2.739 (9.50)	2.790 (9.85)	2.805 (9.47)	2.807 (9.80)	2.833 (10.41)	2.854 (9.75)	2.853 (10.56)	3.794 (10.15)
30%	2.992 (20.47)	3.060 (22.23)	2.979 (17.44)	3.074 (22.92)	3.050 (20.11)	3.074 (19.99)	3.152 (23.31)	3.171 (23.61)	3.162 (21.58)	3.084 (19.51)	4.126 (19.80)
40%	3.248 (30.78)	3.316 (32.46)	3.289 (29.70)	3.305 (32.14)	3.350 (31.91)	3.399 (32.64)	3.384 (32.38)	3.375 (31.55)	3.469 (33.41)	3.381 (31.00)	4.515 (31.08)
Imputation using predictive mean matching method											
10%	2.601 (4.73)	2.511 (0.30)	2.579 (1.67)	2.594 (3.71)	2.583 (1.69)	2.581 (0.74)	2.606 (1.95)	2.596 (1.17)	2.638 (1.45)	2.637 (2.17)	3.559 (3.33)
20%	2.573 (3.62)	2.640 (5.44)	2.660 (4.89)	2.654 (6.13)	2.678 (5.44)	2.673 (4.32)	2.713 (6.12)	2.720 (6.01)	2.741 (5.39)	2.703 (4.73)	3.645 (5.83)
30%	2.838 (14.27)	2.836 (13.30)	2.790 (9.99)	2.906 (16.17)	2.875 (13.22)	2.901 (13.21)	2.947 (15.30)	3.018 (17.64)	2.946 (13.27)	2.855 (10.65)	3.895 (13.08)
40%	2.971 (19.63)	2.948 (17.75)	2.924 (15.27)	2.968 (18.68)	2.983 (17.46)	3.003 (17.20)	3.038 (18.86)	3.012 (17.39)	3.027 (16.40)	3.023 (17.12)	4.129 (19.87)
Imputation using propensity−score method											
10%	2.615 (5.28)	2.534 (1.24)	2.591 (2.14)	2.607 (4.24)	2.592 (2.07)	2.611 (1.89)	2.634 (3.03)	2.612 (1.83)	2.637 (1.40)	2.658 (3.00)	3.574 (3.76)
20%	2.626 (5.75)	2.683 (7.19)	2.706 (6.70)	2.693 (7.67)	2.731 (7.52)	2.719 (6.13)	2.749 (7.54)	2.765 (7.79)	2.767 (6.40)	2.758 (6.86)	3.705 (7.56)
30%	2.861 (15.18)	2.905 (16.05)	2.854 (12.54)	2.971 (18.78)	2.949 (16.13)	2.947 (15.03)	2.986 (16.81)	3.058 (19.19)	3.005 (15.57)	2.932 (13.63)	3.937 (14.29)
40%	2.997 (20.68)	3.063 (22.38)	2.976 (17.32)	3.032 (21.24)	3.100 (22.07)	3.038 (18.56)	3.123 (22.19)	3.072 (19.73)	3.183 (22.41)	3.115 (20.69)	4.168 (21.02)

MR	Many_T2vs.T0	Many_T3vs.T0	Many_T4vs.T0	Many_T5vs.T0	Many_T6vs.T0	Many_T7vs.T0	Many_T8vs.T0	Many_T9vs.T0	Many_T10vs.T0
Scenario I: all means are 0 for T0,T1,...,T10									
0%	3.460	3.478	3.481	3.464	3.495	3.501	3.494	3.493	3.502
Imputation using linear regression									
10%	3.620 (4.63)	3.650 (4.96)	3.688 (5.93)	3.656 (5.56)	3.661 (4.77)	3.653 (4.33)	3.689 (5.59)	3.671 (5.10)	3.684 (5.20)
20%	3.822 (10.48)	3.838 (10.35)	3.810 (9.46)	3.876 (11.91)	3.847 (10.10)	3.893 (11.19)	3.882 (11.12)	3.855 (10.36)	3.817 (8.98)
30%	4.172 (20.60)	4.197 (20.67)	4.202 (20.70)	4.196 (21.15)	4.217 (20.67)	4.181 (19.40)	4.228 (21.02)	4.232 (21.15)	4.212 (20.26)
40%	4.565 (31.96)	4.585 (31.85)	4.693 (34.80)	4.695 (35.54)	4.735 (35.50)	4.665 (33.24)	4.699 (34.50)	4.664 (33.51)	4.750 (35.62)
Imputation using predictive mean matching method									
10%	3.549 (2.59)	3.584 (3.04)	3.605 (3.56)	3.589 (3.61)	3.582 (2.51)	3.583 (2.33)	3.614 (3.44)	3.605 (3.22)	3.612 (3.13)
20%	3.678 (6.30)	3.692 (6.17)	3.652 (4.90)	3.708 (7.04)	3.688 (5.53)	3.726 (6.40)	3.738 (6.98)	3.727 (6.71)	3.725 (6.36)
30%	3.907 (12.94)	3.893 (11.93)	3.942 (13.23)	3.925 (13.33)	3.901 (11.62)	3.903 (11.48)	3.965 (13.48)	3.987 (14.14)	3.896 (11.25)
40%	4.161 (20.27)	4.133 (18.83)	4.145 (19.06)	4.091 (18.09)	4.168 (19.27)	4.119 (17.65)	4.164 (19.18)	4.176 (19.57)	4.249 (21.32)
Imputation using propensity−score method									
10%	3.572 (3.25)	3.598 (3.47)	3.609 (3.67)	3.594 (3.75)	3.602 (3.06)	3.583 (2.34)	3.618 (3.54)	3.608 (3.31)	3.629 (3.61)
20%	3.707 (7.16)	3.711 (6.69)	3.691 (6.03)	3.748 (8.21)	3.747 (7.22)	3.738 (6.75)	3.757 (7.53)	3.735 (6.94)	3.747 (6.98)
30%	3.894 (12.56)	3.980 (14.45)	3.913 (12.41)	4.004 (15.58)	3.950 (13.04)	3.983 (13.75)	4.022 (15.12)	3.981 (13.97)	3.950 (12.77)
40%	4.281 (23.74)	4.267 (22.69)	4.406 (26.57)	4.252 (22.75)	4.317 (23.54)	4.333 (23.75)	4.341 (24.25)	4.343 (24.33)	4.428 (26.44)
Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu10=6.0									
0%	3.447	3.453	3.448	3.464	3.454	3.466	3.467	3.470	3.467
Imputation using linear regression									

MR	Many_T2vs.T0	Many_T3vs.T0	Many_T4vs.T0	Many_T5vs.T0	Many_T6vs.T0	Many_T7vs.T0	Many_T8vs.T0	Many_T9vs.T0	Many_T10vs.T0
10%	3.583 (3.95)	3.624 (4.96)	3.621 (5.02)	3.624 (4.60)	3.611 (4.53)	3.630 (4.75)	3.623 (4.49)	3.657 (5.36)	3.631 (4.71)
20%	3.838 (11.36)	3.821 (10.66)	3.806 (10.36)	3.858 (11.35)	3.865 (11.88)	3.865 (11.52)	3.899 (12.44)	3.892 (12.15)	3.909 (12.74)
30%	4.222 (22.48)	4.103 (18.85)	4.206 (21.96)	4.188 (20.88)	4.223 (22.25)	4.282 (23.56)	4.315 (24.47)	4.289 (23.58)	4.232 (22.04)
40%	4.599 (33.42)	4.574 (32.47)	4.563 (32.32)	4.635 (33.78)	4.684 (35.59)	4.665 (34.60)	4.655 (34.25)	4.762 (37.21)	4.663 (34.47)
Imputation using predictive mean matching method									
10%	3.506 (1.73)	3.546 (2.69)	3.539 (2.65)	3.551 (2.50)	3.533 (2.28)	3.545 (2.28)	3.554 (2.51)	3.567 (2.78)	3.557 (2.58)
20%	3.672 (6.53)	3.688 (6.82)	3.690 (7.01)	3.718 (7.33)	3.689 (6.79)	3.744 (8.03)	3.745 (8.01)	3.736 (7.66)	3.715 (7.14)
30%	3.914 (13.56)	3.835 (11.07)	3.967 (15.04)	3.938 (13.66)	3.980 (15.21)	4.000 (15.42)	4.088 (17.92)	3.975 (14.53)	3.910 (12.77)
40%	4.100 (18.95)	4.078 (18.11)	4.104 (19.01)	4.132 (19.26)	4.142 (19.90)	4.184 (20.72)	4.156 (19.86)	4.156 (19.76)	4.171 (20.29)
Imputation using propensity−score method									
10%	3.531 (2.45)	3.561 (3.13)	3.562 (3.31)	3.563 (2.83)	3.568 (3.29)	3.577 (3.22)	3.573 (3.06)	3.567 (2.78)	3.585 (3.40)
20%	3.722 (7.99)	3.734 (8.16)	3.730 (8.18)	3.777 (9.04)	3.745 (8.42)	3.787 (9.27)	3.801 (9.63)	3.762 (8.39)	3.780 (9.02)
30%	4.020 (16.64)	3.934 (13.95)	4.061 (17.77)	4.047 (16.81)	4.054 (17.37)	4.064 (17.27)	4.155 (19.85)	4.073 (17.35)	4.026 (16.12)
40%	4.252 (23.36)	4.142 (19.95)	4.193 (21.59)	4.290 (23.83)	4.195 (21.44)	4.305 (24.22)	4.244 (22.40)	4.371 (25.95)	4.303 (24.08)